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THE MOTION AND STRUCTURE OF SINGULARITIES
IN GENERAL RELATIVITY

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FOREWORD

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1.0. INTRODUCTION

The study of motion is of great importance and interest in general relativity because of the unique feature of Einstein's field equations which makes them determine the motion of their localized sources. In other field theories such as electromagnetic theory the equations of motion have to be added to the field equations as an independent postulate. In general relativity, however, the equations of motion are derivable from the field equations and may not be postulated independently.

Historically, the postulate of geodesic motion for test bodies was introduced by Einstein¹ alongside his field equations in his original formulation of general relativity in 1916. Eleven years later he and Grommer² showed that the equations of motion for test bodies could actually be derived from the field equations and need not be postulated separately. Then in 1938 the problem of motion for heavy bodies was solved for the first time by Einstein, Infeld, and Hoffmann (EIH)³ who showed that the equations of motion for two bodies could also be determined from the field equations. Independently of EIH, Fock⁴ a year later also obtained the two-body equations of motion but only in the Newtonian approximation. The post-Newtonian equations of motion were later obtained by Papapetrou⁵ with a simplified version of Fock's method and shown to be the same as the results of EIH. Since then a number of refinements and new techniques have been developed to study motion in general relativity.⁶

In order to discuss how equations of motion arise in general relativity, it will be necessary first to give a brief review of the theory.⁷ General relativity is essentially a geometric theory of gravitation in which

space-time is regarded as a four-dimensional, normal, hyperbolic Riemannian manifold with signature(+---). The metric tensor $g_{\mu\nu}$, which is defined by the line element*

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad (1.1)$$

plays the role of gravitational potential and the Riemann tensor $R_{\mu\nu\rho\sigma}$, which is a non-linear function of $g_{\mu\nu}$ and its partial derivatives up to second order, represents the curvature of space-time due to the presence of matter. The relation between the metric field $g_{\mu\nu}$ and its material sources is expressed by Einstein's field equations

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = -4\pi k T_{\mu\nu} \quad (1.2)$$

where k is a universal constant[†], $T_{\mu\nu}$ is the stress-energy-momentum tensor of the matter distribution, and $G_{\mu\nu}$ is the Einstein tensor, with the Ricci tensor $R_{\mu\nu}$ defined by $R_{\mu\nu} = R^\alpha_{\mu\alpha\nu}$ and the scalar curvature R defined by $R = R^\alpha_\alpha$.

Now, the two properties of these field equations which make them impose restrictions on their sources are: general covariance and non-linearity.^{8,9} The covariance of the theory with respect to arbitrary coordinate transformations implies that $G^{\mu\nu}$ satisfy four differential identities, namely, the contracted Bianchi identities**

* Greek indices are to range and sum over 0,1,2,3; lower case Latin indices, over 1,2,3; and upper case Latin indices, over 2,3.

† The constant k is related to the Newtonian gravitational constant G and the velocity of light c by means of the equation $k = 2G/c^4$. In our calculations we shall generally take $c = 1$.

**A semicolon shall denote covariant differentiation; a comma, partial differentiation.

$$G^{\mu\nu};_{\nu} \equiv 0 \quad (1.3)$$

which in turn impose restrictions on $T^{\mu\nu}$ via the field equations, i.e.

$$T^{\mu\nu};_{\nu} = 0 \quad (1.4)$$

It follows from this that the distribution of matter cannot be arbitrarily assigned if a solution of (1.2) is to exist. Thus, by requiring that the field equations be satisfied whenever (1.4) is satisfied, one can obtain the equations of motion of the sources described by $T^{\mu\nu}$. Moreover, the non-linearity of the field equations ensures that the equations of motion so obtained are non-trivial, i.e. they give rise to interactions between mass points, for example. Some linear field theories, which are invariant under a group depending on arbitrary functions, also restrict their sources but they always lead to trivial equations of motion with no interactions.¹⁰

The problem of motion in general relativity is actually concerned with two related, but separate, questions. The first is the motion of a test body (a body with vanishing mass) in a given gravitational field. The second is the n-body problem where the task is to find both the motion of a finite number n of heavy bodies (bodies with finite mass) and the gravitational field produced by these n bodies.

In the case of test bodies, one considers a timelike world tube and assumes that the matter tensor $T^{\mu\nu}$ is small but finite inside the tube and zero outside. Ultimately, the tube is to be shrunk to a single world line by means of a suitable limiting process which makes $T^{\mu\nu}$ zero everywhere except on the world line.^{6c} By applying a perturbation method to (1.2) and (1.4), one then obtains equations of motion in the form of algebraic relations and differential equations for the mass multipole moments defined in terms of

$T^{\mu\nu}$,¹¹⁻¹⁴ Alternatively, one could find the equations of motion of the test particle by analyzing the gravitational field outside the world line of a particle of mass m (treated as a singularity of the field) and then taking the limit $m \rightarrow 0$.^{15,16} In either way, it can be shown that the world line of a test monopole particle is necessarily a geodesic.

For the n -body problem, there are basically two methods that have been used to derive equations of motion: the EIH method which is based on the vanishing of a certain two-dimensional surface integral¹⁷⁻²⁰ and the method of Fock and Papapetrou which makes explicit use of the matter tensor.²¹⁻²⁴ In general, one considers a finite number of localized bodies each of which is enclosed within a timelike world tube and assumes that $T^{\mu\nu}$ vanishes outside these tubes. It is also assumed that the system of bodies is isolated and that the gravitational field is relatively weak so that space-time is considered to be asymptotically flat. Thus, one assumes a flat background space with Minkowski metric $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ and takes the perturbations $h_{\mu\nu}$ on this space,

$$h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu} \quad (1.5)$$

to represent the gravitational field produced by the localized sources within the tubes.

In the EIH method, matter is assumed to be composed of point particles which are represented by singularities of the field. In this way, the choice of a special form for the matter tensor is avoided. From the structure of the field outside the tubes containing the singularities, one then tries to deduce the motion of the singularities producing the field.

Now, the free field equations $G_{\mu\nu} = 0$, which hold outside the tubes, can be expressed equivalently as the vanishing of a two-dimensional surface integral enclosing a singularity, that is,²⁵

$$\int [t_{\mu}^{m} - U_{\mu}^{m\sigma}{}_{,0}] n_m dS = 0, \quad (1.6)$$

where the Einstein pseudo-tensor t_{μ}^{v} and the antisymmetric superpotential $U_{\mu}^{v\sigma}$ are related to the Einstein tensor by $2\sqrt{-g} G_{\mu}^{v} = t_{\mu}^{v} - U_{\mu}^{v\sigma}{}_{,\sigma}$, with $g \equiv \det(g_{\mu\nu})$. With the use of approximation procedures, these integrals can be shown to yield explicit equations of motion.^{6a,b}

In the method of Fock and Papapetrou, matter is assumed to be continuously distributed within the tubes so that there are no singularities in the field. Thus one deals with the combined gravitational and hydrodynamical problem of a finite number of extended bodies. The basis for this approach is $T^{\mu\nu}$; $\nu = 0$, which must be solved simultaneously with the field equations (1.2) by means of a self-consistent approximation method. The usual procedure is to perform a power series expansion on $g_{\mu\nu}$ and $T_{\mu\nu}$ and then solve the resulting linearized equations in stepwise fashion. A sort of compromise between this method and that of EIH has been developed by Infeld and his co-workers²⁶⁻²⁸, who also represent the sources by point singularities but introduce these singularities into the matter tensor by means of a modified Dirac δ -function, namely, $\hat{\delta}(x)$. A similar approach has been used by Havas²⁹ to solve the two-body problem.

It is also possible to derive equations of motion for charged particles in general relativity from the coupled Einstein-Maxwell

field equations

$$\begin{aligned} G^{\mu\nu} &= -4\pi k E^{\mu\nu}, \quad F^{\mu\nu}{}_{;\nu} = 4\pi J^\mu, \quad *F^{\mu\nu}{}_{;\nu} = 0, \\ E^{\mu\nu} &= \frac{1}{4\pi} (F^{\mu\rho} F_{\rho}{}^{\nu} + \frac{1}{4} \varepsilon^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta}), \end{aligned} \quad (1.7)$$

where $E^{\mu\nu}$ is the electromagnetic stress-energy-momentum tensor, $F^{\mu\nu}$ is the electromagnetic field tensor ($*F^{\mu\nu}$ being its dual), and J^μ is the charge-current vector. The resulting equations of motion will then be the curved-space generalization of the flat space Lorentz-Dirac equation³⁰

$$m\ddot{\xi}^\mu = e\dot{\xi}_\nu F^{\nu\mu} + \frac{2}{3} e^2 (\ddot{\xi}^\mu + \ddot{\xi}^\alpha \ddot{\xi}_\alpha \dot{\xi}^\mu), \quad (1.8)$$

where the first term on the right is the Lorentz force on the charge e due to an incoming, external field $F^{\mu\nu}$, the second term is the Abraham radiation reaction force, and a dot denotes differentiation with respect to the proper time. In 1940 Infeld and Wallace³¹ were able to derive (1.8) from (1.7) by using an adaptation of the EIH method in which they neglected non-linear terms in $G^{\mu\nu}$ and gravitational-electromagnetic coupling terms in $E^{\mu\nu}$ and assumed that the electromagnetic field could be divided into an incoming, external field and the field of the charge, with the latter being taken to be a retarded field. Their method is thus unsatisfactory because it contains ambiguities in the identification of the fields and fails to incorporate the full non-linear dynamics of the Einstein-Maxwell theory.

With this historical background to the problem of motion in general relativity, we present our own approach to equations of motion,³² which differs in several respects from the standard approaches described above.

First, our approach is based on an analysis of the structure of the light cone in the neighborhood of a particle. Secondly, we use the Newman-Penrose spin coefficient formulation of Einstein's field equations to derive the equations of motion. And lastly, our approach provides an intrinsic description of the motion of a particle in its own space-time, with no reference to a regular background space.

We adopt the point of view that matter is to be represented by suitably defined singularities in the field, the field being expressed by the Weyl tensor (see Chapter 2.0). Thus we consider a timelike world line, along which the field becomes singular, and attach a null coordinate system with an associated null tetrad to this singular line. A suitable definition of the singularity is then given in terms of the properties of the null "cones" emanating from the line. Applying Einstein's field equations in spin coefficient formalism, we are able to obtain rigorous equations of motion for the singularity in terms of the behavior of the family of null cones associated with the line. Furthermore, we obtain the concomitant result that the point singularity has an internal structure, with an equation governing the time development of this structure. The other major result of this approach is the rigorous derivation of equations of motion for charged particles from the Einstein-Maxwell theory, which reduce in the linear approximation to the Lorentz-Dirac equation.

In this work we have restricted ourselves to one-particle systems of non-spinning, non-twisting singularities having no intrinsic dipole moment. First, we present the general formalism in Chapter 2.0 with a discussion of the Newman-Penrose formalism,³³ the differential operator δ (edth) and a related class of generalized spherical functions³⁴, a special

null coordinate system,³⁵ and the basic assumptions to be used.³² This is followed by Chapter 3.0 where we analyze and interpret the Robinson-Trautman (RT) type II metrics³⁶⁻³⁹ and their charged counterparts, namely, the Robinson-Trautman-Maxwell (RTM) metrics.³² Here we show that these two special classes of solutions lead to equations of motion for "free singularities" (i.e. singularities not interacting with incoming fields), with the unexpected appearance of the Abraham radiation reaction term in the charged case.³² In Chapter 4.0 we then couple these RT-type singularities to incoming fields in a general empty space ($R_{\mu\nu} = 0$) and discuss the resulting equations for the motion and structure of singularities. In Chapter 5.0 we give the derivation of equations of motion in the Einstein-Maxwell theory and show that in the linear approximation these equations reduce to the Lorentz-Dirac equation. Finally, a summary of the work is given in Chapter 6.0.

2.0. GENERAL FORMALISM

2.1. The Spin Coefficient Formalism

We give here a summary of the Newman-Penrose (NP) spin coefficient formulation of general relativity.³³ Into a four-dimensional Riemannian space a set of four null vectors $(\ell^\mu, n^\mu, m^\mu, \bar{m}^\mu)$ is introduced such that the real vectors, ℓ^μ and n^μ , together with m^μ and its complex conjugate $\bar{m}^\mu$ form a null tetrad system, with the orthonormality properties

$$\begin{aligned}\ell_\mu n^\mu &= -m_\mu \bar{m}^\mu = 1, \\ \ell_\mu \ell^\mu &= \bar{m}_\mu \bar{m}^\mu = m_\mu m^\mu = n_\mu n^\mu = 0 \\ \ell_\mu m^\mu &= \ell_\mu \bar{m}^\mu = n_\mu m^\mu = n_\mu \bar{m}^\mu = 0\end{aligned}\tag{2.1}$$

and the completeness relation

$$g^{\mu\nu} = \ell^\mu n^\nu + n^\mu \ell^\nu - m^\mu \bar{m}^\nu - \bar{m}^\mu m^\nu\tag{2.2}$$

The complex null vector m^μ can be defined from a pair of real, orthogonal, spacelike unit vectors b^μ and c^μ by $m^\mu = \frac{1}{\sqrt{2}}(b^\mu + ic^\mu)$.

Now, as mentioned in the Introduction, the curvature of space-time is represented by the Riemann tensor $R_{\mu\nu\rho\sigma}$ which is defined through the Ricci identity⁷

$$A_{\nu;[\rho\sigma]} = \frac{1}{2} A^\mu R_{\mu\nu\rho\sigma}\tag{2.3}$$

for any vector A_ν , with the brackets denoting antisymmetrization. This tensor has the following algebraic symmetries⁷

$$R_{\mu\nu\rho\sigma} = R_{[\mu\nu][\rho\sigma]} = R_{\rho\sigma\mu\nu}, \quad R_{\mu[\nu\rho\sigma]} = 0$$

and satisfies the so-called Bianchi identities

$$R_{\mu\nu}[\rho\sigma;\lambda] = 0 \quad (2.4)$$

The Weyl (conformal curvature) tensor $C_{\mu\nu\rho\sigma}$, which will be our basic field variable, is defined as

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} + \frac{1}{2} (g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} + g_{\nu\sigma}R_{\mu\rho} - g_{\nu\rho}R_{\mu\sigma}) + \frac{R}{6} (g_{\mu\sigma}g_{\nu\rho} - g_{\mu\rho}g_{\nu\sigma}), \quad (2.5)$$

where $R_{\mu\nu} \equiv R^{\alpha}_{\mu\alpha\nu}$ is the Ricci tensor and $R \equiv R^{\alpha}_{\alpha}$ is the scalar curvature. The latter two quantities, of course, are related to the matter tensor $T_{\mu\nu}$ by means of the Einstein field equations given in the Introduction, namely,

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = -4\pi k T_{\mu\nu} \quad (1.2)$$

The idea now is to rewrite the Ricci identity (applied to each of the tetrad vectors) and the Bianchi identities as scalar equations by taking appropriate linear combinations of their scalar products with the tetrad vectors. The resulting set of equations will be expressed in terms of the five complex tetrad components of the Weyl tensor defined as

$$\begin{aligned} \psi_0 &= -C_{\mu\nu\rho\sigma} \ell^{\mu} m^{\nu} \ell^{\rho} m^{\sigma}, \\ \psi_1 &= -C_{\mu\nu\rho\sigma} \ell^{\mu} n^{\nu} \ell^{\rho} m^{\sigma}, \\ \psi_2 &= -C_{\mu\nu\rho\sigma} \bar{m}^{\mu} n^{\nu} \ell^{\rho} m^{\sigma}, \\ \psi_3 &= -C_{\mu\nu\rho\sigma} \bar{m}^{\mu} n^{\nu} \ell^{\rho} n^{\sigma}, \\ \psi_4 &= -C_{\mu\nu\rho\sigma} \bar{m}^{\mu} n^{\nu} \bar{m}^{\rho} n^{\sigma}; \end{aligned} \quad (2.6)$$

the scalar curvature plus the tetrad components of the Ricci tensor defined as

$$\begin{aligned}
 \phi_{00} &= -\frac{1}{2} R_{\mu\nu} \ell^\mu \ell^\nu, & \phi_{10} &= -\frac{1}{2} R_{\mu\nu} \ell^\mu \bar{m}^\nu, \\
 \phi_{01} &= -\frac{1}{2} R_{\mu\nu} \ell^\mu m^\nu, & \phi_{02} &= -\frac{1}{2} R_{\mu\nu} m^\mu m^\nu, \\
 \phi_{20} &= -\frac{1}{2} R_{\mu\nu} \bar{m}^\mu \bar{m}^\nu, & \phi_{21} &= -\frac{1}{2} R_{\mu\nu} n^\mu \bar{m}^\nu, \\
 \phi_{12} &= -\frac{1}{2} R_{\mu\nu} n^\mu m^\nu, & \phi_{22} &= -\frac{1}{2} R_{\mu\nu} n^\mu n^\nu, \\
 \phi_{11} &= -\frac{1}{4} R_{\mu\nu} (\ell^\mu n^\nu + m^\mu \bar{m}^\nu);
 \end{aligned} \tag{2.7}$$

and the twelve spin coefficients defined as

$$\begin{aligned}
 \kappa &= \ell_{\mu;\nu} m^\mu \ell^\nu, & \nu &= -n_{\mu;\nu} \bar{m}^\mu n^\nu, \\
 \rho &= \ell_{\mu;\nu} m^\mu \bar{m}^\nu, & \mu &= -n_{\mu;\nu} \bar{m}^\mu m^\nu, \\
 \sigma &= \ell_{\mu;\nu} m^\mu m^\nu, & \lambda &= -n_{\mu;\nu} \bar{m}^\mu \bar{m}^\nu, \\
 \tau &= \ell_{\mu;\nu} m^\mu n^\nu, & \pi &= -n_{\mu;\nu} \bar{m}^\mu \ell^\nu, \\
 \alpha &= \frac{1}{2} (\ell_{\mu;\nu} n^\mu \bar{m}^\nu - m_{\mu;\nu} \bar{m}^\mu \bar{m}^\nu), \\
 \beta &= \frac{1}{2} (\ell_{\mu;\nu} n^\mu m^\nu - m_{\mu;\nu} \bar{m}^\mu m^\nu), \\
 \gamma &= \frac{1}{2} (\ell_{\mu;\nu} n^\mu n^\nu - m_{\mu;\nu} \bar{m}^\mu n^\nu), \\
 \epsilon &= \frac{1}{2} (\ell_{\mu;\nu} n^\mu \ell^\nu - m_{\mu;\nu} \bar{m}^\mu \ell^\nu).
 \end{aligned} \tag{2.8}$$

We shall be interested here not only in the empty space equations, $R_{\mu\nu} = 0$ (or $R = \phi_{mn} = 0$; $mn = 0,1,2$) but also in the Einstein-Maxwell equations given in the Introduction. The source-free Maxwell equations can be rewritten in terms of the spin coefficients and the three complex tetrad

components of the Maxwell tensor defined as

$$\begin{aligned}\phi_0 &= F_{\mu\nu} \ell^\mu m^\nu, \\ \phi_1 &= \frac{1}{2} F_{\mu\nu} (\ell^\mu n^\nu + \bar{m}^\mu m^\nu), \\ \phi_2 &= F_{\mu\nu} \bar{m}^\mu n^\nu,\end{aligned}\tag{2.9}$$

and the relation $G_{\mu\nu} = -4\pi k E_{\mu\nu}$ can be expressed as

$$\phi_{mn} = k \phi_m \bar{\phi}_n \quad (m, n = 0, 1, 2).\tag{2.10}$$

Before writing down the spin coefficient equations, we shall first choose a suitable coordinate system. Since our approach to the problem of motion will be based on the structure of the null cones emanating from a singular world line in space-time, we introduce null coordinates³⁵ $x^0 = u$, $x^1 = r$, and x^A ($A = 2, 3$) such that $\sqrt{2} u$ is a "retarded time" parameter labeling a family of outgoing null hypersurfaces, $r/\sqrt{2}$ is a standard affine parameter for each of the null geodesics lying on the null hypersurfaces $u = \text{const}$, and x^A are "angular" coordinates labeling the null geodesics. In this null coordinate system we let ℓ^μ be tangent to the null geodesics as well as normal to the hypersurfaces ($\ell^\mu = dx^\mu/dr$, $\ell_\mu = \partial u / \partial x^\mu$), and parallelly propagate the rest of the tetrad system along ℓ^μ . This leads to the following form for the tetrad vectors³⁵

$$\begin{aligned}\ell_\mu &= \delta_\mu^0, \quad \ell^\mu = \delta_1^\mu, \quad m^\mu = \omega \delta_1^\mu + \xi^A \delta_A^\mu, \\ n^\mu &= \delta_0^\mu + U \delta_1^\mu + X^A \delta_A^\mu, \quad A = 2, 3\end{aligned}\tag{2.11}$$

and the following conditions on the spin coefficients

$$\kappa = \varepsilon = \pi = 0, \quad \rho = \bar{\rho}, \quad \tau = \alpha + \beta. \quad (2.12)$$

Using the completeness relation (2.2), we then obtain the following form for the contravariant metric tensor

$$g^{\mu\nu} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & g^{11} & g^{12} & g^{13} \\ 0 & g^{12} & g^{AB} & \\ 0 & g^{13} & & \end{bmatrix} \quad (2.13a)$$

where

$$\begin{aligned} g^{\mu 0} &= \delta_1^\mu, \quad g^{11} = 2(U - \omega\bar{\omega}), \\ g^{1A} &= X^A - (\xi^A \bar{\omega} + \omega \bar{\xi}^A), \\ g^{AB} &= -(\xi^A \bar{\xi}^B + \bar{\xi}^A \xi^B), \quad A = 2, 3. \end{aligned} \quad (2.13b)$$

With this choice of coordinates and tetrad, the two spin coefficients ρ and σ , which will play a significant role in our approach to equations of motion, can be related to the divergence and complex shear,³³ respectively, of the null geodesics, that is,

$$\begin{aligned} \rho &= -\frac{1}{2} l^\mu{}_{;\mu}, \\ \sigma\bar{\sigma} &= \frac{1}{2} [l_{(\mu;\nu)} l^{\mu;\nu} - \frac{1}{2} (l^\mu{}_{;\mu})^2]. \end{aligned} \quad (2.14)$$

At this point the null hypersurfaces are still arbitrary, but later they will be made definite by the imposition of certain conditions on ρ and σ . (See Section 2.4).

Defining the intrinsic derivatives D , δ , $\bar{\delta}$, and Δ acting on a scalar ϕ by

$$\begin{aligned} D\phi &\equiv \phi_{;\mu} l^\mu = \frac{\partial \phi}{\partial r} , \\ \delta\phi &\equiv \phi_{;\mu} m^\mu = \omega \frac{\partial \phi}{\partial r} + \xi^A \frac{\partial \phi}{\partial x^A} , \\ \bar{\delta}\phi &\equiv \phi_{;\mu} \bar{m}^\mu = \omega \frac{\partial \phi}{\partial r} + \bar{\xi}^A \frac{\partial \phi}{\partial x^A} , \\ \Delta\phi &\equiv \phi_{;\mu} n^\mu = U \frac{\partial \phi}{\partial r} + \frac{\partial \phi}{\partial u} + X^A \frac{\partial \phi}{\partial x^A} , \end{aligned} \quad (2.15)$$

we can now write down the NP spin coefficient equations for the Einstein-Maxwell theory.⁴⁰

I. Metric Equations:

$$D\xi^A = \rho\xi^A + \sigma\bar{\xi}^A , \quad (2.16a)$$

$$D\omega = \rho\omega + \sigma\bar{\omega} - \tau , \quad (2.16b)$$

$$DX^A = \tau\bar{\xi}^A + \bar{\tau}\xi^A , \quad (2.16c)$$

$$DU = \tau\bar{\omega} + \bar{\tau}\omega - (\gamma + \bar{\gamma}) , \quad (2.16d)$$

$$\delta X^A - \Delta\xi^A = (\mu + \bar{\gamma} - \gamma)\xi^A + \bar{\lambda}\bar{\xi}^A , \quad (2.17a)$$

$$\delta\bar{\xi}^A - \bar{\delta}\xi^A = (\bar{\beta} - \alpha)\xi^A + (\bar{\alpha} - \beta)\bar{\xi}^A \quad (2.17b)$$

$$\delta\bar{\omega} - \bar{\delta}\omega = (\bar{\beta} - \alpha)\omega + (\bar{\alpha} - \beta)\bar{\omega} + (\mu - \bar{\mu}) , \quad (2.17c)$$

$$\delta U - \Delta\omega = (\mu + \bar{\gamma} - \gamma)\omega + \bar{\lambda}\bar{\omega} - \bar{\nu} . \quad (2.17d)$$

II. Field Equations:

$$D\rho = \rho^2 + \sigma\bar{\sigma} + k\phi_0\bar{\phi}_0, \quad (2.18a)$$

$$D\sigma = 2\rho\sigma + \psi_0, \quad (2.18b)$$

$$D\alpha = \sigma\rho + \beta\bar{\sigma} + k\phi_1\bar{\phi}_0, \quad (2.18c)$$

$$D\beta = \beta\rho + \alpha\sigma + \psi_1, \quad (2.18d)$$

$$D\tau = \tau\rho + \bar{\tau}\sigma + \psi_1 + k\phi_0\bar{\phi}_1, \quad (2.18e)$$

$$D\gamma = \tau\alpha + \bar{\tau}\beta + \psi_2 + k\phi_1\bar{\phi}_1, \quad (2.18f)$$

$$D\lambda = \lambda\rho + \mu\bar{\sigma} + k\phi_2\bar{\phi}_0, \quad (2.18g)$$

$$D\mu = \mu\rho + \lambda\sigma + \psi_2, \quad (2.18h)$$

$$D\nu = \tau\lambda + \bar{\tau}\mu + \psi_3 + k\phi_2\bar{\phi}_1, \quad (2.18i)$$

$$\Delta\lambda - \bar{\delta}\nu = 2\alpha\nu + (\bar{\gamma}-3\gamma-\mu-\bar{\mu})\lambda - \psi_4, \quad (2.19a)$$

$$\delta\rho - \bar{\delta}\sigma = \tau\rho + (\bar{\beta}-3\alpha)\sigma - \psi_1 + k\phi_0\bar{\phi}_1, \quad (2.19b)$$

$$\delta\alpha - \bar{\delta}\beta = \mu\rho - \lambda\sigma - 2\alpha\beta + \alpha\bar{\alpha} + \beta\bar{\beta} - \psi_2 + k\phi_1\bar{\phi}_1, \quad (2.19c)$$

$$\delta\lambda - \bar{\delta}\mu = \bar{\tau}\mu + (\bar{\alpha}-3\beta)\lambda - \psi_3 + k\phi_2\bar{\phi}_1, \quad (2.19d)$$

$$\delta\nu - \Delta\mu = \gamma\mu - 2\nu\beta + \bar{\gamma}\mu + \mu^2 + \lambda\bar{\lambda} + k\phi_2\bar{\phi}_2, \quad (2.19e)$$

$$\delta\gamma - \Delta\beta = \tau\mu - \sigma\nu + (\mu-\gamma+\bar{\gamma})\beta + \bar{\lambda}\alpha + k\phi_1\bar{\phi}_2, \quad (2.19f)$$

$$\delta\tau - \Delta\sigma = 2\tau\beta + (\bar{\gamma}+\mu-3\gamma)\sigma + \bar{\lambda}\rho + k\phi_0\bar{\phi}_2, \quad (2.19g)$$

$$\Delta\rho - \bar{\delta}\tau = (\gamma+\bar{\gamma}-\bar{\mu})\rho - 2\alpha\tau - \lambda\sigma - \psi_2, \quad (2.19h)$$

$$\Delta\alpha - \bar{\delta}\gamma = \rho\nu - \tau\lambda - \lambda\beta + (\bar{\gamma}-\gamma-\bar{\mu})\alpha - \psi_3 \quad (2.19i)$$

III. Bianchi identities:

$$\begin{aligned}
D\psi_1 - \bar{\delta}\psi_0 &= 4\rho\psi_1 - 4\alpha\psi_0 + k(\bar{\phi}_1 D\phi_0 - \bar{\phi}_0 \delta\phi_0) \\
&\quad + 2k(\bar{\phi}_0 \phi_0 \beta - \bar{\phi}_0 \phi_1 \sigma), \quad (2.20a)
\end{aligned}$$

$$\begin{aligned}
D\psi_2 - \bar{\delta}\psi_1 &= 3\rho\psi_2 - 2\alpha\psi_1 - \lambda\psi_0 + k(\bar{\phi}_1 \bar{\delta}\phi_0 - \bar{\phi}_0 \Delta\phi_0) \\
&\quad + 2k(\bar{\phi}_0 \phi_0 \gamma - \bar{\phi}_1 \phi_0 \alpha - \bar{\phi}_0 \phi_1 \tau + \bar{\phi}_1 \phi_1 \rho), \quad (2.20b)
\end{aligned}$$

$$\begin{aligned}
D\psi_3 - \bar{\delta}\psi_2 &= 2\rho\psi_3 - 2\lambda\psi_1 + k(\bar{\phi}_1 D\phi_2 - \bar{\phi}_0 \delta\phi_2) \\
&\quad + 2k(\bar{\phi}_0 \phi_1 \mu - \bar{\phi}_0 \phi_2 \beta), \quad (2.20c)
\end{aligned}$$

$$\begin{aligned}
D\psi_4 - \bar{\delta}\psi_3 &= \rho\psi_4 + 2\alpha\psi_3 - 3\lambda\psi_2 + k(\bar{\phi}_1 \bar{\delta}\phi_2 - \bar{\phi}_0 \Delta\phi_2) \\
&\quad + 2k(\bar{\phi}_0 \phi_1 \nu - \bar{\phi}_1 \phi_1 \lambda - \bar{\phi}_0 \phi_2 \gamma + \bar{\phi}_1 \phi_2 \alpha), \quad (2.20d)
\end{aligned}$$

$$\begin{aligned}
\Delta\psi_0 - \delta\psi_1 &= (4\gamma - \mu)\psi_0 - (4\tau + 2\beta)\psi_1 + 3\sigma\psi_2 \\
&\quad + k(\bar{\phi}_1 \delta\phi_0 - \bar{\phi}_2 D\phi_0) + 2k(\bar{\phi}_1 \phi_1 \sigma - \bar{\phi}_1 \phi_0 \beta), \quad (2.21a)
\end{aligned}$$

$$\begin{aligned}
\Delta\psi_1 - \delta\psi_2 &= (2\gamma - 2\mu)\psi_1 - 3\tau\psi_2 + \nu\psi_0 + 2\sigma\psi_3 \\
&\quad + k(\bar{\phi}_1 \Delta\phi_0 - \bar{\phi}_2 \bar{\delta}\phi_0) + 2k(\bar{\phi}_2 \phi_0 \alpha + \bar{\phi}_1 \phi_1 \tau \\
&\quad - \bar{\phi}_1 \phi_0 \gamma - \bar{\phi}_2 \phi_1 \rho), \quad (2.21b)
\end{aligned}$$

$$\begin{aligned}
\Delta\psi_2 - \delta\psi_3 &= 2\nu\psi_1 - 3\mu\psi_2 - 2\bar{\alpha}\psi_3 + \sigma\psi_4 \\
&\quad + k(\bar{\phi}_1 \delta\phi_2 - \bar{\phi}_2 D\phi_2) + 2k(\bar{\phi}_1 \phi_2 \beta - \bar{\phi}_1 \phi_1 \mu), \quad (2.21c)
\end{aligned}$$

$$\begin{aligned}
\Delta\psi_3 - \delta\psi_4 &= 3\nu\psi_2 - (2\gamma + 4\mu)\psi_3 + (4\beta - \tau)\psi_4 \\
&\quad + k(\bar{\phi}_1 \Delta\phi_2 - \bar{\phi}_2 \bar{\delta}\phi_2) + 2k(\bar{\phi}_2 \phi_1 \lambda + \bar{\phi}_1 \phi_2 \gamma \\
&\quad - \bar{\phi}_1 \phi_1 \nu - \bar{\phi}_2 \phi_2 \alpha) \quad (2.21d)
\end{aligned}$$

IV. Maxwell's source-free equations:

$$D\phi_1 - \bar{\delta}\phi_0 = 2\rho\phi_1 - 2\alpha\phi_0, \quad (2.22a)$$

$$D\phi_2 - \bar{\delta}\phi_1 = \rho\phi_2 - \lambda\phi_0, \quad (2.22b)$$

$$\Delta\phi_0 - \delta\phi_1 = (2\gamma - \mu)\phi_0 - 2\tau\phi_1 + \sigma\phi_2, \quad (2.23a)$$

$$\Delta\phi_1 - \delta\phi_2 = \nu\phi_0 - 2\mu\phi_1 + (2\beta - \tau)\phi_2 \quad (2.23b)$$

To integrate these equations, the quantities ψ_0 and ϕ_0 , which are respectively the gravitational and electromagnetic null-data for a null hypersurface $u = \text{const}$ ⁴⁰, are specified as arbitrary (analytic) functions of r on an initial null hypersurface. The integration of the "radial" equations (2.16), (2.18), (2.20), and (2.22) then yields the r dependence of all the other variables, with "constants" of integration which are in general functions of the other three coordinates u, x^2, x^3 . From the "non-radial" equations (2.17) and (2.19), relationships between these integration "constants" are obtained. The remaining equations (2.21) and (2.23) determine the propagation of the components of the Weyl and Maxwell tensors off the initial null hypersurface. The analysis of the resulting expressions can be greatly simplified by the introduction of a differential operator $\eth$, called "edth". It will therefore be useful to give a brief discussion of this operator and the related class of generalized spherical functions called spin- s spherical harmonics.³⁴

2.2. $\mathfrak{S}$ and Spin-s Spherical Harmonics

Consider a rotation of the spacelike vectors $\text{Re}(m^\mu)$ and $\text{Im}(m^\mu)$ in their plane given by

$$m^{\mu'} = e^{i\chi} m^\mu, \quad (\chi \text{ is real}). \quad (2.24)$$

We define a quantity η to be of spin weight s if it transforms under (2.24) as

$$\eta' = e^{is\chi} \eta, \quad (s = \text{integral, in general}) \quad (2.25)$$

An example of such a quantity is $A_{\alpha_1 \dots \alpha_s} m^{\alpha_1} \dots m^{\alpha_s}$, where $A_{\alpha_1 \dots \alpha_s}$ is any covariant tensor of rank s . This concept of spin weight applies to functions on any two-dimensional abstract surface having a Riemannian structure, with $\text{Re}(m^\mu)$ and $\text{Im}(m^\mu)$ regarded as orthogonal tangent vectors at each point of the surface. From the definitions (2.6) and (2.9) we see that the tetrad components of the Weyl and Maxwell tensors have the following spin weights:

s	quantity	
+2	ψ_0	
+1	ψ_1, ϕ_0	
0	ψ_2, ϕ_1	(2.26)
-1	ψ_3, ϕ_2	
-2	ψ_4	

Now let the two-dimensional surface have the line element

$$dq^2 = \frac{1}{P^2} d\zeta d\bar{\zeta} \quad , \quad P = P(\zeta, \bar{\zeta}). \quad (2.27)$$

and choose $\text{Re}(m^\mu)$ and $\text{Im}(m^\mu)$ to be tangential, respectively, to the curves $\text{Im}(\zeta) = \text{const}$ and $\text{Re}(\zeta) = \text{const}$. We define the operators, $\eth$ and $\bar{\eth}$, acting on a quantity η of spin weight s by

$$\eth \eta = 2P^{1-s} \frac{\partial}{\partial \zeta} (P^s \eta) \quad (2.28a)$$

and

$$\bar{\eth} \eta = 2P^{1+s} \frac{\partial}{\partial \bar{\zeta}} (P^{-s} \eta). \quad (2.28b)$$

The fundamental property of $\eth$ is its invariance (with spin weight + 1) with respect to coordinate transformations that preserve the metric (2.27). Another important property is that of raising and lowering spin weights by unity under the transformation (2.24), i.e.

$$(\eth \eta)' \equiv \eth' \eta' = e^{i(s+1)\chi} \eth \eta, \quad (2.29)$$

$$(\bar{\eth} \eta)' \equiv \bar{\eth}' \eta' = e^{i(s-1)\chi} \bar{\eth} \eta,$$

We also have the commutation relation

$$(\eth \bar{\eth} - \bar{\eth} \eth) \eta = 2s \eta K, \quad (2.30)$$

where $K = \eth \bar{\eth} \log P$ is the Gaussian curvature of the 2-surface. The operator $\eth$ is effectively a covariant differentiation operator on the surface.

If the surface is taken to be the unit sphere ($K = 1$) with polar coordinates (θ, ϕ) , a natural choice for m^μ is to make $\text{Re}(m^\mu)$ and $\text{Im}(m^\mu)$ tangential, respectively, to the curves $\phi = \text{const}$ and $\theta = \text{const}$. The operators $\mathcal{V}$ and $\bar{\mathcal{V}}$ are then defined by

$$\begin{aligned}\mathcal{V}\eta &= -(\sin\theta)^s \left(\frac{\partial}{\partial\theta} + \frac{1}{\sin\theta} \frac{\partial}{\partial\phi} \right) \{ (\sin\theta)^{-s} \eta \}, \\ \bar{\mathcal{V}}\eta &= -(\sin\theta)^{-s} \left(\frac{\partial}{\partial\theta} - \frac{1}{\sin\theta} \frac{\partial}{\partial\phi} \right) \{ (\sin\theta)^s \eta \}.\end{aligned}\quad (2.31)$$

By using complex stereographic coordinates ζ , related to (θ, ϕ) by

$$\zeta = e^{i\phi} \cot \frac{\theta}{2}, \quad (2.32)$$

we can put the metric of the sphere into the form (2.27) and define $\mathcal{V}$ and $\bar{\mathcal{V}}$ as in (2.28), with

$$P = \frac{1}{2} (1 + \zeta \bar{\zeta}). \quad (2.33)$$

We now introduce a class of generalized spherical functions through the action of $\mathcal{V}$ and $\bar{\mathcal{V}}$ on ordinary spherical harmonics, $Y_{\ell m}(\theta, \phi)$, ($-\ell \leq m \leq \ell$, $\ell = 0, 1, 2, \dots$). The spin s spherical harmonics ${}_s Y_{\ell m}$ (for integral s) are defined by

$${}_s Y_{\ell m}(\theta, \phi) = \begin{cases} \sqrt{\frac{(\ell-s)!}{(\ell+s)!}} \mathcal{V}^s Y_{\ell m}(\theta, \phi), & 0 \leq s \leq \ell, \\ \sqrt{\frac{(\ell-s)!}{(\ell+s)!}} (-)^s \bar{\mathcal{V}}^{-s} Y_{\ell m}(\theta, \phi), & -\ell \leq s \leq 0, \end{cases} \quad (2.34)$$

with ${}_0 Y_{\ell m} \equiv Y_{\ell m}$. The ${}_s Y_{\ell m}$ (which are not defined for $|s| > \ell$) have the

following properties:

$${}_s \bar{Y}_{lm} = (-)^{m+s} {}_{-s} Y_{l,-m}, \quad (2.35a)$$

$$\mathcal{S}({}_s Y_{lm}) = \sqrt{(l-s)(l+s-1)} {}_{s+1} Y_{lm} \quad (2.35b)$$

$$-\mathcal{S}({}_s Y_{lm}) = \sqrt{(l-s)(l+s-1)} {}_{s-1} Y_{lm}, \quad (2.35c)$$

$$\mathcal{S}^2({}_s Y_{lm}) = -(l-s)(l+s+1) {}_s Y_{lm}. \quad (2.35d)$$

The last equation shows that the ${}_s Y_{lm}$ are eigenfunctions of $\mathcal{S}^2$ with eigenvalue $-(l-s)(l+s+1)$. In addition, these spin-weighted functions form a complete, orthonormal set of functions on the sphere for each s , i.e.,

$$\sum_{l=|s|}^{\infty} \sum_{m=-l}^{+l} {}_s Y_{lm}(\theta, \phi) {}_s Y_{lm}(\theta', \phi') = \delta(\phi - \phi') \delta(\cos\theta - \cos\theta'), \quad (2.36a)$$

$$\int {}_s Y_{lm} {}_s Y_{l'm'} d\Omega = \delta_{ll'} \delta_{mm'}, \quad (2.36b)$$

where $d\Omega = \sin\theta d\theta d\phi$. Hence, if $\eta(\theta, \phi)$ is a function which has spin weight s and is suitably regular on the sphere, there exist constants η_{lm} such that η can be expressed as

$$\eta(\theta, \phi) = \sum_{l,m} \eta_{lm} {}_s Y_{lm}(\theta, \phi). \quad (2.37)$$

Putting it another way, given any suitably regular η on the sphere, of (integral) spin weight $s > 0$, there exists ξ of spin weight zero such that

$$\eta = \mathcal{S}^s \xi \quad (2.38)$$

This result enables us to define the "electric" and "magnetic" parts of η by

$$\eta_e = \vartheta^s \operatorname{Re}(\xi) , \quad \eta_m = \vartheta^s \operatorname{Im}(\xi) \quad (2.39)$$

and write η as

$$\eta = \eta_e + \eta_m . \quad (2.40)$$

Due to the invariance properties of ϑ , the splitting (2.40) is invariant under rotations of the sphere. For $s < 0$ the corresponding results are obtained by replacing ϑ by $\bar{\vartheta}$ and s by $-s$ in (2.38) and (2.39).

2.3. A Special Flat Space Null Coordinate System

In our point of view a particle is to be represented by a singular, timelike world line in space-time. To see how this description can be given a geometric invariance, it is essential for interpretive reasons to consider first an alternative description of an arbitrary timelike world line in flat space, which is based on the family of light cones emanating from the line.³⁵

Let the parametric form of an arbitrary timelike world line in Minkowski coordinates (y^μ) be given by $y^\mu = \xi^\mu(u)$ and consider the transformation from y^μ to null coordinates $x^\mu = (u, r, x^2, x^3)$ given by

$$y^\mu = \xi^\mu(u) + r \ell^\mu(u, x^A) , \quad A = 2, 3, \quad (2.41)$$

where $x^0 = u$ is a measure of proper time on the world line and also labels the light cones, $x^1 = r$ is an affine parameter for each null geodesic on the cones as well as a measure of radial distance for the sphere at each $u = \text{const}$, x^2, x^3 are "angular coordinates" which can be related to the complex stereographic coordinates $\zeta, \bar{\zeta}$ by $\zeta = x^2 + ix^3$, and ℓ^μ is a null vector field which sweeps out the directions of the full cone at each u as x^A vary. Since ℓ^μ is null it must satisfy the conditions

$$\ell^\mu \ell_\mu = \ell^\mu \dot{\ell}_\mu = \ell^\mu \ell_{\mu,A} = 0, \quad (2.42)$$

where a dot denotes $\partial/\partial u$. In addition we require the velocity vector $\dot{\xi}^\mu$ of the world line to satisfy the normalizations*

$$\dot{\xi}^\mu \ell_\mu = 1, \quad \dot{\xi}^\mu \dot{\xi}_\mu = 2. \quad (2.43)$$

To specify how null directions are to be propagated as u varies, we set

$$\ell^\mu = \frac{b^\mu}{P_0}, \quad b^\mu = b^\mu(x^A), \quad P_0 = P_0(u, x^A) \quad (2.44)$$

and then differentiate ℓ^μ with respect to u . This yields the propagation law

$$\dot{\ell}^\mu = - \frac{\dot{P}_0}{P_0} \ell^\mu \quad (2.45)$$

which simply expresses the parallel transfer of the direction of ℓ^μ .

*The 2 implies that u is $\sqrt{2}/2$ times the proper time, a choice of normalization adopted to make the line element agree with E.T. Newman and R. Penrose, Proc. Roy. Soc (London) A305, 175 (1968).

Note that from (2.43)-(2.45) we get

$$P_0 = b^\mu \dot{\xi}_\mu \quad (2.46a)$$

and

$$\frac{\dot{P}_0}{P_0} = - \dot{\xi}^\mu \dot{\xi}_\mu = \dot{\xi}^\mu \ddot{\xi}_\mu \quad (2.46b)$$

The metric tensor $g_{\mu\nu}$ in the new coordinates (x^μ) can now be calculated from (2.41) and the transformation law

$$g_{\mu\nu} = \frac{\partial y^\alpha}{\partial x^\mu} \frac{\partial y^\beta}{\partial x^\nu} \eta_{\alpha\beta},$$

where $\eta_{\alpha\beta}$ is the usual Minkowski metric $\eta_{\alpha\beta} = \text{diag}(1, -1, -1, -1)$. With the help of (2.42)-(2.46), we obtain

$$\begin{aligned} g_{\mu 1} &= \delta_\mu^0, & g_{00} &= 2\left(1 - \frac{\dot{P}_0}{P_0} r\right) \\ g_{0A} &= 0, & g_{AB} &= -\frac{r^2}{2} \overset{\circ}{g}_{AB}, \end{aligned} \quad (2.47)$$

where

$$\overset{\circ}{g}_{AB} = -2\dot{\xi}^\alpha{}_{,A} \xi^\alpha{}_{,B} = -2 \frac{b^\alpha{}_{,A} b_{\alpha,B}}{P_0^2}. \quad (2.48a)$$

Since P_0 and b^μ were defined in (2.44) only up to an arbitrary u -independent factor, we can choose this factor such that

$$b^\alpha{}_{,A} b_{\alpha,B} = -\frac{\delta_{AB}}{2}. \quad (2.49)$$

It can easily be checked that a solution of this equation is

$$b^a = \frac{\sqrt{2}}{4} (1 + \zeta \bar{\zeta}, 1 - \zeta \bar{\zeta}, \zeta + \bar{\zeta}, \frac{\zeta - \bar{\zeta}}{i}), \quad \zeta = x^2 + ix^3 \quad (2.50)$$

Substituting (2.49) into (2.48a), we obtain

$$g_{AB}^0 = + \frac{1}{P_0^2} \delta_{AB} \quad (2.48b)$$

The flat space line element in the x^μ -coordinates can now be written as

$$ds^2 = 2(1 - \frac{\dot{P}_0}{P_0} r) du^2 + 2du dr - \frac{r^2}{2} \frac{d\zeta d\bar{\zeta}}{P_0^2} \quad (2.51)$$

Therefore, given the velocity $\dot{\xi}^\mu$ and acceleration $\ddot{\xi}^\mu$ of the world line, one can construct uniquely from (2.46) and (2.50) the line element (2.51); conversely, given the line element (2.51) or in particular $P_0(u, \zeta, \bar{\zeta})$, one can determine all the properties of the world line. In the static case when the world line is straight, $\ddot{\xi}^\mu = 0$ and $\dot{\xi}^\mu = \sqrt{2} \delta_0^\mu$ with $P_0 = \frac{1}{2} (1 + \zeta \bar{\zeta})$, and the 2-surfaces, u and r constant, are ordinary spheres with $r/\sqrt{2}$ measuring radial distance. In the general case, on the other hand, the axes of the sphere at each $u = \text{const}$, $r = \text{const}$ are comoving with the arbitrary timelike world line.

Several useful properties of $\frac{\dot{P}_0}{P_0}$ will now be derived.³² First, we choose the set of four linearly independent vectors $(\dot{\xi}^\mu, l^\mu; l^\mu, A)$ as a basis for Minkowski space such that any Minkowskian tensor can be expressed in terms of these vectors. In particular we can write $\eta_{\mu\nu}$

as

$$\eta_{\mu\nu} = \dot{\ell}_\mu \dot{\xi}_\nu + \dot{\xi}_\mu \dot{\ell}_\nu - 2\ell_\mu \ell_\nu + 2g^{AB} \ell_{\mu,A} \ell_{\nu,B}, \quad (2.52a)$$

where $g^{AB} = + P_0^2 \delta^{AB}$. We also have the expression

$$\ell^\mu{}_{,AB} = \frac{1}{2} g^{AB} (\dot{\xi}^\mu - 2\ell^\mu) + \Gamma_{AB}^C \ell^\mu{}_{,C}, \quad (2.53)$$

where $\Gamma_{AB}^C = \frac{1}{2} g^{CD} (g_{AD,B} + g_{BD,A} - g_{AB,D})$. Multiplying (2.53) by g^{AB} we get

$$g^{AB} \ell^\mu{}_{,AB} = (\dot{\xi}^\mu - 2\ell^\mu), \quad (2.54a)$$

which can be written as

$$4 P_0^2 \frac{\partial^2 \ell^\mu}{\partial \zeta \partial \bar{\zeta}} = -2(\ell^\mu - \frac{1}{2} \dot{\xi}^\mu) \quad (2.55b)$$

Defining the spin-weight zero vector a^μ by

$$a^\mu \equiv \sqrt{2} (\ell^\mu - \frac{1}{2} \dot{\xi}^\mu) \quad (2.56)$$

we may rewrite (2.55b) in terms of $\eth_0$ and $\bar{\eth}_0$ as*

$$\eth_0 \bar{\eth}_0 a^\mu = -2a^\mu. \quad (2.57)$$

*The subscript 0 in $\eth_0, \bar{\eth}_0$ means that these operators are to be defined in terms of P_0 . For spin-weight zero quantities χ , $\eth_0 \eth_0 \chi = 4 P_0^2 \frac{\partial^2 \chi}{\partial \zeta \partial \bar{\zeta}}$.

Comparing this with the eigenvalue equation (2.35d) for $s = 0$, we see that a^μ is an $\ell = 1$ quantity. Moreover, a^μ is a unit, radial, spacelike vector orthogonal to $\dot{\xi}^\mu$, i.e.

$$a_\mu a^\mu = -1, \quad a^\mu \dot{\xi}_\mu = 0 \quad (2.58)$$

We shall call a^μ an $\ell=1$ projection operator because by taking the scalar product of (2.57) with any vector attached to the world line we can project that part of the vector which is orthogonal to the world line into an $\ell=1$ spherical harmonic. In particular, since

$$\frac{\dot{P}_0}{P_0} = \ell^\mu \ddot{\xi}_\mu = \frac{1}{\sqrt{2}} a^\mu \ddot{\xi}_\mu, \quad (2.59)$$

we find from (2.57) that

$$\bar{\sigma}_0 \bar{\sigma}_0 \frac{\dot{P}_0}{P_0} = -2 \frac{\dot{P}_0}{P_0} \quad (2.60)$$

thus showing that $\dot{P}_0/P_0$ is an $\ell = 1$ spherical harmonic. This result also follows from the fact that the Gaussian curvature K_0 of the 2-surface defined by g_{AB}^0 (essentially the unit sphere) is given by

$$K_0 = \bar{\sigma}_0 \bar{\sigma}_0 \log P_0 = 1, \quad (2.61)$$

where $\bar{\sigma}_0 \bar{\sigma}_0 = 4 P_0^2 \partial^2 / \partial \zeta \partial \bar{\zeta}$. By differentiating this equation with respect to u , we obtain (2.60). (It is possible to construct out of a^μ projection operators for any ℓ -value⁴¹, but we shall be interested here only in a^μ .)

In terms of a^μ the expression (2.52a) for $\eta_{\mu\nu}$ can be rewritten

as

$$\eta_{\mu\nu} = \frac{1}{2} \dot{\xi}_\mu \dot{\xi}_\nu - a_\mu a_\nu - \frac{1}{2} (\bar{\sigma}_0 a_\mu \bar{\sigma}_0 a_\nu + \bar{\sigma}'_0 a_\nu \bar{\sigma}'_0 a_\mu), \quad (2.52b)$$

and when this is multiplied with $\ddot{\xi}^\mu \ddot{\xi}^\nu$ the following relation is obtained

$$\bar{\sigma}_0 \frac{\dot{P}_0}{P_0} \cdot \bar{\sigma}_0 \frac{\dot{P}_0}{P_0} = - \left[\left(\frac{\dot{P}_0}{P_0} \right)^2 + \frac{\ddot{\xi}^2}{2} \right], \quad \ddot{\xi}^2 = \ddot{\xi}^\mu \ddot{\xi}_\mu \quad (2.62)$$

Applying $\bar{\sigma}_0 \bar{\sigma}_0$ to $(\dot{P}_0/P_0)^2$ we then find, with the help of (2.60) and (2.62), that

$$\bar{\sigma}_0 \bar{\sigma}_0 \left[\left(\frac{\dot{P}_0}{P_0} \right)^2 + \frac{\ddot{\xi}^2}{6} \right] = -6 \left[\left(\frac{\dot{P}_0}{P_0} \right)^2 + \frac{\ddot{\xi}^2}{6} \right]. \quad (2.63)$$

By comparison of (2.63) with (2.35d) for $s = 0$, we therefore conclude that $[(\dot{P}_0/P_0)^2 + \ddot{\xi}^2/6]$ is an $\ell=2$ spherical harmonic. Differentiating (2.60) with respect to u , one can also prove that

$$\bar{\sigma}_0 \bar{\sigma}_0 \left[\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right] = -2 \left[\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right] \quad (2.64)$$

which shows that $[\ddot{P}_0/P_0 + \ddot{\xi}^2/2]$ is an $\ell=1$ spherical harmonic.

We will now show how alternative expressions for equations of motion in Minkowski space can be given in terms of $\dot{P}_0/P_0$. First we note from (2.59) that the maximum value of $\dot{P}_0/P_0$ on the sphere at each

u is proportional to the magnitude of the acceleration, and the direction $\zeta_{\max}$ in which this maximum occurs is the direction of the acceleration, i.e.,

$$\max \left(\frac{\dot{P}_0}{P_0} \right) = \sqrt{\frac{\ddot{\xi}^2}{2}} \quad , \quad \ddot{\xi}^2 = \ddot{\xi}^\mu \ddot{\xi}_\mu \quad , \quad (2.65)$$

$$\zeta_{\max} = \text{direction of } \ddot{\xi}^\mu$$

This follows immediately from the fact that, since a^μ is a unit radial vector normal to $\dot{\xi}^\mu$, $\dot{P}_0/P_0$ has its maximum when a^μ is in the direction of $\ddot{\xi}^\mu$. Therefore, we can give a unique statement of equivalence between $\ddot{\xi}^\mu$ and $\dot{P}_0/P_0$, writing this as

$$\ddot{\xi}^\mu \longleftrightarrow \frac{\dot{P}_0}{P_0} \quad (2.66)$$

A similar statement can be made for that part of the derivative of the acceleration which is normal to the velocity vector, namely

$$\ddot{\xi}^\mu + \frac{1}{2} \ddot{\xi}^2 \dot{\xi}^\mu \longleftrightarrow \frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \quad . \quad (2.67)$$

This can be seen by noting that the u -derivative of (2.59) yields

$$\ddot{\xi}_\mu \dot{\xi}^\mu = \frac{\ddot{P}_0}{P_0} \quad ,$$

or in terms of a^μ

$$\frac{1}{\sqrt{2}} \left(\ddot{\xi}_\mu + \frac{1}{2} \ddot{\xi}^2 \dot{\xi}_\mu \right) a^\mu = \frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2}$$

Equations of motion in Minkowski space can now be expressed in an alternative way by means of the equivalence statements (2.66) and (2.67), e.g.,

(a) free-particle or geodesic motion:

$$m \ddot{\xi}^\mu = 0 \longleftrightarrow m \frac{\dot{P}_0}{P_0} = 0, \quad (2.68a)$$

(b) motion with the Abraham radiation reaction force*:

$$\sqrt{2} m \ddot{\xi}^\mu = \frac{2}{3} e^2 (\ddot{\xi}^\mu + \frac{1}{2} \ddot{\xi}^2 \dot{\xi}^\mu) \longleftrightarrow \sqrt{2} m \frac{\dot{P}_0}{P_0} = \frac{2}{3} e^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right) \quad (2.68b)$$

(c) motion with a general force:

$$\frac{1}{2} m \ddot{\xi}^\mu = F^\mu \longleftrightarrow \frac{1}{2} m \frac{\dot{P}_0}{P_0} = F(u, \zeta, \bar{\zeta}), \quad (2.68c)$$

where $F = \frac{1}{\sqrt{2}} F^\mu a_\mu$ is an $\ell = 1$ spherical harmonic.

Actually, these equivalences apply as well to regular timelike world lines in an arbitrary Riemannian space, where a similar coordinate system can be constructed at least locally. The equations of motion that we shall derive later will be essentially the generalizations to curved space of the right-hand sides of (2.68), and hence the above equivalences will be useful for the interpretation of our results.

*The $\sqrt{2}$ is due to the fact that our u is $\sqrt{2}/2$ times the proper time.

An interesting by-product of this alternative description of acceleration is the possibility of measuring $\dot{P}_0/P_0$ (and hence the acceleration of the world line) directly by means of Doppler shift observations of light emanating from the world line.³² Consider a family of geodesic observers completely surrounding the world line and let their velocity field be $v^\mu(\tau, \zeta, \bar{\zeta})$, where τ is the proper time along an observer trajectory and $(\zeta, \bar{\zeta})$ gives the angular position of the observer. The ratio of source frequency f_s to observed frequency f_o is given by

$$\sqrt{2} \quad \frac{f_s}{f_o} = \frac{d\tau}{du} = \frac{\xi_\mu l^\mu}{v_\mu l^\mu} = \frac{1}{v_\mu l^\mu}$$

Differentiating this with respect to τ , we get

$$\begin{aligned} \sqrt{2} \quad \frac{d}{d\tau} \left(\frac{f_s}{f_o} \right) &= - \left(\frac{d\tau}{du} \right)^2 \frac{d}{d\tau} (v_\mu l^\mu) = - \left(\frac{d\tau}{du} \right)^2 (v_{\mu;\nu} v^\nu l^\mu + l_{\mu;\nu} v^\mu v^\nu) \\ &= - \left(\frac{d\tau}{du} \right)^2 l_{\mu;\nu} v^\mu v^\nu, \end{aligned}$$

where the last equality comes from the assumption that the observers are all moving along geodesics (i.e., $v_{\mu;\nu} v^\nu = 0$). Now, in the x^μ -coordinates, $l_\mu = \delta_\mu^0$; hence,

$$\begin{aligned} \sqrt{2} \quad \frac{d}{d\tau} \left(\frac{f_s}{f_o} \right) &= \left(\frac{d\tau}{du} \right)^2 \Gamma_{\mu\nu}^0 v^\mu v^\nu = r_{\mu\nu}^0 \frac{dx^\mu}{du} \frac{dx^\nu}{du} \\ &= - \frac{1}{2} \epsilon_{\mu\nu,1} \frac{dx^\mu}{du} \frac{dx^\nu}{du} = \frac{\dot{P}_0}{P_0} + \frac{r}{2P_0} \frac{d\zeta}{du} \frac{d\bar{\zeta}}{du} \end{aligned}$$

Since $d\zeta/du$ measures angular velocity, it behaves as $1/r$. Therefore, for observers at infinity,

$$\sqrt{2} \frac{d}{d\tau} \left(\frac{f_s}{f_o} \right) = \frac{\dot{P}_o}{P_o} \quad (2.69)$$

2.4. Basic Assumptions

We now would like to see how the alternative description of a timelike world line in flat space given in the previous section could be generalized to a singular, timelike world line in an arbitrary Riemannian space. Since the flat space description was based on the null cones issuing from the line, the obvious thing to do is to make the arbitrary null hypersurfaces introduced in Section 2.1 behave like "cones" in the neighborhood of the origin $r = 0$.

In the null coordinate system and null tetrad system adopted in Section 2.1, each of the null hypersurfaces, $u = \text{const}$, is generated by a congruence of null geodesics to which ℓ^μ is tangent. Hence, the geometry of each null hypersurface can be completely characterized by the spin coefficients ρ and σ , given in (2.14), which determine the properties of the null geodesics. Now, in flat space light cones are characterized by $\rho = -1/r$ and $\sigma = 0$. We are therefore led to impose the following restrictive condition on the null hypersurfaces,³² namely, that near $r = 0$ *

$$\begin{aligned} \rho &= -\frac{1}{r} + O(r), \\ \sigma &= O(r). \end{aligned} \quad (2.70)$$

*The order symbol used here is defined as follows: $f(r, u, x^A) = O[g(r)]$ implies $|f(r, u, x^A)| < g(r)F(u, x^A)$ for some function F independent of r and for all small r .

This is the condition for the null hypersurfaces to behave like cones in the neighborhood of $r = 0$. Condition (2.70) also fixes the origin of r . When (2.70) is satisfied and the Weyl tensor is singular at $r = 0$, the singularity shall be called an elementary singularity.

From condition (2.70) we can show (without use of the field equations) that the two-surfaces, u and r constant (i.e., the cross-section of the cones), possess a metric given by

$$g_{AB} = -\frac{r^2}{2} g_{AB}^0 + O(r^4) \quad , \quad g_{AB}^0 = g_{AB}^0(u, x^C), \quad (2.71)$$

The proof (see Appendix A for details) consists in first noting that from (2.13)

$$g_{AB} g^{BC} = \delta_A^C$$

and then recalling that ξ^A satisfies (2.16a), namely,

$$\frac{\partial \xi^A}{\partial r} = \rho \xi^A + \sigma \bar{\xi}^A.$$

From this it can be shown that

$$\xi^A = \frac{\xi^{0A}}{r} + O(r) \quad , \quad \xi^{0A} = \xi^{0A}(u, x^B), \quad (2.72)$$

and, by taking the inverse of g^{AB} , the result (2.71) is obtained.

We define the fundamental 2-surface (F2S) metric by

$$g_{AB}^0 = \lim_{r \rightarrow 0} \left(-2 \frac{g_{AB}}{r^2} \right) \quad (2.73)$$

One can prove that if the line $r = 0$ is a regular, timelike (null, spacelike) line, then the F2S has a positive (zero, negative) constant Gaussian curvature.

The F2S can be represented most conveniently in conformally flat coordinates. The coordinate freedom to relabel the null geodesics⁴² enables us to make the F2S metric conformally flat, i.e.

$$g^{AB} = \frac{(\xi^{OA}\xi^{OB} + \bar{\xi}^{OA}\bar{\xi}^{OB})}{2} = P^2 \delta^{AB}, \quad (2.74)$$

by choosing

$$\xi^{02} = i\xi^{03} = -P = -\bar{P} \quad (2.75)$$

Thus the F2S line element can be written as

$$dq^2 = g_{AB}^0 dx^A dx^B = \frac{1}{P^2} d\zeta d\bar{\zeta}, \quad (2.76)$$

where $\zeta = x^2 + ix^3$. The quantity $P = P(u, \zeta, \bar{\zeta})$ will be our fundamental variable; as we shall see later, it carries all information concerning the motion and internal structure of singularities.

With the restriction (2.70) which we have imposed on the family of null hypersurfaces, the only coordinate freedom that remains is the freedom of renaming the null surfaces,⁴² i.e.

$$u' = f(u), \quad r' = \frac{r}{f}, \quad \zeta' = g(u, \zeta). \quad (2.77)$$

Thus the F2S is defined up to an arbitrary factor f^2 . A conventional choice of this factor is to make the surface area of the F2S constant, i.e.

$$\frac{d}{du} \int \frac{dx^2 dx^3}{P^2} = 0.$$

Here, however, we shall make a different choice of this factor.

We impose the important regularity restriction that $P = P_0 H$, where P_0 is defined by (2.46a) and (2.50) and H is a regular function on the sphere whose metric is P_0 . The arbitrary factor in the definition of P can then be used to set the $\ell = 0$ part of H equal to one and to incorporate the $\ell = 1$ part into the definition of P_0 . The latter can be done because any variation in P_0 is such that $\delta P_0 / P_0$ is an $\ell = 1$ harmonic. This is seen by varying $\delta \bar{\delta} \log P_0 = 1$, which yields

$$\delta \bar{\delta} \frac{\delta P_0}{P_0} = -2 \frac{\delta P_0}{P_0}.$$

Therefore, the regularity condition can be written as³²

$$P = P_0(1+I) \quad (2.78)$$

where I is expandable in terms of $\ell \geq 2$ spherical harmonics. This condition simply states that the F2S is a deformed sphere. (The additional condition $I > -1$ should probably be imposed also in order to keep the F2S a deformed sphere). We shall interpret I as internal degrees of freedom for the elementary singularities.

When the world line $r = 0$ has an elementary singularity, the field equations yield differential equations for $\dot{P}_0 / P_0$ and I , which give, respectively, the equations of motion and the time development of the internal structure.

Finally, we should point out that the method of measuring $\dot{P}_0/P_0$ (and hence $\ddot{\xi}^\mu$), which was discussed at the end of Section 2.3, would probably remain applicable to the case of a singular timelike world line in a general curved space. However, it will no longer be $\dot{P}_0/P_0$ but rather $\dot{P}/P$ which will be measured. In addition, the frequency ratio $f_g/f_0 = d\tau/\sqrt{2} du$ will no longer be the ratio of two proper times but rather the ratio of a proper time to our "retarded time" $\sqrt{2} u$. Presumably, u can be regarded as a physically significant parameter which plays the role of the proper time at the singularity.

3.0. MOTION OF FREE SINGULARITIES

We shall now show how the alternative description of motion developed in Chapter II can be applied when there exists a singularity in the field. In particular, we shall consider the special case when condition (2.70) becomes

$$\rho = -\frac{1}{r}, \quad \sigma = 0 \quad (3.1)$$

This condition leads uniquely to the two classes of algebraically special metrics which we shall analyze in Sections 3.1 and 3.2, namely, the Robinson-Trautman type II (RT) solutions and their Einstein-Maxwell counterparts, the Robinson-Trautman-Maxwell (RTM) solutions. As we shall see, these solutions are characterized by the vanishing of the quantities ψ_0 and ϕ_0 which represent incoming gravitational and electromagnetic fields, respectively. Hence the singularities in these solutions correspond to free singularities, i.e., not interacting with incoming background fields.

3.1. Analysis of the Robinson-Trautman Solutions

The RT solutions can be easily obtained in the NP formalism by taking (3.1) and integrating the equations (2.16)-(2.21) for empty space ($\phi_{mn} = R = 0$). The integration, which is carried out in Ref. 37, yields the following results:

A. Spin coefficients:

$$\begin{aligned}
\sigma &= \tau = \lambda = \kappa = \pi = \epsilon = 0 \\
\rho &= -\frac{1}{r}, \quad \gamma = \gamma^0 - \frac{1}{2} \frac{\psi_2^0}{r^2}, \\
\alpha &= \frac{\alpha^0}{r}, \quad \mu = \frac{\mu^0}{r} - \frac{\psi_2^0}{r^2}, \\
\beta &= -\frac{\alpha^0}{r}, \quad \nu = \nu^0 - \frac{\psi_3^0}{r},
\end{aligned} \tag{3.3}$$

where an 0-superscript denotes r -independence.

B. Metric variables:

$$\begin{aligned}
\omega &= x^A = 0, \\
\xi^A &= -\frac{(P, -iP)}{r}, \quad P = \bar{P} \\
U &= U^0 - 2\gamma^0 r - \frac{\psi_2^0}{r}
\end{aligned} \tag{3.4}$$

C. Tetrad components of the Weyl tensor:

$$\begin{aligned}
\psi_0 &= \psi_1 = 0, \\
\psi_2 &= \frac{\psi_2^0}{r^3}, \\
\psi_3 &= \frac{\psi_3^0}{r^2}, \\
\psi_4 &= \frac{\psi_4^0}{r} + \frac{\bar{\delta}\psi_3^0}{r^2}.
\end{aligned} \tag{3.5}$$

D. Relationships among "constants":

$$\begin{aligned}
\alpha^0 &= -\frac{1}{2} \mathcal{S} \log P, & \gamma^0 &= -\frac{1}{2} \frac{\dot{P}}{P}, \\
\nu^0 &= +\mathcal{S} \frac{\dot{P}}{P}, & U^0 = \mu^0 &= -K = -\mathcal{S} \log P \\
\psi_2^0 &= \bar{\psi}_2^0, & \psi_3^0 &= \mathcal{S} K, \quad \psi_4^0 = -\mathcal{S}^2 \frac{\dot{P}}{P} \\
\mathcal{S} \psi_2^0 &= 0, & \dot{\psi}_2^0 - 3 \frac{\dot{P}}{P} \psi_2^0 &= -\mathcal{S} \psi_3^0
\end{aligned} \tag{3.6}$$

E. Components of the metric tensor:

$$\begin{aligned}
g^{\mu 0} &= \delta_1^\mu, & g^{AB} &= -\frac{2P^2}{r^2} \delta^{AB}, \quad (A, B = 2, 3) \\
g^{11} &= -2\left(K - \frac{\dot{P}}{P} r + \frac{\psi_2^0}{r}\right).
\end{aligned} \tag{3.7}$$

Note that $\mathcal{S}$ above is defined with respect to the F2S metric P , i.e., for any spin- s function η we have

$$\mathcal{S}\eta \equiv 2P^{1-s} \frac{\partial}{\partial \zeta} (P^s \eta).$$

By imposing the regularity condition $P = P_0(1+I)$, we can express $\mathcal{S}$ in terms of the operator $\mathcal{S}'_0$, which is defined with respect to the metric of the sphere P_0 . This relationship is given by

$$\mathcal{S}\eta = (1+I)\mathcal{S}'_0\eta + s\eta\mathcal{S}'_0 I, \tag{3.8a}$$

with

$$\bar{\mathcal{S}}\eta = (1+I)\bar{\mathcal{S}}'_0\eta - s\eta\bar{\mathcal{S}}'_0 I.$$

Similarly, for a spin weight 0 function χ we have

$$\partial\bar{\partial}\chi = \bar{\partial}\partial\chi = (1+I)^2 \partial_0\bar{\partial}_0\chi = 4P_0^2(1+I)^2 \frac{\partial^2\chi}{\partial\zeta\partial\bar{\zeta}} \quad (3.9)$$

Thus the Gaussian curvature $K = \partial\bar{\partial} \log P$ of the F2S can be written as

$$K = (1+I)^2 \partial_0\bar{\partial}_0 \log \{P_0(1+I)\}$$

and since $\partial_0\bar{\partial}_0 \log P_0 = 1$, this becomes

$$K = (1+I)^2 + (1+I)\partial_0\bar{\partial}_0 I - \partial_0 I \cdot \bar{\partial}_0 I \quad (3.10)$$

In all the work to follow we shall assume that all the tetrad variables have well-defined spin weights and are regular functions on the sphere, i.e., they are expandable in spin- s spherical harmonics Y_{lm} . In addition to the spin weights (2.26) for the tetrad components of the Weyl and Maxwell tensors and the spin weight 0 for the functions of P_0 and I , we assign the spin weight 1 to the quantity Z defined by

$$Z = \frac{(X^2 + iX^3)}{P} \quad (3.11)$$

Going back to (3.6) we find from $\partial\psi_2^0 = 0$ (or $\partial_0\psi_2^0 = 0$) that ψ_2^0 is a pure $l = 0$ quantity (see 2.35b), i.e. $\psi_2^0 = -M(u)$. As we shall see later, M is related to the "mass". With this result, the RT line element can be written as

$$ds^2 = 2(K - \frac{\dot{P}}{P} r - \frac{M}{r})du^2 + 2du dr - \frac{r^2}{2P^2} d\zeta d\bar{\zeta} \quad (3.12)$$

and all the tetrad variables can be expressed in terms of just the two basic quantities M and P which satisfy the differential equation

$$\dot{M} - 3M \frac{\dot{P}}{P} = \mathfrak{g}\bar{\mathfrak{g}}K \quad (3.13)$$

It is this equation which will give us the time dependence of the "mass", the equations of motion, and the time development of the internal structure for the singularity. In the generalizations to a general $R_{\mu\nu} = 0$ and to the full Einstein-Maxwell theory, (3.13) will be modified by the addition of extra terms representing the interactions of the singularity with background fields.

If we now substitute $P = P_0(1+I)$ into (3.13), we obtain

$$\dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M \frac{\dot{I}}{1+I} = (1+I)^2 \mathfrak{g}_0 \bar{\mathfrak{g}}_0 K ,$$

where K is given by (3.10). Carrying out the differentiation of K , we then get

$$\begin{aligned} \dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M \frac{\dot{I}}{(1+I)} &= (1+I)^3 (\mathfrak{g}_0 \bar{\mathfrak{g}}_0 \mathfrak{g}_0 \bar{\mathfrak{g}}_0 I + 2\mathfrak{g}_0 \bar{\mathfrak{g}}_0 I) \\ &- (1+I)^2 \mathfrak{g}_0^2 I \cdot \bar{\mathfrak{g}}_0^2 I. \end{aligned} \quad (3.14)$$

This equation can in principle (though not in practice due to its extreme non-linearity) be expanded in spherical harmonics such that the $\ell = 0$ part gives the time dependence of M , the $\ell = 1$ part gives the the equation of motion in terms of the "acceleration" $\dot{P}_0/P_0$ of the form (2.68c), and the $\ell \geq 2$ parts give the time dependence of the internal degrees of freedom I .

In the particular case when $I = 0$, we find that $\dot{M} = 0$ and $\dot{P}_0/P_0 = 0$, i.e., the singularity is unaccelerated with constant mass. This static solution leads to the line element

$$ds^2 = 2(1 - \frac{M}{r})du^2 + 2 du dr - \frac{r^2}{2P_0^2} d\tilde{r} d\tilde{r} \quad (3.15)$$

which is just the Schwarzschild solution in null coordinates with M related to the Schwarzschild mass m by $M = \sqrt{2} \ k m$ and with $P_0 = \frac{1}{2}(1 + \tilde{r})$.

So far no exact solution to (3.13) other than Schwarzschild has been found which satisfies the regularity condition $P = P_0(1+I)$. Here we shall go only as far as the linearization of (3.14). Thus, taking $\dot{P}_0/P_0$ and I to be small quantities in (3.14), we obtain

$$\dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M\dot{I} = (\mathcal{F}_0 \bar{\mathcal{F}}_0 \mathcal{F}_0 \bar{\mathcal{F}}_0 + 2\mathcal{F}_0 \bar{\mathcal{F}}_0)I \quad (3.16)$$

Because of the linearity of (3.16), we can (without loss of generality) assume I to have a definite ℓ value, i.e., $\mathcal{F}_0 \bar{\mathcal{F}}_0 I = -\ell(\ell+1)I$. Eq. (3.16) then becomes

$$\dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M\dot{I} = \ell(\ell+1) [\ell(\ell+1) - 2]I \quad (3.17)$$

By differentiating $\mathcal{F}_0 \bar{\mathcal{F}}_0 I = -\ell(\ell+1)I$ with respect to u and neglecting $I(\dot{P}_0/P_0)$ as second order, we see that $\mathcal{F}_0 \bar{\mathcal{F}}_0 \dot{I} = -\ell(\ell+1)\dot{I}$. Hence, the decomposition of (3.17) into different ℓ values yields

$$\ell = 0: \quad \dot{M} = 0, \text{ i.e. } M = \sqrt{2} \text{ k.m} \quad (3.18a)$$

$$\ell = 1: \quad M \frac{\dot{P}_0}{P_0} = 0 \quad (3.18b)$$

$$\ell \geq 2: \quad \dot{I} = -\frac{1}{3M} \ell(\ell+1)[\ell(\ell+1) - 2]I,$$

$$\text{or} \quad I = I_0 e^{-\frac{u}{u_0}} \quad (3.19)$$

where $I_0 = I_0(\zeta, \bar{\zeta})$ is the value of I at $u = 0$ and the time constant u_0 is given by

$$u_0 = \frac{c^2}{6\sqrt{2} Gm} [\ell(\ell+1) - 2]\ell(\ell+1) \quad (3.20)$$

For the sun, with u measured in seconds and $\ell = 2$, u_0 is of the order of 10^{-16} seconds.

Therefore, to first order, M is conserved, there is no acceleration, and the internal degrees of freedom decay to zero very rapidly. It seems very likely that all solutions of (3.14) decay towards the Schwarzschild solution, but we have no proof of this conjecture.

We conclude our analysis of the RT solutions with a discussion of the concept of mass. There seems to be at least four candidates for the title, "mass," all involving M times some function of u . In each case the function is such that all the different "masses" reduce to the Schwarzschild mass m in the static case:

If we let the area of the F2S be

$$A(u) = \oint \frac{dx^2 dx^3}{P^2},$$

then four possible definitions of mass are³²:

$$(1) \quad m_0 = \frac{MA^2}{16\sqrt{2} \, k\pi^2} \quad (3.21a)$$

which appears to be a good candidate for the neutral inertial mass of the singularity due to its appearance in the ponderomotive law (see Section 3.2);

$$(2) \quad m_B = \frac{M}{4\sqrt{2} \, k\pi} \left\{ \oint \frac{\frac{1}{2}(\zeta + \bar{\zeta}) \, d\zeta^2 dx^3}{p^3} \right\} \quad (3.21b)$$

which appears to be equivalent to the Bondi mass⁴³ since it has the property that $\dot{m}_B < 0$;

$$(3) \quad m_c = \frac{MA^{3/2}}{\sqrt{2} \, k(4\pi)^{3/2}} \quad (3.21c)$$

which satisfies the conservation law $\dot{m}_c = 0$ (valid for the RF solutions); and

$$(4) \quad m_e = \frac{MA}{4\sqrt{2} \, k\pi} \quad , \quad (3.21d)$$

which actually has nothing to recommend it other than its analogy with the definition of charge (see Section 3.2).

The proof of the conservation law $\dot{m}_c = 0$ is straightforward. Differentiation of (3.21c) yields

$$\dot{m}_c = \frac{A^{1/2}}{\sqrt{2} k(4\pi)^{3/2}} \oint \left(\dot{M} - 3M \frac{\dot{P}}{P} \right) \frac{dx^2 dx^3}{P^2}.$$

From (3.13) we then find

$$\dot{m}_c = \frac{4A^{1/2}}{\sqrt{2} k(4\pi)^{3/2}} \oint \frac{\partial Q}{\partial \bar{\zeta}} dx^2 dx^3,$$

where $Q = \partial K / \partial \bar{\zeta}$. Since

$$2 \oint \frac{\partial Q}{\partial \bar{\zeta}} dx^2 dx^3 = \int dx^3 \int dx^2 \frac{\partial Q}{\partial x^2} - i \int dx^2 \int dx^3 \frac{\partial Q}{\partial x^3} = 0, \quad (3.22)$$

therefore $\dot{m}_c = 0$.

To prove $\dot{m}_B < 0$, we set $P = \frac{W}{2} (1 + \zeta \bar{\zeta})$, where W is a positive quantity.

Eq. (3.13) can then be written as

$$\dot{M} - 3M \frac{\dot{W}}{W} = W^3 (1 + \zeta \bar{\zeta})^2 \frac{\partial V}{\partial \bar{\zeta}} - W^2 |J|^2 \quad (3.23)$$

where $V = \frac{\partial}{\partial \bar{\zeta}} \left[2W + (1 + \zeta \bar{\zeta})^2 \frac{\partial^2 W}{\partial \zeta \partial \bar{\zeta}} \right]$ and $J = \frac{\partial}{\partial \bar{\zeta}} \left[(1 + \zeta \bar{\zeta}) \frac{\partial W}{\partial \bar{\zeta}} \right]$. Dividing (3.23) by $W/2$ and integrating over the F2S area we obtain

$$\begin{aligned} \oint \frac{2}{W} \left(\dot{M} - 3M \frac{\dot{W}}{W} \right) \frac{dx^2 dx^3}{P^2} &= \oint 2W^2 (1 + \zeta \bar{\zeta}) \frac{\partial V}{\partial \bar{\zeta}} \frac{dx^2 dx^3}{P^2} \\ &- \oint 2W |J|^2 \frac{dx^2 dx^3}{P^2}, \end{aligned} \quad (3.24)$$

where the left-hand side can be rewritten as

$$\frac{d}{du} \oint \frac{M(1+\zeta\bar{\zeta})}{p^3} dx^2 dx^3 = 8\sqrt{2} \pi k \frac{dm_B}{du} .$$

Simplifying the right-hand side, (3.24) then becomes

$$8\sqrt{2} \pi k \dot{m}_B = 8 \int \frac{\partial V}{\partial \zeta} dx^2 dx^3 - 8 \int \frac{|J|^2}{W(1+\zeta\bar{\zeta})^2} dx^2 dx^3 .$$

The first integral on the right vanishes as in (3.22) while the second integral is always positive because W is positive. Therefore, $\dot{m}_B$ is always negative.

3.2. Analysis of the RTM Solutions

In this section we will present and analyze a class of solutions of the Einstein-Maxwell equations which are characterized by the coincidence of a principal null direction of the Maxwell field with a doubly degenerate principal null direction of the Weyl tensor, these null directions being hypersurface orthogonal. These solutions are the Einstein-Maxwell counterparts of the RT solutions; hence the name, Robinson-Trautman-Maxwell solutions.³²

Adopting the conditions $\rho = -1/r$ and $\sigma = 0$, we immediately obtain from (2.18a) and (2.18b) the results $\psi_0 = \phi_0 = 0$. By the Goldberg-Sachs Theorem³³, $\sigma = \kappa = 0$ also implies that $\psi_1 = 0$. Under these conditions the integration of the Einstein-Maxwell equations (2.16)-(2.22) which we outline here, is relatively easy.

As a general procedure in integrating the NP equations, we start with the radial equations (2.16), (2.18), (2.20), and (2.22), obtaining at each step a "constant" of integration independent of r which we denote by a superscript 0. We then substitute the results into the non-radial equations and, by comparing powers of r , obtain relationships among these "constants". Thus, writing the equation number on the left for easy reference to Section 2.1., we have:

$$(2.16a) \quad D\xi^A + r^{-1}\xi^A = 0 \rightarrow \xi^A = \xi^{0A}r^{-1}$$

As in (2.75) we choose

$$\xi^{02} = i\xi^{03} = -P = -\bar{P}$$

$$(2.18c) \quad D\alpha + r^{-1}\alpha = 0 \rightarrow \alpha = \alpha^0 r^{-1}.$$

$$(2.18d) \quad D\beta + r^{-1}\beta = 0 \rightarrow \beta = \beta^0 r^{-1}.$$

$$(2.18e) \quad D\tau + r^{-1}\tau = 0 \rightarrow \tau = \tau^0 r^{-1}.$$

Since $\tau = \bar{\alpha} + \beta$ in our coordinate system, $\tau^0 = \bar{\alpha}^0 + \beta^0$. By means of the tetrad freedom called "null rotation",⁴² we can set $\tau^0 = 0$ and hence

$$\bar{\alpha} = -\beta, \quad \tau = 0$$

$$(2.16b) \quad D\omega + r^{-1}\omega = 0 \rightarrow \omega = \omega^0 r^{-1}$$

$$(2.19b) \quad \delta\rho = 0 \rightarrow \omega D\rho = 0 \rightarrow \omega^0 = 0$$

$$(2.18g) \quad D\lambda + r^{-1}\lambda = 0 \rightarrow \lambda = \lambda^0 r^{-1}$$

$$(2.19g) \quad \bar{\lambda}\rho = 0 \rightarrow \lambda^0 = 0$$

$$(2.16c) \quad DX^A = 0 \rightarrow X^A = X^{0A}$$

$$(2.22a) \quad D\phi_1 + 2r^{-1}\phi_1 = 0 \rightarrow \phi_1 = \phi_1^0 r^{-2}$$

$$(2.22c) \quad \delta\phi_1^0 = 0 \rightarrow 2P \cdot \frac{\partial\phi_1^0}{\partial\zeta} = \delta\phi_1^0 = 0$$

Applying $\bar{\delta}$ to this result, we obtain $\bar{\delta}\phi_1^0 = 0$ which implies $\delta_0\bar{\delta}_0\phi_1^0 = 0$. From (2.35d) we then see that ϕ_1^0 is a pure $l = 0$ spherical harmonic, i.e. a "constant" independent of $\zeta, \bar{\zeta}$. Hence, we can write

$$\phi_1^0 = E(u) \dots$$

$$(2.22b) \quad D\phi_2 + r^{-1}\phi_2 = 0 \rightarrow \phi_2 = \phi_2^0 r^{-1}$$

$$(2.20b) \quad D\psi_2 + 3r^{-1}\psi_2 = -2k\phi_1\bar{\phi}_1 r^{-1} \rightarrow$$

$$\psi_2 = \psi_2^0 r^{-3} + 2k|\phi_1^0|^2 r^{-4}$$

$$(2.18h) \quad D\mu + r^{-1}\mu = \psi_2 \rightarrow \mu = \mu^0 r^{-1} - \psi_2^0 r^{-2} - k|\phi_1^0|^2 r^{-3}$$

$$(2.18f) \quad D\gamma = \psi_2 + k\phi_1\bar{\phi}_1 \rightarrow \gamma = \gamma^0 - \frac{1}{2}\psi_2^0 r^{-2} - k|\phi_1^0|^2 r^{-3}$$

$$(2.17c) \quad \mu - \bar{\mu} = 0 \rightarrow \mu^0 = \bar{\mu}^0$$

$$(2.16d) \quad DU = -(\gamma + \bar{\gamma}) \rightarrow$$

$$U = U^0 - (\gamma^0 + \bar{\gamma}^0)r - \psi_2^0 r^{-1} - k|\phi_1^0|^2 r^{-2}$$

$$(2.21b) \quad \delta\psi_2 = -2k\phi_1\bar{\phi}_2 r^{-1} \rightarrow \delta\psi_2^0 = 2k\phi_1^0\bar{\phi}_2^0$$

$$(2.20c) \quad D\psi_3 + 2r^{-1}\psi_3 = \bar{\delta}\psi_2 + k\bar{\phi}_1 D\phi_2 \rightarrow$$

$$\psi_3 = \psi_3^0 r^{-2} + 3k\bar{\phi}_1^0\phi_2^0 r^{-3}$$

$$(2.18i) \quad Dv = \psi_3 + k\bar{\phi}_1\phi_2 \rightarrow v = v^0 - \psi_3^0 r^{-1} - 2k\bar{\phi}_1^0\phi_2^0 r^{-2}$$

$$(2.17b) \quad \delta\bar{\xi}^A - \bar{\delta}\xi^A = 2\bar{\alpha}\bar{\xi}^A - 2\alpha\xi^A \rightarrow \alpha^0 = -\frac{\partial P}{\partial \bar{\xi}}$$

$$(2.20d) \quad D\psi_4 + r^{-1}\psi_4 = (\bar{\delta}+2\alpha)\psi_3 + k\bar{\phi}_1(\bar{\delta}+2\alpha)\phi_2 \rightarrow$$

$$\psi_4 = \psi_4^0 r^{-1} + \bar{\theta}\psi_3^0 r^{-2} + 2k\bar{\phi}_1^0\bar{\theta}\phi_2^0 r^{-3}$$

$$(2.16e) \quad \delta X^A - \Delta\xi^A = (\mu+\bar{\gamma}-\gamma)\xi^A \rightarrow$$

$$U^0 = \mu^0 \quad \gamma^0 = -\frac{1}{2}\frac{\dot{P}}{P} \quad \frac{\partial X^0}{\partial \bar{\xi}} = 0$$

The last equation implies that $X^0 \equiv (X^{02} + iX^{03})$ is analytic, i.e.

$X^0 = X^0(u, \bar{\zeta})$. Because of this we can use the coordinate freedom

$\zeta' = g(u, \zeta)$ to make⁴²

$$X^{0A} = 0.$$

$$(2.19c) \quad \delta\alpha + \bar{\delta}\bar{\alpha} = \mu\rho + 4\alpha\bar{\alpha} - \psi_2 + k\phi_1\bar{\phi}_1 \rightarrow$$

$$\mu^0 = -\bar{\theta}\bar{\theta}\log P = -K$$

$$(2.19d) \quad \bar{\delta}\mu = \psi_3 - k\phi_2\bar{\phi}_1 \rightarrow \psi_3^0 = -\bar{\theta}\mu^0 = \bar{\theta}K$$

$$(2.17d) \quad \delta U = -v \rightarrow v^0 = -2\bar{\theta}\gamma^0 = \bar{\theta}\frac{\dot{P}}{P}$$

$$(2.19a) \quad \bar{\delta}v = \psi_4 - 2\alpha v \rightarrow \psi_4^0 = -\bar{\theta}v^0 = -\bar{\theta}^2\frac{\dot{P}}{P}$$

$$(2.22d) \quad \Delta\phi_1 + 2\mu\phi_1 = (\delta-2\bar{\alpha})\phi_2 \rightarrow$$

$$\dot{\phi}_1^0 - 2\frac{\dot{P}}{P}\phi_1^0 = -\bar{\theta}\phi_2^0$$

$$\begin{aligned}
(2.21c) \quad \Delta\psi_2 + 3\mu\psi_2 &= (\delta-2\bar{\alpha})\psi_3 + k\bar{\phi}_1(\delta-2\bar{\alpha})\phi_2 \\
&\quad -k\bar{\phi}_2 D\phi_2 - 2k\phi_1\bar{\phi}_1\mu \rightarrow \\
\dot{\psi}_2^0 - 3\frac{\dot{P}}{P}\psi_2^0 &= -\delta\psi_3^0 + k\phi_2^0\bar{\phi}_2^0
\end{aligned}$$

This completes the integration of the Einstein-Maxwell equations. The remaining equations are found to contain only identities and redundant information.

For convenience we now give a summary of the results.

A. Tetrad components of the Weyl tensor:

$$\psi_0 = \psi_1 = 0 \quad (3.25a)$$

$$\psi_2 = \psi_2^0 r^{-3} + 2k|\phi_1^0|^2 r^{-4} \quad (3.25b)$$

$$\psi_3 = \psi_3^0 r^{-2} + 3k\bar{\phi}_1^0\phi_2^0 r^{-3} \quad (3.25c)$$

$$\psi_4 = \psi_4^0 r^{-1} + \bar{\delta}\psi_3^0 r^{-2} + 2k\bar{\phi}_1^0\bar{\delta}\phi_2^0 r^{-3} \quad (3.25d)$$

B. Tetrad components of the Maxwell tensor:

$$\phi_0 = 0 \quad (3.26a)$$

$$\phi_1 = \phi_1^0 r^{-2} \quad (3.26b)$$

$$\phi_2 = \phi_2^0 r^{-1} \quad (3.26c)$$

C. Spin coefficients:

$$\sigma = \tau = \lambda = \kappa = \epsilon = \pi = 0 \quad (3.27a)$$

$$\rho = -r^{-1} \quad (3.27b)$$

$$\alpha = \alpha^0 r^{-1} \quad (3.27c)$$

$$\beta = -\bar{\alpha}^0 r^{-1} \quad (3.27d)$$

$$\gamma = \gamma^0 - \frac{1}{2} \psi_2^0 r^{-2} - k |\phi_1^0|^2 r^{-3} \quad (3.27e)$$

$$\mu = \mu^0 r^{-1} - \psi_2^0 r^{-2} - k |\phi_1^0|^2 r^{-3} \quad (3.27f)$$

$$\nu = \nu^0 - \psi_3^0 r^{-1} - 2k \phi_1^0 \phi_2^0 r^{-2} \quad (3.27g)$$

D. Metric variables:

$$\omega = X^A = 0 \quad (3.28a)$$

$$\xi^A = -(P, -iP) r^{-1} \quad (3.28b)$$

$$U = U^0 - 2\gamma^0 r - \psi_2^0 r^{-1} - k |\phi_1^0|^2 r^{-2} \quad (3.28c)$$

E. Relationships among the "constants":

$$\alpha^0 = -\frac{\partial P}{\partial \zeta} = -\frac{1}{2} \bar{\psi} \log P \quad (3.29a)$$

$$U^0 = \mu^0 = -K = -\bar{\psi} \log P \quad (3.29b)$$

$$\gamma^0 = -\frac{1}{2} \frac{\dot{P}}{P}, \quad \nu^0 = \bar{\psi} \frac{\dot{P}}{P} \quad (3.29c)$$

$$\psi_2^0 = \bar{\psi}_2^0, \quad \phi_1^0 = E(u) \quad (3.29d)$$

$$\psi_3^0 = \bar{\psi}_3^0 K, \quad \psi_4^0 = -\bar{\psi}^2 \frac{\dot{P}}{P} \quad (3.29e)$$

F. Differential equations:

$$\dot{\psi}_2^0 = 2k E \bar{\psi}_2^0 \quad (3.30a)$$

$$\dot{E} - 2 \frac{\dot{P}}{P} E = -\bar{\psi} \phi_2^0 \quad (3.30b)$$

$$\dot{\psi}_2^0 - 3 \frac{\dot{P}}{P} \psi_2^0 = -\bar{\psi} K + k \phi_2^0 \bar{\phi}_2^0 \quad (3.30c)$$

G. Components of the metric tensor:

$$g^{\mu 0} = \delta_1^\mu, \quad g^{AB} = \frac{2P^2}{r^2} \delta^{AB} \quad (3.31a)$$

$$g^{11} = -2 \left(K - \frac{\dot{P}}{P} r + \frac{\psi_2^0}{r} + k \frac{|E|^2}{r^2} \right) \quad (3.31b)$$

H. Line element:

$$ds^2 = 2(K - \frac{\dot{P}}{P} r + \frac{\psi_2^0}{r} + k \frac{|E|^2}{r^2}) du^2 + 2 du dr - \frac{r^2}{2P^2} d\zeta d\bar{\zeta} \quad (3.32)$$

We again impose the regularity condition $P = P_0(1+I)$ on the above results and try to obtain from them the equations of motion and the time dependence of the internal degrees of freedom. First we note that by integrating (3.30b) over the F2S we obtain

$$\oint (\dot{E} - 2 \frac{\dot{P}}{P} E) \frac{dx^2 dx^3}{P^2} = -2 \oint \frac{\partial}{\partial \zeta} \left(\frac{\phi_2^0}{P} \right) dx^2 dx^3.$$

The right-hand side vanishes as in (3.22) and the left-hand side can be expressed as a u-derivative of E times the F2S area A, i.e.

$$\frac{d}{du} (EA) = \frac{d}{du} \int E \frac{dx^2 dx^3}{P^2} = 0 \quad (3.33)$$

Therefore, since EA is conserved we can take the electric charge e of the singularity as

$$e = \frac{1}{8\pi} (E + \bar{E})A \quad (3.34a)$$

and the magnetic monopole moment q as

$$q = \frac{1}{8\pi i} (E - \bar{E})A \quad (3.34b)$$

$\text{Re}(E)$ can thus be regarded as essentially a surface charge density and $\text{Im}(E)$, as the magnetic monopole per unit area. We had hoped that the Einstein-Maxwell theory would yield the result $q = 0$, but

our analysis has failed to show this. Nevertheless, we shall assume that E is real or $q = 0$.

Now, consider the three differential equations (3.30). If we let $\phi_2^0 = \bar{\delta}H$, where H is a spin weight zero quantity, Eq. (3.30b) becomes

$$\delta\bar{\delta}H = -(\dot{E} - 2\frac{\dot{P}}{P}E) . \quad (3.35)$$

Then by applying $\bar{\delta}$ to (3.30a) and using (3.30b) we obtain

$$\delta\bar{\delta}\psi_2^0 = -2k E(\dot{E} - 2\frac{\dot{P}}{P}E), \quad (3.36)$$

which looks very similar to (3.35). The procedure now is to try to solve for H and ψ_2^0 as functions of P_0 and I in (3.35) and (3.36) and then to substitute these into (3.30c) to determine P_0 and I . As we shall see, however, Equations (3.30) are quite complicated. The only known exact solution is the Reissner-Nordstrom metric which can be obtained explicitly by setting $I = 0$, $P_0 = \frac{1}{2}(1+\zeta\bar{\zeta})$, $K = 1$, $\psi_2^0 = -\sqrt{2} \text{ km}$, $\phi_1^0 = e$, and $\phi_2^0 = 0$. Although in the general case of $I \neq 0$ there seems to be little hope of solving (3.30) exactly, nevertheless we can do it partially and obtain a startling result.

Rewriting (3.35) and (3.36) in terms of P_0 and I , we get

$$(1+I)^2 \delta_0 \bar{\delta}_0 H = -\dot{E} + 2E \frac{\dot{P}_0}{P_0} + 2E \frac{\dot{I}}{1+I} \quad (3.37)$$

$$(1+I)^2 \delta_0 \bar{\delta}_0 \psi_2^0 = -2kE\dot{E} + 4kE^2 \frac{\dot{P}_0}{P_0} + 4kE^2 \frac{\dot{I}}{1+I} \quad (3.38)$$

Note that when $I = 0$ we can solve these two equations exactly, obtaining

$$\dot{E} = 0 \quad (3.39a)$$

$$H = -E \frac{\dot{P}_0}{P_0} \quad (3.39b)$$

$$\psi_2^0 = -M(u) - 2k E^2 \frac{\dot{P}_0}{P_0} \quad (3.39c)$$

This leads us to make the following ansatz for the solutions of (3.37) and (3.38) with $I \neq 0$:

$$H = -E \frac{\dot{P}_0}{P_0} + ET \quad (3.40a)$$

$$\psi_2^0 = -M(u) - 2k E^2 \frac{\dot{P}_0}{P_0} + 2k E^2 S, \quad (3.40b)$$

where $\dot{E} = T = S = 0$ when $I = 0$, that is, T and S contain the I -dependent parts of H and ψ_2^0 . Putting (3.40a) and (3.40b) back into (3.37) and (3.38) we obtain the following equations for T and S :

$$(1+I)^2 \mathcal{F}_0 \mathcal{F}_0 T = -2(2I+I^2) \frac{\dot{P}_0}{P_0} + 2 \frac{\dot{I}}{1+I} - \frac{\dot{E}}{E} \quad (3.41a)$$

$$(1+I)^2 \mathcal{F}_0 \mathcal{F}_0 S = -2(2I+I^2) \frac{\dot{P}_0}{P_0} + 2 \frac{\dot{I}}{1+I} - \frac{\dot{E}}{E} \quad (3.41b)$$

If we now substitute (3.40a) and (3.40b) into (3.30c), we get (after some regrouping of terms)

$$\ddot{M} + 2kE^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right) - 3k \left(\frac{M}{k} - \frac{4}{3} E\dot{E} \right) \frac{\dot{P}_0}{P_0} - 9kE^2 \left[\left(\frac{\dot{P}_0}{P_0} \right)^2 + \frac{\ddot{\xi}^2}{6} \right] = \mathcal{F}, \quad (3.42)$$

where $\ddot{\xi}^2$ comes from (2.62) and $\mathfrak{F}$ is given by

$$\begin{aligned}\mathfrak{F} = & 2kE^2\dot{S} + 2kS(2E\dot{E} - 3E^2\frac{\dot{P}_0}{P_0}) + kE^2(2I+I^2)\left[\left(\frac{\dot{P}_0}{P_0}\right)^2 + \frac{\ddot{\xi}^2}{6}\right] \\ & + 3\frac{\dot{I}}{1+I}(M + 2kE^2\frac{\dot{P}_0}{P_0} - 2kE^2S) + (1+I)^2[\mathfrak{F}_0\bar{\mathfrak{F}}_0K \\ & + kE^2\mathfrak{F}_0T\bar{\mathfrak{F}}_0\frac{\dot{P}_0}{P_0} + \bar{\mathfrak{F}}_0T\mathfrak{F}_0\frac{\dot{P}_0}{P_0} - \mathfrak{F}_0\bar{T}\bar{\mathfrak{F}}_0T)]\end{aligned}\quad (3.43)$$

When $I = 0$, we have $S = T = \dot{E} = K - 1 = 0$, and hence $\mathfrak{F} = 0$. In principle $\mathfrak{F}$ can be decomposed as a sum of different spherical harmonic terms $\mathfrak{F}_\ell$ such that the different ℓ -terms in (3.42) can be separated out as follows

$$\ell = 0, \quad \dot{M} = \mathfrak{F}_0 \quad (3.44a)$$

$$\ell = 1, \quad 2kE^2\left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2}\right) - k\left(\frac{M}{k} - \frac{4}{3}E\dot{E}\right)\frac{\dot{P}_0}{P_0} = \mathfrak{F}_1 \quad (3.44b)$$

$$\ell = 2, \quad -9kE^2\left[\left(\frac{\dot{P}_0}{P_0}\right)^2 + \frac{\ddot{\xi}^2}{6}\right] = \mathfrak{F}_2 \quad (3.44c)$$

$$\ell \geq 3, \quad 0 = \mathfrak{F}_\ell \quad (3.44d)$$

If we multiply (3.44b) by $-A^2/48k\pi^2$ and use the definitions of the neutral mass m_0 and electric charge e given in (3.21a) and (3.34a), we can express (3.44b) as

$$\sqrt{2} \left(m_0 + \frac{4}{3\sqrt{2}} e^2 \frac{\dot{A}}{A} \right) \frac{\dot{P}_0}{P_0} - \frac{2}{3} e^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right) = F, \quad (3.45)$$

where F is given by

$$F = - \frac{3_1 A^2}{48k\pi^2} . \quad (3.46)$$

We see from (3.45) that the "inertial mass" m_i of the singularity can be defined as

$$m_i \equiv m_o + m_{EM} , \quad (3.47)$$

where

$$m_{EM} \equiv \frac{4}{3\sqrt{2}} e^2 \frac{\dot{A}}{A} \quad (3.48)$$

can be taken as its "electromagnetic mass".³⁰ (Note: in the static case, $m_{EM} = 0$.) Rewriting (3.45) in terms of this "inertial mass" we get

$$\sqrt{2} \ m_i \frac{\dot{P}_o}{P_o} = F + \frac{2}{3} e^2 \left(\frac{\ddot{P}_o}{P_o} + \frac{\ddot{\xi}^2}{2} \right) , \quad (3.49)$$

where the second term on the right is precisely the Abraham radiation reaction force [see (2.68b)]. Thus, without the use of ad hoc models or mass renormalization, we have been able to derive the Abraham radiation reaction force rigorously. In Chapter 5.0 it will be shown that if incoming fields are allowed to interact with the charged singularity the same result plus the Lorentz force can be obtained.

From (3.44c) we also get the following result: if $I = 0$, then $\mathfrak{F}_2 = 0$ which implies that $\dot{P}_0/P_0 = 0$; hence if there is acceleration, then the internal degrees of freedom must be excited.

It appears as if not much more can be done with the exact equations. So we consider now the linear approximation in which we take $\dot{P}_0/P_0$ and I to be small quantities and keep only first order terms. The linearization of (3.41) yields

$$\mathfrak{F}_0 \bar{\mathfrak{F}}_0 T = 2\dot{I}, \quad \mathfrak{F}_0 \bar{\mathfrak{F}}_0 S = 2\dot{I}, \quad \dot{E} = 0, \quad (3.50)$$

while that of (3.42) gives

$$\begin{aligned} \dot{M} - 3M \frac{\dot{P}_0}{P_0} + 2kE^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{E}^2}{2} \right) &= 2kE^2 \dot{S} + 3M\dot{I} \\ \mathfrak{F}_0 \bar{\mathfrak{F}}_0 \mathfrak{F}_0 \bar{\mathfrak{F}}_0 I + 2\mathfrak{F}_0 \bar{\mathfrak{F}}_0 I. & \end{aligned} \quad (3.51)$$

If I is assumed to have a definite ℓ -value, i.e. $\mathfrak{F}_0 \bar{\mathfrak{F}}_0 I = -LI$ where $L = \ell(\ell+1)$, then $\dot{I}$ will also be a spherical harmonic with the same ℓ -value because of the linearization. So from (3.50) we obtain

$$T = -2 \frac{\dot{I}}{L}, \quad S = -2 \frac{\dot{I}}{L}. \quad (3.52)$$

Substituting this into (3.51) and breaking up the result into the different ℓ terms, we find

$$\ell = 0, \quad \dot{M} = 0 \quad (3.53)$$

$$\ell = 1, \quad 2E^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right) - 3M \frac{\dot{P}_0}{P_0} = 0 \quad (3.54)$$

$$\ell > 2, \quad \frac{4kE^2}{L} \ddot{I} - 3M\dot{I} - L(L-2)I = 0 \quad (3.55)$$

The general solutions of (3.54) and (3.55) have exponential growth, which means that the Reissner-Nordström solution is unstable (at least initially) to small perturbations I and to small accelerations. This unpleasant behavior could possibly be reversed or stabilized by the non-linear terms in (3.41) and (3.42), but having no consistent approximation method so far, we can only hope that the non-linear terms will actually have a stabilizing effect on the solutions.

For completeness we should also consider the gravitational and electromagnetic radiation fields which are given by ψ_4^0 and ϕ_2^0 respectively. If we linearize the expressions for ψ_4^0 and ϕ_2^0 , we obtain

$$\psi_4^0 = -\bar{r}^2 \frac{\dot{P}}{P} = -\bar{r}_0 [(1+I)^2 \bar{r}_0 \left(\frac{\dot{P}_0}{P_0} + \frac{\dot{I}}{1+I} \right)] = -\bar{r}_0^2 \dot{I} \quad (3.56)$$

$$\phi_2^0 = \bar{r}H = (1+I)\bar{r}_0 (E\bar{r} - E \frac{\dot{P}_0}{P_0}) = -\frac{2E}{L} \bar{r}_0 \dot{I} - E\bar{r}_0 \frac{\dot{P}_0}{P_0} \quad (3.57)$$

The last term in ϕ_2^0 is precisely the Lienard-Wiechert radiation field³⁵; the first term is what one would expect from an electric 2^ℓ -pole field ($\ell \geq 2$). The result for ψ_4^0 corresponds to radiation from a mass 2^ℓ -pole.⁴⁴ We are therefore led to interpret I as representing some type of distribution from which both the gravitational and electric multipole moments can be calculated.

4.0 MOTION IN A GENERAL EMPTY SPACE

In this chapter we shall consider the generalization of the Robinson-Trautman solutions to a general empty space $R_{\mu\nu} = 0$, in which the elementary singularities are coupled to an incoming gravitational background field represented by ψ_0 . We first integrate the vacuum field equation under the condition (2.70) and then analyze the results for the motion and structure of the singularities.

4.1 Integration of the Empty Space Equations

Consider the two empty-space radial equations for ρ and σ *

$$D\rho = \rho^2 + \sigma\bar{\sigma}, \quad (2.18a)'$$

$$D\sigma = 2\rho\sigma + \psi_0, \quad (2.18b)'$$

and write condition (2.70) as

$$\rho = -r^{-1} + g, \quad g = O(r), \quad (4.1a)$$

$$\sigma = h, \quad h = O(r). \quad (4.1b)$$

To determine the general form that ψ_0 must have near $r=0$ in order to be compatible with condition (4.1), let us substitute (4.1b) into (2.18b). This gives us

$$Dh + 2r^{-1}h = 2hg + \psi_0 = \psi_0 + O(r^2),$$

*The prime after the equation numbers given in 2.1 shall indicate the empty-space NP equations ($\phi_{mn} = R = 0$).

and integrating** this equation, we get

$$h = r^{-2} \{h^0(u, x^A) + \int r^2 [\psi_0 + O(r^2)] dr\}, \quad (4.2)$$

from which it follows that since $h = O(r)$, $h^0 = 0$ and $\psi_0 = O(1)$, i.e. ψ_0 must be a regular function of r near the origin $r = 0$. Thus condition (2.70) excludes, for example, r^{-5} and higher singularities whose presence in ψ_0 , in linear theory, corresponds to the presence of intrinsic quadrupole and higher multipole moments.

Let us now assume an explicit form for ψ_0 , namely,

$$\psi_0 = \psi_0^0 + \psi_0^1 r + O(r^2), \quad (4.3)$$

with the additional condition

$$D\psi_0 = \psi_0^1 + O(r). \quad (4.4)$$

Substituting (4.3) into (4.2) we get

$$h = r^{-2} \int r^2 [\psi_0^0 + \psi_0^1 r + O(r^2)] dr,$$

or

$$h = \frac{1}{3} \psi_0^0 r + \frac{1}{4} \psi_0^1 r^2 + O(r^3). \quad (4.5)$$

Now, using (4.1a) and (4.1b) in (2.18a)' we find

$$Dg + 2r^{-1}g = g^2 + h\bar{h} = O(r^2)$$

with the solution $g = O(r^3)$. Again, let $\rho = -r^{-1} + g$ in (2.18a)'

where this time $g = O(r^3)$, and use (4.5) to solve for g . This gives us

**Order symbols can be integrated, but cannot be differentiated.

$$Dg + 2r^{-1}g = \frac{1}{9} |\psi_0^0|^2 r^2 + \frac{1}{6} \text{Re}(\psi_0^0 \psi_0^1) r^3 + O(r^4),$$

with the solution

$$g = \frac{r^3}{45} |\psi_0^0|^2 + \frac{r^4}{36} \text{Re}(\psi_0^0 \psi_0^1) + O(r^5). \quad (4.6)$$

Therefore, with the form (4.3) of ψ_0 given explicitly to $O(r^2)$, we obtain from (4.5) and (4.6) the following expressions for ρ and σ :

$$\rho = -r^{-1} + \frac{r^3}{45} |\psi_0^0|^2 + \frac{r^4}{36} \text{Re}(\psi_0^0 \psi_0^1) + O(r^5) \quad (4.7)$$

$$\sigma = \frac{1}{3} \psi_0^0 r + \frac{1}{4} \psi_0^1 r + O(r^3). \quad (4.8)$$

Note that it was the condition (2.70) on ρ and σ which led us to the result that ψ_0 must be regular around $r = 0$. In Appendix B we show that it is also possible to go the other way, that is, obtain (2.70) as a special case of the assumption $\psi_0 = O(1)$.

To find the r -dependence of the other NP quantities, we shall require the following lemma from the theory of ordinary differential equations⁴⁵, which is given in NP (ref. 33):

Lemma: Consider the differential equation

$$dy = (As^{-1} + B)y + b \quad (4.9)$$

where $d \equiv d/ds$, y is a column matrix with n rows, A is a complex $(n \times n)$ matrix independent of s , and the complex $(n \times n)$ matrix B and the column matrix b are given functions of s . If A has no eigenvalue with a positive real part and if B and b have the asymptotic behavior

$$B = \hat{O}(s^{-2}), \quad b = \hat{O}(s^{-2}),$$

then all the solutions of (4.7) are bounded as $s \rightarrow \infty$, i.e. $y = \hat{O}(1)$.*

If we now let $r = s^{-1}$, we can express D as $D = -s^2 d$ and write ρ and σ from (4.7) and (4.8) as

$$\rho = -s + \hat{O}(s^{-3})$$

$$\sigma = \frac{1}{3} \psi_0^0 s^{-1} + \frac{1}{4} \psi_0^1 s^{-2} + \hat{O}(s^{-3}). \quad (4.10)$$

Making these substitutions in the radial equation (2.16a)' for ξ^A , we can rewrite this equation and its complex conjugate as

$$\begin{aligned} d\xi^A &= [s^{-1} + \hat{O}(s^{-5})]\xi^A - [\frac{1}{3} \psi_0^0 s^{-3} + \hat{O}(s^{-4})]\bar{\xi}^A \\ d\bar{\xi}^A &= [s^{-1} + \hat{O}(s^{-5})]\bar{\xi}^A - [\frac{1}{3} \bar{\psi}_0^0 s^{-3} + \hat{O}(s^{-4})]\xi^A \end{aligned} \quad (4.11)$$

By letting $\xi^A = \xi^{*A} s$, (4.11) can be expressed in matrix form as

$$d \begin{bmatrix} \xi^{*A} \\ \bar{\xi}^{*A} \end{bmatrix} = \begin{bmatrix} \hat{O}(s^{-5}) & \hat{O}(s^{-3}) \\ \hat{O}(s^{-3}) & \hat{O}(s^{-5}) \end{bmatrix} \begin{bmatrix} \xi^{*A} \\ \bar{\xi}^{*A} \end{bmatrix}$$

from which we see that the lemma applies with $A = b = 0$. Therefore, it follows that $\xi^{*A} = \hat{O}(1)$, i.e. $\xi^A = \tilde{O}(s)$ or

*Here, $f(s, u, x^A) = \tilde{O}(f(s))$ means $|f(s, u, x^A)| < g(s)F(u, x^A)$ for some function F independent of s and for all large s .

$$\xi^A = O(r^{-1}) \quad (4.12)$$

With this information we may now integrate (2.16a)' in steps to solve for ξ^A in terms of ψ_0 . Thus, using (4.12) in (2.16a)' we have

$$(2.16a)' \quad D\xi^A = \rho \xi^A + \sigma \bar{\xi}^A \rightarrow D\xi^A + r^{-1}\xi^A = O(1) \rightarrow \xi^A = \xi^{OA} r^{-1} + O(r), \quad (4.13)$$

where the "constant" of integration ξ^{OA} shall be chosen as usual to be

$$\xi^{O2} = i \xi^{O3} = -P(u, r, \bar{\xi}), \quad P = \bar{P} \quad (4.14)$$

By letting $\xi^A = \xi^{OA} r^{-1} + x^A$, $x^A = O(r)$, and putting this back into (2.16a)' we obtain

$$Dx^A + r^{-1}x^A = \frac{1}{3} \psi_0^O \bar{\xi}^{OA} + \frac{1}{4} \psi_0^1 \bar{\xi}^{OA} r + O(r^2) \rightarrow$$

$$x^A = \frac{1}{6} \psi_0^O \bar{\xi}^{OA} r + \frac{1}{12} \psi_0^1 \bar{\xi}^{OA} r^2 + O(r^3).$$

Therefore, ξ^A is given by

$$\xi^A = \xi^{OA} r^{-1} + \frac{1}{6} \psi_0^O \bar{\xi}^{OA} r + \frac{1}{12} \psi_0^1 \bar{\xi}^{OA} r^2 + O(r^3). \quad (4.15)$$

The lemma can be applied once more to determine the r -behavior of ω , α , β , τ , and ψ_1 near $r = 0$. Using (4.3), (4.4), and (4.10) in the radial equations (2.16b)', (2.18c,d,e)', and (2.20a)', these equations for ω , α , β , τ , and ψ_1 can be written in terms of $s = 1/r$ as

$$(2.16b)' \quad d\omega = [s^{-1} + \hat{O}(s^{-5})]\omega - \left[\frac{1}{3} \psi_0^O s^{-3} + \hat{O}(s^{-4}) \right] \bar{\omega} + \tau s^{-2} \quad (4.16a)$$

$$(2.18c)' \quad d\alpha = [s^{-1} + \hat{O}(s^{-5})]\alpha - \left[\frac{1}{3} \bar{\psi}_0^O s^{-3} + \hat{O}(s^{-4}) \right] \beta \quad (4.16b)$$

$$(2.18d)' \quad d\beta = [s^{-1} + \hat{O}(s^{-5})]\beta - \left[\frac{1}{3} \psi_0^0 s^{-3} + \hat{O}(s^{-4})\right]\alpha - \psi_1 s^{-2} \quad (4.16c)$$

$$(2.18e)' \quad d\tau = [s^{-1} + \hat{O}(s^{-5})]\tau - \left[\frac{1}{3} \psi_0^0 s^{-3} + \hat{O}(s^{-4})\right]\bar{\tau} - \psi_1 s^{-2} \quad (4.16d)$$

$$(2.20a)' \quad d\psi_i = [4s^{-1} + \hat{O}(s^{-4})]\psi_i + [4\psi_0^0 s^{-2} + \hat{O}(s^{-3})]\alpha + \hat{O}(s^{-2})\bar{\omega} + \hat{O}(s^{-1}) \quad (4.16e)$$

If we let

$$\omega = \omega^* s^2, \quad \alpha = \alpha^* s^2, \quad \beta = \beta^* s^3, \quad \tau = \tau^* s^3, \quad \psi_1 = \psi_1^* s^3 \quad (4.17)$$

then equations (4.16) and their complex conjugates can be expressed in the form of (4.9) with

$$y = \begin{bmatrix} \omega^* \\ \bar{\omega}^* \\ \alpha^* \\ \bar{\alpha}^* \\ \beta^* \\ \bar{\beta}^* \\ \tau^* \\ \bar{\tau}^* \\ \psi_1^* \\ \bar{\psi}_1^* \end{bmatrix} \quad A = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (4.18)$$

Since A has no eigenvalue with positive real part, it therefore follows from the lemma that

$$\omega^*, \alpha^*, \beta^*, \tau^*, \psi_1^* = \hat{O}(1)$$

or by (4.17)

$$\omega, \alpha = O(r^{-2}), \quad \beta, \tau = O(r^{-3}), \quad \psi_1 = O(r^{-4}) \quad (4.19)$$

Using this information to integrate (2.16b)', (2.18c,d,e)', and (2.20a)', we obtain

$$(2.20a)' \quad D\psi_1 + 4r^{-1}\psi_1 = O(r^{-2}) \rightarrow \psi_1 = \psi_1^0 r^{-4} + O(r^{-4}), \quad (4.20a)$$

where ψ_1^0 is again a "constant" of integration, which, in linear theory, can be related to the intrinsic mass dipole moment and angular momentum of the singularity.⁴⁴ Here we shall adopt the assumption

$$\psi_1^0 = 0, \quad (4.21)$$

but it will be shown in Appendix C that this assumption is actually unnecessary. Integrating the other equations with the assumption (4.21), we then have

$$(2.18a)' \quad D\beta + r^{-1}\beta = O(r^{-1}) \rightarrow \beta = \beta^0 r^{-1} + O(1) \quad (4.20b)$$

$$(2.18c)' \quad D\alpha + r^{-1}\alpha = \frac{1}{3} \beta^0 \bar{\psi}_0^0 + O(r) \rightarrow \alpha = \alpha^0 r^{-1} + \frac{1}{6} \beta^0 \bar{\psi}_0^0 r + O(r^{-2}) \quad (4.20c)$$

$$(2.18e)' \quad D\tau + r^{-1}\tau = O(r^{-1}) \rightarrow \tau = \tau^0 r^{-1} + O(1), \quad (4.20d)$$

where again $\tau^0 = \bar{\alpha}^0 + \beta^0$ is made zero by means of the null rotations,⁴² that is,

$$\tau^0 = 0, \quad \beta^0 = -\bar{\alpha}^0 \quad (4.22)$$

$$(2.16b)' \quad D\omega + r^{-1}\omega = O(r^{-1}) \rightarrow \omega = \omega^0 r^{-1} + O(1) \quad (4.20e)$$

Before proceeding further, let us substitute these results into the non-radial equation (2.17b)', namely,

$$\omega D \bar{\xi}^A + \bar{\xi}^B \bar{\xi}_{,A} - \bar{\omega} D \xi^A - \bar{\xi}^B \xi_{,B}^A + (\alpha - \bar{\beta}) \xi^A + (\beta - \bar{\alpha}) \bar{\xi}^A = 0 \quad (2.17b)'$$

and set the coefficients of the various powers of r equal to zero.*

From the coefficient of r^{-3} we find that

$$\omega^0 = 0 \quad (4.23)$$

and from the coefficient of r^{-2} we get

$$\alpha^0 = - \frac{\partial P}{\partial \bar{\xi}} \quad (4.24)$$

Note that from (4.14) and (4.24) the operator $\mathcal{S}$ defined in Section 2.2 can be expressed as

$$\mathcal{S}\eta = -(\xi^{0A} \eta_{,A} + 2s\bar{\alpha}^0 \eta), \quad \bar{\mathcal{S}}\eta = -(\bar{\xi}^{0A} \eta_{,A} - 2s\alpha^0 \eta) \quad (4.25)$$

for any spin-weight s quantity η .

If we now go back to the radial equations (2.16b)', (2.18c,d,e)', and (2.20a)' with the information given by (4.23) and (4.24), we can integrate these equations further and obtain the final results

$$\psi_1 = -\frac{1}{4} \bar{\mathcal{S}}\psi_0^0 - \frac{1}{5} \bar{\mathcal{S}}\psi_0^1 r + O(r^2) \quad (4.26a)$$

*If, for example, we have the expression $Ar^{-1} + Br \log r + Cr + O(r^2) = 0$, where A, B, C are all independent of r , we get $A = 0$ by multiplying by r and taking the limit $r \rightarrow 0$. Similarly, multiplying by $(r \log r)^{-1}$ and then taking the limit $r \rightarrow 0$ give $B = 0$; etc. Thus $A, B, C = 0$ separately.

$$\alpha = \alpha^0 r^{-1} - \frac{1}{6} \bar{\alpha}^0 \bar{\psi}_0^0 r - \frac{1}{12} \bar{\alpha}^0 \bar{\psi}_0^1 r^2 + O(r^3) \quad (4.26b)$$

$$\beta = -\bar{\alpha}^0 r^{-1} + \left(\frac{1}{6} \alpha^0 \psi_0^0 - \frac{1}{8} \bar{\alpha} \psi_0^0\right) r + \left(\frac{1}{12} \alpha^0 \psi_0^1 - \frac{1}{15} \bar{\alpha} \psi_0^1\right) r + O(r^3) \quad (4.26c)$$

$$\tau = -\frac{1}{8} \bar{\alpha} \psi_0^0 r - \frac{1}{15} \bar{\alpha} \psi_0^1 r^2 + O(r^3) \quad (4.26d)$$

$$\omega = \frac{1}{24} \bar{\alpha} \psi_0^0 r^2 + \frac{1}{60} \bar{\alpha} \psi_0^1 r^3 + O(r^4) \quad (4.26e)$$

Then with the help of (4.15) and (4.26d) we find from (2.16c)' that

$$X^A = X^{0A} - \frac{1}{8} (\bar{\xi}^{0A} \bar{\alpha} \psi_0^0 + \xi^{0A} \bar{\alpha} \psi_0^0) r - \frac{1}{30} (\bar{\xi}^{0A} \bar{\alpha} \psi_0^1 + \xi^{0A} \bar{\alpha} \psi_0^1) r^2 + O(r^3) \quad (4.26f)$$

Let us next apply the lemma to determine the r -dependence of λ , μ , and ψ_2 . In terms of s , the radial equations (2.18g,h)' and (2.20b)' can be written as

$$(2.18g)' \quad d\lambda = [s^{-1} + \hat{O}(s^{-5})]\lambda - \left[\frac{1}{3} \bar{\psi}_0^0 s^{-3} + \hat{O}(s^{-4})\right]$$

$$(2.18h)' \quad d\mu = [s^{-1} + \hat{O}(s^{-5})]\mu - \left[\frac{1}{3} \psi_0^0 s^{-3} + \hat{O}(s^{-4})\right]\lambda - \psi_2 s^{-2}$$

$$(2.20b)' \quad d\psi_2 = [3s^{-1} + \hat{O}(s^{-5})]\psi_2 + [\psi_0^0 s^{-2} + \hat{O}(s^{-3})]\lambda + \hat{O}(s^{-1})$$

By putting $\lambda = \lambda^* s$, $\mu = \mu^* s^2$, $\psi_2 = \psi_2^* s^3$, the lemma can be applied to these equations and their complex conjugates, with

$$Y = \begin{bmatrix} \lambda^* \\ \bar{\lambda}^* \\ \mu^* \\ \bar{\mu}^* \\ \psi_2^* \\ \bar{\psi}_2^* \end{bmatrix} \quad A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (4.27)$$

Thus, λ^* , μ^* , $\psi_2^* = \hat{O}(1)$, that is

$$\lambda = O(r^{-1}), \quad \mu = O(r^{-2}), \quad \psi_2 = O(r^{-3}) \quad (4.28)$$

With this information we can again integrate (2.18g,h)' and (2.20b)' and obtain

$$\lambda = \lambda^0 r^{-1} - \frac{1}{3} \psi_2^0 \bar{\psi}_0^0 + \left(\frac{1}{6} \mu^0 \bar{\psi}_0^0 - \frac{1}{8} \psi_2^0 \bar{\psi}_0^1 \right) r + O(r^2) \quad (4.29a)$$

$$\begin{aligned} \mu = & -\psi_2^0 r^{-2} + \mu^0 r^{-1} + \frac{1}{24} \bar{\psi}_0^2 \psi_0^0 + \left(\frac{1}{60} \bar{\psi}_0^2 \psi_0^1 - \frac{1}{90} \psi_2^0 |\psi_0^0|^2 \right) r^2 \\ & + O(r^3) \end{aligned} \quad (4.29b)$$

$$\begin{aligned} \psi_2 = & \psi_2^0 r^{-3} + \frac{1}{12} (\bar{\psi}_0^2 \psi_0^0 - 4\lambda^0 \psi_0^0) + \frac{1}{20} (\bar{\psi}_0^2 \psi_0^1 + 2\psi_2^0 |\psi_0^0|^2 - 5\lambda^0 \psi_0^1) r \\ & + O(r^2) \end{aligned} \quad (4.29c)$$

Then substituting these results into (2.18f)' we find that

$$\begin{aligned} \gamma = & -\frac{1}{2} \psi_2^0 r^{-2} + \gamma^0 + \frac{1}{12} [\bar{\psi}_0^2 \psi_0^0 - 4\lambda^0 \psi_0^0 - 3 \operatorname{Im}(\alpha^0 \bar{\psi}_0^0)] r \\ & + \frac{1}{40} [\bar{\psi}_0^2 \psi_0^1 + 2\psi_2^0 |\psi_0^0|^2 - 5\lambda^0 \psi_0^1 - \frac{8}{3} \operatorname{Im}(\alpha^0 \bar{\psi}_0^1)] r^2 + O(r^3) \end{aligned} \quad (4.29d)$$

This in turn is used in (2.16d)' to obtain

$$\begin{aligned} U = & -\frac{1}{2} (\psi_2^0 + \bar{\psi}_2^0) r^{-1} + U^0 - (\gamma^0 + \bar{\gamma}^0) r - \frac{1}{24} [\bar{\psi}_0^2 \psi_0^0 + \bar{\psi}_0^2 \bar{\psi}_0^0 \\ & - 4\lambda^0 \psi_0^0 - 4\bar{\lambda}^0 \bar{\psi}_0^0] r^2 - \frac{1}{120} [\bar{\psi}_0^2 \psi_0^1 + \bar{\psi}_0^2 \bar{\psi}_0^1 + 2(\psi_2^0 + \bar{\psi}_2^0) |\psi_0^0|^2 \\ & - 5\lambda^0 \psi_0^1 - 4\bar{\lambda}^0 \bar{\psi}_0^1] r^3 + O(r^4) \end{aligned} \quad (4.29e)$$

If we now consider the non-radial equation (2.19g)', we find from the coefficient of r^{-2} that

$$\lambda^0 = 0. \quad (4.30)$$

Finally, we consider the r -dependence of ψ_3 , ψ_4 , and v . If we let $\psi_3 = \psi_3^* s^4$, the radial equation (2.20c)' can be written as

$$d\psi_3^* = [-2s^{-1} + O(s^{-5})]\psi_3^* + \hat{O}(s^{-5}),$$

which obviously satisfies the lemma. Therefore, $\psi_3^* = \hat{O}(1)$ or

$$\psi_3 = O(r^{-4}). \quad (4.31)$$

Putting this back into (2.20c)', the integration yields

$$\begin{aligned} \psi_3 = & \bar{\sigma}\psi_2^0 r^{-3} + \psi_3^0 r^{-2} - \left(\frac{1}{8} \psi_2^0 \bar{\sigma}\psi_0^0 + \frac{1}{6} \bar{\psi}_0^0 \bar{\sigma}\psi_2^0 \right) r^{-1} \\ & - \left(\frac{1}{40} \psi_2^0 \bar{\sigma}\bar{\psi}_0^1 + \frac{1}{24} \bar{\sigma}^3 \psi_0^0 + \frac{1}{24} \bar{\psi}_0^1 \bar{\sigma}\psi_2^0 \right) + O(r) \end{aligned} \quad (4.32a)$$

Then integrating (2.18i)' we obtain

$$\begin{aligned} v = & -\frac{1}{2} \bar{\sigma}\psi_2^0 r^{-2} - \psi_3^0 r^{-1} - \frac{1}{6} \bar{\psi}_0^0 \bar{\sigma}\psi_2^0 \log r + v^0 \\ & + \frac{1}{24} (\psi_2^0 \bar{\sigma}\bar{\psi}_0^1 - \bar{\psi}_0^1 \bar{\sigma}\psi_2^0 - \bar{\sigma}^3 \psi_0^0 - 3\bar{\psi}_0^0 \bar{\sigma}\bar{\psi}_0^1) r + O(r^2) \end{aligned} \quad (4.32b)$$

The radial equation (2.20d)' can now be written as

$$d\psi_4^* = [-3s^{-1} + \hat{O}(s^{-5})]\psi_4^* + O(s^{-2}),$$

where $\psi_4 = \psi_4^* s^4$. Thus the lemma applies and we have $\psi_4^* = \hat{O}(1)$ or

$$\psi_4 = O(r^{-4}). \quad (4.33)$$

With this information we then obtain from (2.20d)' the result

$$\begin{aligned}
\psi_4 = & \frac{1}{2} \bar{\psi}_2 \psi_2^0 r^{-3} + [\bar{\psi} \psi_3^0 - (\psi_2^0)^2 \bar{\psi}_0^0] r^{-2} + [\frac{1}{6} \bar{\psi} \psi_2^0 \bar{\psi}_0^0 \\
& + \frac{1}{8} \psi_2^0 (\bar{\psi} \bar{\psi}_0^0 - 4 \mu^0 \bar{\psi}_0^0 + 3 \psi_2^0 \bar{\psi}_0^1) r^{-1} \log r + \psi_4^0 r^{-1} \\
& + O(1)
\end{aligned} \tag{4.32c}$$

The remaining relationships among the "constants" of integrations can now be found by substituting the results of the radial integration into the non-radial equations and comparing coefficients of equal powers of r . Indicating on the left-hand side the equation from which the relationship was obtained, we have the following:

$$(2.16e)' \quad U^0 = \mu^0 \tag{4.34}$$

$$\bar{\psi} Z^0 = -\frac{4}{3} \psi_2^0 \psi_0^0, \quad Z^0 \equiv \frac{X^{02} + iX^{03}}{P} \tag{4.34b}$$

$$\gamma^0 = -\frac{1}{2} \frac{\dot{P}}{P} + \frac{1}{4} \bar{\psi} Z^0 - \text{Im}(\alpha^0 Z^0) \tag{4.34c}$$

$$(2.17c)' \quad \psi_2^0 = \bar{\psi}_2^0 \tag{4.34d}$$

$$\mu^0 = \bar{\mu}^0 \tag{4.34e}$$

$$(2.17d)' \quad \bar{\psi} \psi_2^0 = 0 \tag{4.34f}$$

$$\psi_3^0 = -\bar{\psi} U^0 \tag{4.34g}$$

$$v^0 = -\bar{\psi}(\gamma^0 + \bar{\gamma}^0) - \frac{1}{6} \psi_2^0 \bar{\psi}_0^0 \tag{4.34h}$$

$$(2.19a)' \quad \psi_4^0 = -\bar{\psi} v^0 + \frac{1}{6} \psi_2^0 (8U^0 \bar{\psi}_0^0 - 3\psi_2^0 \bar{\psi}_0^1) \tag{4.34i}$$

$$(2.19c)' \quad \mu^0 = -\bar{\psi} \log P = -K \tag{4.34j}$$

$$(2.21a)' \quad \bar{\psi} \bar{\psi}_0^0 = -3\psi_2^0 \psi_0^1 \tag{4.34k}$$

$$(2.21c)' \quad \dot{\psi}_2^0 + 3(\gamma^0 + \bar{\gamma}^0) \psi_2^0 = -\bar{\psi} \psi_3^0 \tag{4.34l}$$

The other equations not indicated here yield results similar to (4.34) or lead to identities.

We now give a summary of the final results together with the metric obtained through (2.13).

A. Tetrad components of the Weyl tensor:

$$\psi_0 = \psi_0^0 + \psi_0^1 r + O(r^2) \quad (4.35a)$$

$$\psi_1 = -\frac{1}{4} \bar{\sigma} \psi_0^0 - \frac{1}{5} \bar{\sigma} \psi_0^1 + O(r^2) \quad (4.35b)$$

$$\psi_2 = \psi_2^0 r^{-3} + \frac{1}{12} \bar{\sigma}^2 \psi_0^0 + \frac{1}{20} (\bar{\sigma}^2 \psi_0^1 + 2\psi_2^0 |\psi_0^0|^2) r + O(r^2) \quad (4.35c)$$

$$\psi_3 = \psi_3^0 r^{-2} - \frac{1}{8} \psi_2^0 \bar{\sigma} \psi_0^0 r^{-1} - \left(\frac{1}{24} \bar{\sigma}^3 \psi_0^0 + \frac{1}{40} \psi_2^0 \bar{\sigma} \psi_0^1 \right) + O(r) \quad (4.35d)$$

$$\psi_4 = [\bar{\sigma} \psi_3^0 - (\psi_2^0)^2 \bar{\psi}_0^0] r^{-2} + \psi_4^0 r^{-1} + O(1) \quad (4.35e)$$

B. Spin coefficients:

$$\rho = -r^{-1} + \frac{1}{45} |\psi_0^0|^2 r + \frac{1}{36} \text{Re}(\psi_0^0 \bar{\psi}_0^1) r^4 + O(r^5) \quad (4.36a)$$

$$\sigma = \frac{1}{3} \psi_0^0 r + \frac{1}{4} \psi_0^1 r^2 + O(r^3) \quad (4.36b)$$

$$\alpha = \alpha^0 r^{-1} - \frac{1}{6} \bar{\alpha}^0 \bar{\psi}_0^0 r - \frac{1}{12} \bar{\alpha}^0 \bar{\psi}_0^1 r^2 + O(r^3) \quad (4.36c)$$

$$\beta = -\bar{\alpha}^0 r^{-1} + \left(\frac{1}{6} \alpha^0 \psi_0^0 - \frac{1}{8} \bar{\sigma} \psi_0^0 \right) r + \left(\frac{1}{12} \alpha^0 \psi_0^1 - \frac{1}{15} \bar{\sigma} \psi_0^1 \right) r^2 + O(r^3) \quad (4.36d)$$

$$\tau = -\frac{1}{8} \bar{\sigma} \psi_0^0 r - \frac{1}{15} \bar{\sigma} \psi_0^1 r^2 + O(r^3) \quad (4.36e)$$

$$\lambda = -\frac{1}{3} \psi_2^0 \bar{\psi}_0^0 + \left(\frac{1}{6} \mu^0 \bar{\psi}_0^0 - \frac{1}{8} \psi_2^0 \bar{\psi}_0^1 \right) r + O(r^2) \quad (4.36f)$$

$$\mu = -\psi_2^0 r^{-2} + \mu^0 r^{-1} + \frac{1}{24} \bar{\vartheta}^2 \psi_0^0 r + \left(\frac{1}{60} \bar{\vartheta}^2 \psi_0^1 - \frac{1}{90} \psi_2^0 |\psi_0^0|^2 \right) r^2 + O(r^3)$$

$$\begin{aligned} \gamma = & -\frac{1}{2} \psi_2^0 r^{-2} + \gamma^0 + \left[\frac{1}{12} \bar{\vartheta}^2 \psi_0^0 - \frac{1}{4} \text{Im}(\alpha^0 \bar{\vartheta} \psi_0^0) \right] r \\ & + \left[\frac{1}{40} \bar{\vartheta}^2 \psi_0^1 + \frac{1}{20} \psi_2^0 |\psi_0^0|^2 - \frac{1}{15} \text{Im}(\alpha^0 \bar{\vartheta} \psi_0^1) \right] r^2 + O(r^3) \end{aligned} \quad (4.36h)$$

$$\nu = -\psi_3^0 r^{-1} + \nu^0 + \frac{1}{24} (\psi_2^0 \bar{\vartheta} \psi_0^1 - \bar{\vartheta}^3 \psi_0^0 - 3\mu^0 \bar{\vartheta} \psi_0^0) r + O(r^2) \quad (4.36i)$$

C. Metric variables

$$\omega = \frac{1}{24} \bar{\vartheta} \psi_0^0 r^2 + \frac{1}{60} \bar{\vartheta} \psi_0^1 r^3 + O(r^4) \quad (4.37a)$$

$$\begin{aligned} U = & -\psi_2^0 r^{-1} + U^0 - (\gamma^0 + \bar{\gamma}^0) r - \frac{1}{12} \text{Re}(\bar{\vartheta}^2 \bar{\psi}_0^0) r^2 \\ & - \frac{1}{60} \text{Re}(\bar{\vartheta}^2 \bar{\psi}_0^1 + 2\psi_2^0 |\psi_0^0|^2) r^3 + O(r^4) \end{aligned} \quad (4.37b)$$

$$\xi^A = \xi^{0A} r^{-1} + \frac{1}{6} \psi_0^0 \bar{\xi}^{0A} r + \frac{1}{12} \psi_0^1 \bar{\xi}^{0A} r^2 + O(r^3) \quad (4.37c)$$

$$X^A = X^{0A} - \frac{1}{4} \text{Re}(\xi^{0A} \bar{\vartheta} \bar{\psi}_0^0) r - \frac{1}{15} \text{Re}(\xi^{0A} \bar{\vartheta} \bar{\psi}_0^1) r^2 + O(r^3) \quad (4.37d)$$

D. Contravariant components of the metric tensor:

$$g^{00} = g^{0A} = 0, \quad g^{01} = 1 \quad (4.38a)$$

$$\begin{aligned} g^{11} = & -2[\psi_2^0 r^{-1} + K + (\gamma^0 + \bar{\gamma}^0) r + \frac{1}{12} \text{Re}(\bar{\vartheta}^2 \bar{\psi}_0^0) r^2 + \\ & \frac{1}{60} \text{Re}(\bar{\vartheta}^2 \bar{\psi}_0^1 + 2\psi_2^0 |\psi_0^0|^2) r^3 + O(r^4)] \end{aligned} \quad (4.38b)$$

$$g^{12} = X^{02} + \frac{P}{3} \text{Re}(\bar{\vartheta} \psi_0^0) r + \frac{P}{10} \text{Re}(\bar{\vartheta} \psi_0^1) r^2 + O(r^3) \quad (4.38c)$$

$$g^{13} = X^{03} + i \frac{P}{3} \text{Im}(\bar{\vartheta} \psi_0^0) r + i \frac{P}{10} \text{Im}(\bar{\vartheta} \psi_0^1) r^2 + O(r^3) \quad (4.38d)$$

$$g^{22} = -2P^2 r^{-2} - \frac{P^2}{3} (\psi_0^0 + \bar{\psi}_0^0) - \frac{P^2}{6} (\psi_0^1 + \bar{\psi}_0^1) r + O(r^2) \quad (4.38e)$$

$$g^{23} = -i \frac{P^2}{3} (\psi_0^0 - \bar{\psi}_0^0) - i \frac{P^2}{6} (\psi_0^1 - \bar{\psi}_0^1)r + O(r^2) \quad (4.38f)$$

$$g^{33} = -2P^2 r^{-2} + \frac{P^2}{3} (\psi_0^0 + \bar{\psi}_0^0) + \frac{P^2}{6} (\psi_0^1 + \bar{\psi}_0^1)r + O(r^2) \quad (4.38g)$$

E. Line element:

$$ds^2 = 2 \left[K - \left(\frac{\dot{P}}{P} - \frac{1}{4} \bar{\theta} Z^0 - \frac{1}{4} \theta \bar{Z}^0 \right) r + \frac{\psi_0^0}{r^2} \right] du^2 \\ + 2dudr - \frac{r^2}{2P^2} d\zeta d\bar{\zeta} + O(r^2) \quad (4.39)$$

F. Relationships among the "constants":

$$\omega^0 = \lambda^0 = \tau^0 = \psi_1^0 = 0 \quad (4.40a)$$

$$\xi^{02} = i\xi^{03} = -P(u, \zeta, \bar{\zeta}), \quad P = \bar{P} \quad (4.40b)$$

$$\alpha^0 = -\frac{\partial P}{\partial \bar{\zeta}} = -\frac{1}{2} \bar{\theta} \log P, \quad \beta^0 = -\bar{\alpha}^0 \quad (4.40c)$$

$$U^0 = \mu^0 = \bar{\mu}^0 = -K = -\theta \bar{\theta} \log P \quad (4.40d)$$

$$\theta Z^0 = -\frac{4}{3} \psi_2^0 \psi_0^0, \quad Z^0 \equiv \frac{1}{P} (X^{02} + iX^{03}) = \frac{X^0}{P} \quad (4.40e)$$

$$\gamma^0 = -\frac{1}{2} \frac{\dot{P}}{P} + \frac{1}{4} \bar{\theta} Z^0 - \text{Im}(\alpha^0 Z^0) \quad (4.40f)$$

$$\nu^0 = -\bar{\theta}(\gamma^0 + \bar{\gamma}^0) - \frac{1}{6} \psi_2^0 \theta \bar{\psi}_0^0 \quad (4.40g)$$

$$\bar{\theta} \theta \psi_0^0 = -3\psi_2^0 \psi_0^1 \quad (4.40h)$$

$$\psi_3^0 = -\bar{\theta} U^0 = \bar{\theta} K \quad (4.40i)$$

$$\psi_4^0 = -\bar{\theta} \nu^0 + \frac{1}{6} \psi_2^0 (\bar{\theta} \theta \bar{\psi}_0^0 - 4K \bar{\psi}_0^0) \quad (4.40j)$$

$$\theta \psi_2^0 = 0, \quad \psi_2^0 = \bar{\psi}_2^0 \quad (4.40k)$$

$$\dot{\psi}_2^0 + 3(\gamma^0 + \bar{\gamma}^0)\psi_2^0 = -\mathcal{E}\psi_3^0 \quad (4.40\ell)$$

4.2 Analysis of the Empty-Space Solutions

The results of **Section 4.1** will again be analyzed with the use of the regularity condition $P = P_0(1+I)$ and of the properties of edth. In particular, Eq. (4.40 ℓ) will once more be shown to yield the equations of motion of the singularity and the time dependence of its internal degrees of freedom.

First, we note that (4.40k) implies that $\mathcal{E}_0\psi_2^0 = 0$, which in turn implies that ψ_2^0 is an $\ell = 0$ spherical harmonic, i.e

$$\psi_2 = -M(u) . \quad (4.41)$$

Now, since Z^0 and ψ_0^0 are respectively $s = 1$ and $s = 2$ quantities, we can write

$$Z^0 = \mathcal{E}V, \quad \psi_0^0 = \mathcal{E}^2R , \quad (4.42)$$

where V and R are both $s = 0$ quantities. Substituting (4.42) into (4.40e), we get

$$\mathcal{E}^2(V - \frac{4}{3}MR) = 0 \quad (4.43)$$

with the solution

$$V = \frac{4}{3}MR + w , \quad (4.44)$$

where w is an $s = 0$ quantity satisfying the equation

$$\mathcal{E}^2w = 0 \quad (4.45)$$

If we let $\bar{w} = f/P$, we can express (4.45) as $\partial f/\partial \zeta = 0$, which shows that f and hence w can be transformed away by means of the coordinate freedom $\zeta' = g(u, \zeta)$.⁴² Therefore, we have

$$V = \frac{4}{3} MR. \quad (4.46)$$

With the help of (4.41), (4.40f), (4.40i), and (4.46), we can now write (4.40l) as

$$\dot{M} - 3M \left[\frac{\dot{P}}{P} - \frac{M}{3} \bar{\sigma} \bar{\sigma} (R - \bar{R}) \right] = \bar{\sigma} \bar{\sigma} K \quad (4.47)$$

or

$$\dot{M} - 3M \frac{\dot{P}}{P} = \bar{\sigma} \bar{\sigma} [K - M(R + \bar{R})] \quad (4.48)$$

Using the regularity condition $P = P_0(1+I)$ and the expression (3.10) for K , (4.48) becomes

$$\begin{aligned} \dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M \frac{\dot{I}}{1+I} &= (1+I)^2 \bar{\sigma}_0 \bar{\sigma}_0 [(1+I)^2 + (1+I) \bar{\sigma}_0 \bar{\sigma}_0 I \\ &\quad - \bar{\sigma}_0 I \cdot \bar{\sigma}_0 I - M^2(R + \bar{R})] \end{aligned} \quad (4.49)$$

Then, carrying out the differentiation of the terms inside the bracket, we obtain

$$\begin{aligned} \dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M \frac{\dot{I}}{1+I} &= (1+I)^3 (\bar{\sigma}_0 \bar{\sigma}_0 \bar{\sigma}_0 \bar{\sigma}_0 I + 2 \bar{\sigma}_0 \bar{\sigma}_0 I) \\ &\quad - (1+I)^2 \bar{\sigma}_0^2 I \cdot \bar{\sigma}_0^2 I - M^2(1+I)^2 \bar{\sigma}_0 \bar{\sigma}_0 (R + \bar{R}) \end{aligned} \quad (4.50)$$

In principle, this equation can be decomposed into different spherical harmonics such that the $l = 0$ part gives the time dependence of M , the

$\ell = 1$ part gives the equation for $\dot{P}_0/P_0$, i.e. the equation of motion, and the $\ell \geq 2$ parts give the time development of the internal degrees of freedom. Due to its extreme non-linearity, however, we are again forced to consider only the linearized version of (4.50). Under the assumption that $\dot{P}_0/P_0$, I , and R are small quantities, the linearization of (4.50) yields

$$\dot{M} - 3M \frac{\dot{P}_0}{P_0} - 3M\dot{I} = \sigma_0 \bar{\sigma}_0 \sigma_0 \bar{\sigma}_0 I + 2\sigma_0 \bar{\sigma}_0 I - M^2 \sigma_0 \bar{\sigma}_0 (R + \bar{R}) \quad (4.51)$$

Note that since ψ_0° is expandable in spin-2 spherical harmonics starting with $\ell = 2$, R must be expandable in $\ell \geq 2$ spherical harmonics in order to be uniquely defined by $\psi_0^\circ = \sigma^2 R \approx \sigma_0^2 R$. If we now let

$I = \sum_{\ell=2} I_\ell$ and $R = \sum_{\ell=2} R_\ell$, then (4.51) decomposes into

$$\ell = 0 \quad \dot{M} = 0 \quad (4.52)$$

$$\ell = 1 \quad M \frac{\dot{P}_0}{P_0} = 0 \quad (4.53)$$

$$\ell \geq 2 \quad \dot{I}_\ell + \frac{1}{3M} \ell(\ell+1)[\ell(\ell+1) - 2]I_\ell = \frac{M}{3} \ell(\ell+1)(R_\ell + \bar{R}_\ell) \quad (4.54)$$

From (4.52) and (4.53) we see that M is constant (i.e. $M = \sqrt{2} \text{ km}$) and the acceleration $\dot{P}_0/P_0$ is zero, while from (4.54) we obtain the general solution

$$I_\ell = e^{-\frac{u}{u_0}} \left\{ I_\ell^0 - \frac{\sqrt{2} \text{ km} \ell(\ell+1)}{3} \int (R_\ell + \bar{R}_\ell) e^{\frac{u}{u_0}} du \right\}, \quad (4.55)$$

where $I_\ell^0 = I_\ell^0(\zeta, \bar{\zeta})$ is a "constant" of integration and u_0 is given by

$$u_0 = \frac{1}{3\sqrt{2} \text{ km}} \ell(\ell+1)[\ell(\ell+1) - 2] \quad (4.56)$$

Thus, in the linear approximation, the particle moves with constant mass and zero acceleration and with its internal degrees of freedom being driven by the incoming gravitational field ψ_0^0 or R . In general, when the non-linear terms in (4.50) are taken into account, the time development of I will be governed by R , but $\dot{M}$ and $\dot{P}_0/P_0$ can no longer be expected to remain zero.

We now consider how the incoming field ψ_0^0 and the radiation field ψ_4^0 are affected by the motion and structure of the singularity. With the help of the commutation relation $(\bar{\mathcal{S}}\mathcal{S} - \mathcal{S}\bar{\mathcal{S}})\eta = 2sK\eta$, we can write (4.40k) in terms of R as

$$\bar{\mathcal{S}}\mathcal{S}^3R = \mathcal{S}^2\bar{\mathcal{S}}\mathcal{S}R + 6K\mathcal{S}^2R = 3M\psi_0^1 \quad (4.57)$$

which shows how the incoming field is affected by the presence of the singularity. Linearizing this and letting $\psi_0^1 = \mathcal{S}^2S$, we obtain

$$\mathcal{S}_0^2(\mathcal{S}_0\bar{\mathcal{S}}_0R + 6R - 3MS) = 0 \quad (4.58)$$

Next, from (4.40j) we have

$$\begin{aligned} \psi_4^0 &= -\bar{\mathcal{S}}[-\bar{\mathcal{S}}(\gamma^0 + \bar{\gamma}^0) - \frac{1}{6}\psi_2^0\bar{\mathcal{S}}\psi_0^0] + \frac{1}{6}\psi_2^0(\bar{\mathcal{S}}\mathcal{S}\psi_0^0 - 4K\bar{\psi}_0^0) \\ &= \bar{\mathcal{S}}^2 \left[-\frac{P}{P} + \frac{1}{4}(\bar{\mathcal{S}}Z^0 + \mathcal{S}\bar{Z}^0) \right] + \frac{1}{3}\psi_2^0(\mathcal{S}\bar{\mathcal{S}}\psi_0^0 - 6K\bar{\psi}_0^0), \end{aligned}$$

where use was made of (4.40f,g) and the commutation relation for $\mathcal{S}$ and $\bar{\mathcal{S}}$. With the help of (4.40h), (4.41), (4.42), and (4.46) we can express ψ_4^0 further as

$$\psi_4^0 = \bar{\mathcal{S}}^2 \left[\frac{M}{3}\mathcal{S}\bar{\mathcal{S}}(R + \bar{R}) - \frac{\dot{P}_0}{P_0} - \frac{\dot{I}}{1+I} \right] + M(2K\bar{\psi}_0^0 - M\bar{\psi}_0^1). \quad (4.59)$$

In the linear approximation this becomes

$$\psi_4^0 \approx \bar{\sigma}_0^2 [2M\bar{R} + \frac{M}{3} \sigma_0 \bar{\sigma}_0 (R + \bar{R}) - M^2 \bar{S} - \dot{I}] ,$$

or, with the help of (4.58),

$$\psi_4^0 \approx \bar{\sigma}_0^2 \left[\frac{M}{3} \sigma_0 \bar{\sigma}_0 R - \dot{I} \right] . \quad (4.60)$$

The second term in (4.60) represents radiation from a 2^l -pole source⁴⁴ while the first term represents the contribution of the incoming field R .

In the test-particle limit $m \rightarrow 0$, we of course obtain from (4.52), (4.53), and (4.55) the usual results $\dot{P}_0/P_0 = 0$ and $I = 0$, i.e. the test-particle moves along a straight world line or geodesic in the regular background space. However, if we constrain I to be zero before going to the test-particle limit, we get a very interesting result: the world line is not only a geodesic but also part of a rigid geodesic congruence. To show this, let us put $I = 0$ in (4.50) and decompose the resulting equation into spherical harmonics. Then we get

$$l = 0 \quad \dot{M} = 0 \quad \text{or} \quad M = \text{const} \quad (4.61)$$

$$l = 1 \quad M \frac{\dot{P}_0}{P_0} = 0 \quad \text{or} \quad \frac{\dot{P}_0}{P_0} = 0 \quad (4.62)$$

$$l \geq 2 \quad M^2(R + \bar{R}) = 0 \quad \text{or} \quad R + \bar{R} = 0 \quad (4.63)$$

In the limit $M \rightarrow 0$, (4.62) implies that the world line is a geodesic, while (4.63) implies that R is pure imaginary, i.e.

$$R = i\chi \quad , \quad (4.64)$$

where χ is real. It follows from this that

$$\psi_0^0 = i\chi^2 \quad , \quad (4.65)$$

which shows, by comparison with (2.39) and (2.40), that ψ_0^0 is pure magnetic. As we shall now show, the vanishing of the electric part of ψ_0^0 implies that neighboring geodesics have zero relative acceleration with respect to the world line $r = 0$.

Given a timelike geodesic congruence, the relative acceleration between any two neighboring geodesics, L_0 and L_1 , is expressed by the equation of geodesic deviation⁴⁶

$$\frac{D^2 \eta^\mu}{Du^2} = R^{\mu\nu\rho\sigma} t_\nu t_\rho \eta_\sigma \quad , \quad (4.66)$$

where t^μ is the tangent vector to the congruence of timelike geodesics parametrized by the proper time u ($t^\mu t_\mu = 1$), η^μ is a vector orthogonal to t^μ which connects points of equal u value on L_0 and L_1 , $R^{\mu\nu\rho\sigma}$ is the Riemann tensor, and D denotes absolute differentiation along L_0 . The tangent vector t^μ may be written in terms of the null tetrad vectors ($\ell^\mu, n^\mu, m^\mu, \bar{m}^\mu$) as

$$t^\mu = \frac{1}{\sqrt{2}} (\ell^\mu + n^\mu) \quad , \quad (4.67)$$

and in empty space $R^{\mu\nu\rho\sigma}$ may be replaced by the Weyl tensor $C^{\mu\nu\rho\sigma}$.

Hence, the equation for geodesic deviation can be rewritten in empty space as

$$\frac{D^2 \eta^\mu}{Du^2} = \frac{1}{2} C^{\mu\sigma} \eta_\sigma \quad (4.68)$$

where

$$C^{\mu\sigma} = C^{\mu\nu\rho\sigma} (\ell_\nu + n_\nu) (\ell_\rho + n_\rho) \quad (4.69)$$

Note that the symmetry properties of $C^{\mu\nu\rho\sigma}$, namely,

$$C_{\mu\nu\rho\sigma} = C_{[\mu\nu][\rho\sigma]} = C_{\rho\sigma\mu\nu} = C_{\mu[\nu\rho\sigma]} = 0, \quad C^a_{\nu\alpha\sigma} = 0 \quad (4.70)$$

imply that $C^{\mu\sigma}$ is symmetric and trace-free, i.e.

$$C_{\mu\nu} = C_{(\mu\nu)}, \quad C^\mu_\mu = 0 \quad (4.71)$$

Moreover, from the definition $C_{\mu\nu} = 2C_{\alpha\beta\gamma\nu} t^\alpha t^\beta$ and the antisymmetry of $C_{\alpha\beta\gamma\nu}$ over the pairs of indices $[\alpha\beta]$ and $[\gamma\nu]$, it follows that $C_{\mu\nu} t^\mu = C_{\mu\nu} t^\nu = 0$, which means that $C_{\mu\nu}$ has no time-like part. With these properties, $C_{\mu\nu}$ is found to have only five real independent components. We choose these components to be the two complex scalars, Ω_1 and Ω_2 , and the real scalar, Ω_3 , defined by

$$\Omega_1 = -C_{\mu\nu} m^\mu m^\nu, \quad \Omega_2 = -C_{\mu\nu} \ell^\mu m^\nu, \quad \Omega_3 = -C_{\mu\nu} \bar{m}^\mu m^\nu \quad (4.72)$$

Remembering the definitions (2.6) and (4.69), we can express (4.72)

as

$$\Omega_1 = -(\psi_0 + \bar{\psi}_4), \quad \Omega_2 = \psi_1 - \bar{\psi}_3, \quad \Omega_3 = \psi_2 + \bar{\psi}_2 \quad (4.73)$$

Now, let $I = 0$ and $M \rightarrow 0$ so that the world line $r = 0$ becomes a timelike geodesic, which we identify with L_0 , and ψ_0^0 becomes pure magnetic, $\psi_0^0 = i\phi_0^2$. Going back to (4.58) we find that in the limit

$M \rightarrow 0$,

$$\mathcal{S}_0 \bar{\mathcal{S}}_0 \chi = -6\chi \quad , \quad (4.74)$$

which shows that χ is an $\ell = 2$ spherical harmonic. Then substituting $\psi_0^0 = i\mathcal{S}_0^2 \chi$ into (4.35a)-(4.35e) we obtain the following values for the ψ 's on the worldline $r = 0$.

$$\psi_0 = \psi_0^0 = i\mathcal{S}_0^2 \chi = i\mathcal{S}_0^2 \chi \quad (4.75a)$$

$$\psi_1 = -\frac{1}{4} \bar{\mathcal{S}}_0 \psi_0^0 = -\frac{1}{4} \bar{\mathcal{S}}_0 \mathcal{S}_0^2 \chi = i\mathcal{S}_0 \chi \quad (4.75b)$$

$$\psi_2 = \frac{1}{12} \bar{\mathcal{S}}_0^2 \psi_0^0 = \frac{1}{12} \bar{\mathcal{S}}_0^2 \mathcal{S}_0^2 \chi = 2i\chi \quad (4.75c)$$

$$\psi_3 = -\frac{1}{24} \mathcal{S}_0^3 \psi_0^0 = -\frac{1}{24} \mathcal{S}_0^3 \mathcal{S}_0^2 \chi = -i\bar{\mathcal{S}}_0 \chi \quad (4.75d)$$

$$\psi_4 = \frac{1}{24} \bar{\mathcal{S}}_0^4 \psi_0^0 = \frac{1}{24} \bar{\mathcal{S}}_0^4 \mathcal{S}_0^2 \chi = i\bar{\mathcal{S}}_0^2 \chi \quad , \quad (4.75e)$$

where we made use of (4.74) and the commutation relation $(\bar{\mathcal{S}}_0 \mathcal{S}_0 - \mathcal{S}_0 \bar{\mathcal{S}}_0)\eta = 2s\eta$. (Note that ψ_4 was given only to $O(1)$ in (4.35e). By adding another term to ψ_0 , i.e. $\psi_0 = \psi_0^0 + \psi_0^1 r + \psi_0^2 r^2 + O(r^3)$, we can integrate the radial equations further and evaluate the $O(1)$ in ψ_4 explicitly. When $I = M = 0$, this $O(1)$ term in ψ_4 is just the expression given in (4.75e).)

Substituting (4.75) back into (4.73), we find that

$$\Omega_1 = \Omega_2 = \Omega_3 = 0 \quad , \quad (4.76)$$

which means that $C_{\mu\nu} = 0$, i.e.

$$\frac{D^2 \eta^\mu}{Du^2} = 0 \quad . \quad (4.77)$$

Therefore, the vanishing of the electric part of ψ_0^0 (under the constraint $I = 0$ and in the test-particle limit $M \rightarrow 0$) implies that neighboring geodesics have zero geodesic deviation from the geodesic $r = 0$.

As a final remark, we point out that the quantity m_c , which was defined in Section 3.1 as

$$m_c = \frac{M}{\sqrt{2} k} \left(\frac{A}{4\pi} \right)^{3/2} \quad (4.78)$$

remains conserved in the presence of the incoming gravitation field ψ_0^0 or R . This can be proven easily by differentiating m_c with respect to u and using (4.48). Thus,

$$\begin{aligned} \dot{m}_c &= \frac{A^{1/2}}{\sqrt{2} k (4\pi)^{3/2}} \oint \left(\dot{M} - 3M \frac{\dot{P}}{P} \right) \frac{dx^2 dx^3}{P^2} \\ &= \frac{4A^{1/2}}{\sqrt{2} k (4\pi)^{3/2}} \oint \frac{\partial^2}{\partial \bar{\zeta} \partial \zeta} [K - M^2 (R + \bar{R})] dx^2 dx^3 = 0 \end{aligned}$$

5.0. MOTION IN THE EINSTEIN-MAXWELL THEORY

We now apply our analysis of motion to the Einstein-Maxwell theory in which the elementary singularity interacts with both incoming gravitational and electromagnetic fields. The Einstein-Maxwell equations are integrated first under the conditions, $\rho = -r^{-1} + O(r)$ and $\sigma = O(r)$, and the solutions are then analyzed with the use of the regularity condition $P = P_0(1+I)$. The analysis here will be seen to be much more complicated than in the previous cases, but nevertheless it will be shown that by means of a linear approximation the Lorentz-Dirac equation of motion can be obtained.

5.1. Integration of the Einstein-Maxwell Equations

As in Chapter 4.0, we first consider the radial equations

$$D\rho = \rho^2 + \sigma\bar{\sigma} + k\phi_0\bar{\phi}_0, \quad (2.18a)$$

$$D\sigma = 2\rho\sigma + \psi_0, \quad (2.18b)$$

and write condition (2.70) as

$$\rho = -r^{-1} + g, \quad g = O(r) \quad (5.1a)$$

$$\sigma = h, \quad h = O(r) \quad (5.1b)$$

Substituting (5.1) into (2.18a) and (2.18b) and solving for g and h , we obtain

$$g = r^{-2} \{g^0(u, x^A) + \int r^2 [k \phi_0 \bar{\phi}_0 + O(r^2)] dr\} \quad (5.2a)$$

$$h = r^{-2} \{h^0(u, x^A) + \int r^2 [\psi_0 + O(r^2)] dr\} \quad (5.2b)$$

But since $g, h = O(r)$, it follows from (5.2) that $g^0 = h^0 = 0$ and

$\psi_0, \phi_0 = O(1)$ (see also Appendix B). We shall adopt the assumption that ψ_0 and ϕ_0 have the following explicit form to $O(r^3)$:

$$\psi_0 = \psi_0^0 + \psi_0^1 r + \psi_0^2 r^2 + O(r^3) \quad (5.3)$$

$$\phi_0 = \phi_0^0 + \phi_0^1 r + \phi_0^2 r^2 + O(r^3) \quad (5.4)$$

With this assumption, (2.18a) and (2.18b) can then be integrated to yield the following expressions for ρ and σ :

$$\begin{aligned} \rho = & -r^{-1} + \frac{k}{3} |\phi_0^0|^2 r + \frac{k}{4} (\phi_0^0 \bar{\phi}_0^1 + \bar{\phi}_0^0 \phi_0^1) r^2 \\ & + \frac{1}{45} (|\psi_0^0|^2 + k^2 |\phi_0^0|^4 + 9k \phi_0^0 \bar{\phi}_0^2 + 9k \bar{\phi}_0^0 \phi_0^2 + 9k |\phi_0^1|^2) r^3 \\ & + O(r^4) \end{aligned} \quad (5.5)$$

$$\sigma = \frac{1}{3} \psi_0^0 r + \frac{1}{4} \psi_0^1 r^2 + \frac{1}{5} (\psi_0^2 + \frac{2}{9} k \psi_0^0 |\phi_0^0|^2) r^3 + O(r^4) \quad (5.6)$$

To find the r -dependence of the other variables we again make use of the lemma given in Chapter 4.0, together with (5.5) and (5.6).

The procedure, as in Chapter 4.0, is to let $s = r^{-1}$ and express certain groups of the radial equation in the form of the matrix equation (4.9) of the lemma.

Thus, writing (5.5) and (5.6) as

$$\rho = -s + \hat{O}(s^{-1}) \quad (5.7a)$$

$$\sigma = \hat{O}(s^{-1}), \quad (5.7b)$$

we can express the radial equation (2.16a) for ξ^A and $\bar{\xi}^A$ as

$$d \begin{bmatrix} \xi^{*A} \\ \bar{\xi}^{*A} \end{bmatrix} = \begin{bmatrix} \hat{O}(s^{-3}) \end{bmatrix} \begin{bmatrix} \xi^{*A} \\ \bar{\xi}^{*A} \end{bmatrix}, \quad (5.8)$$

where $\xi^{*A} = \xi^A s^{-1}$ and $d \equiv d/ds$. The lemma applies to (5.8) with $A = b = 0$, and so we have $\xi^{*A} = \hat{O}(1)$ or

$$\xi^A = O(r^{-1}) \quad (5.9)$$

Next, consider the group of radial equations (2.16b), (2.18c, d, e), (2.20a), and (2.22a). The lemma again applies to these equations, with y and A given by

$$y = \begin{bmatrix} \omega^* \\ \omega_1^* \\ \alpha^* \\ \alpha_1^* \\ \beta^* \\ \bar{\beta}^* \\ \tau^* \\ \bar{\tau}^* \\ \psi_1^* \\ \bar{\psi}_1^* \\ \phi_1^* \\ \bar{\phi}_1^* \end{bmatrix} \quad A = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (5.10)$$

where $\omega^* = \omega s^{-2}$, $\alpha^* = \alpha s^{-2}$, $\beta^* = \beta s^{-3}$, $\tau^* = \tau s^{-3}$, $\psi_1^* = \psi_1 s^{-4}$, and $\phi_1^* = \phi_1 s^{-2}$. Hence, we see that

$$\omega, \alpha, \phi_1 = O(r^{-2}) \quad \beta, \tau = O(r^{-3}) \quad \psi_1 = O(r^{-4}) \quad (5.11)$$

By using (5.9) and (5.11) in (2.16c), we then obtain

$$x^A = O(r^{-3}) \quad (5.12)$$

The r -dependence of $U, \gamma, \lambda, \mu, \psi_2$, and ϕ_2 can be obtained next by applying the lemma to the radial equations (2.16d), (2.18f,g,h), (2.20b), and (2.22b). If we let $U = U^* s^5$, $\gamma = \gamma^* s^6$, $\lambda = \lambda^* s^5$, $\mu = \mu^* s^6$, $\psi_2 = \psi_2^* s^7$, and $\phi_2 = \phi_2^* s^5$, these radial equations can be

expressed in the form of (4.9) of the lemma, with y and A given

by

$$y = \begin{bmatrix} U^* \\ \bar{U}^* \\ \gamma^* \\ \bar{\gamma}^* \\ \lambda^* \\ \bar{\lambda}^* \\ \mu^* \\ \bar{\mu}^* \\ \psi_2^* \\ \bar{\psi}_2^* \\ \phi_2^* \\ \bar{\phi}_2^* \end{bmatrix} \quad A = \begin{bmatrix} -5 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -5 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -6 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -5 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -5 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -4 \end{bmatrix} \quad (5.13)$$

Therefore, $U^*, \gamma^*, \lambda^*, \mu^*, \psi_2^*, \phi_2^* = \hat{O}(1)$ or

$$U, \lambda, \phi_2 = O(r^{-5}) \quad \gamma, \mu = O(r^{-6}) \quad \psi_2 = O(r^{-7}) \quad (5.14)$$

Setting $\psi_3 = \psi_3 s^{10}$ we can write (2.20c) as

$$d\psi_3^* = [-8s^{-1} + \hat{O}(s^{-3})]\psi_3^* + \hat{O}(s^{-2}) \quad (5.15)$$

which shows that $\psi_3^* = \hat{O}(1)$ or

$$\psi_3 = O(r^{-10}) . \quad (5.16)$$

Then, from (2.18i) we find that

$$v = O(r^{-9}) \quad (5.17)$$

Finally, if we let $\psi_4 = \psi_4^* s^{13}$ we can write (2.20d) as

$$d\psi_4^* = [-10s^{-1} + \hat{O}(s^{-3})]\psi_4^* + \hat{O}(s^{-2}), \quad (5.18)$$

from which we see that $\psi_4^* = \hat{O}(1)$ or

$$\psi_4 = O(r^{-13}). \quad (5.19)$$

As in Chapter 4.0, we again put back the information obtained above into the radial equations and integrate them with the assumption that $\psi_1^0 = 0$. Then, by substituting the radial expressions into the non-radial equations and equating coefficients of equal powers of r , we obtain the relationships between the "constants" of integration. The following is a summary of the final results, with $\pm \equiv \log r$.

A. Tetrad components of the Weyl tensor:

$$\psi_0 = \psi_0^0 + \psi_0^1 r + \psi_0^2 r^2 + O(r^3) \quad (5.20a)$$

$$\begin{aligned} \psi_1 = & -k\psi_0^0\phi_0^0\phi_1^0\pm + \frac{1}{4}[2k\psi_0^0\phi_0^0\phi_1^0 + 2k\phi_0^2\phi_1^0 - \bar{\phi}\psi_0^0] \\ & + O(r\pm) \end{aligned} \quad (5.20b)$$

$$\begin{aligned}
\psi_2 = & 2k|\phi_1^o|^2 r^{-4} + \psi_2^o r^{-3} + k[\phi_1^o \bar{\phi}_0^o - \frac{11}{3} k|\phi_0^o|^2 |\phi_1^o|^2] r^{-2} \\
& + \psi_0^o (k\bar{\phi}_0^o \phi_1^o)^2 \mathbb{E}^2 + [\frac{k}{2} \bar{\phi}_0^o \phi_1^o \bar{\phi}_0^o \psi_0^o - \psi_0^o (k\bar{\phi}_0^o \phi_1^o)^2 \\
& - k^2 \phi_0^o \bar{\phi}_0^o |\phi_1^o|^2] \mathbb{E} + [\frac{1}{12} \bar{\phi}_0^o \psi_0^o + \frac{k}{10} |\psi_0^o|^2 |\phi_1^o|^2 + \frac{1}{2} \psi_0^o (k\bar{\phi}_0^o \phi_1^o)^2 \\
& + \frac{k^2}{3} \phi_1^o |\phi_0^o|^2 \bar{\phi}_0^o + \frac{7}{5} k^2 \bar{\phi}_0^o \phi_0^o |\phi_1^o|^2 - \frac{11}{10} k^2 \phi_0^o \bar{\phi}_0^o |\phi_1^o|^2 - \frac{k}{3} \bar{\phi}_1^o \bar{\phi}_0^o \\
& + \frac{k}{6} \phi_1^o \bar{\phi}_0^o - \frac{k}{3} \bar{\phi}_0^o \phi_1^o \bar{\phi}_0^o \psi_0^o - \frac{k}{2} \psi_0^o \bar{\phi}_0^o \bar{\phi}_0^o - \frac{53}{45} k^3 |\phi_0^o|^4 |\phi_1^o|^2] \\
& + O(r \mathbb{E}^2) \quad (5.20c)
\end{aligned}$$

$$\begin{aligned}
\psi_3 = & -6k^2 \bar{\phi}_0^o \phi_1^o |\phi_1^o|^2 r^{-4} \mathbb{E} + 2k\bar{\phi}_1^o \bar{\phi}_0^o r^{-4} - 3k\psi_2^o \bar{\phi}_0^o \phi_1^o r^{-3} \mathbb{E} \\
& + 3k\phi_2^o \bar{\phi}_1^o r^{-3} + [11k^3 \bar{\phi}_0^o \phi_1^o |\phi_0^o|^2 |\phi_1^o|^2 - 3k^2 \bar{\phi}_0^o (\phi_1^o)^2 \bar{\phi}_0^o] r^{-2} \mathbb{E} \\
& + \psi_3^o r^{-2} + O(r^{-1} \mathbb{E}^2) \quad (5.20d)
\end{aligned}$$

$$\begin{aligned}
\psi_4 = & 12 k^3 (\bar{\phi}_0^o \phi_1^o)^2 |\phi_1^o|^2 r^{-4} \mathbb{E}^2 - 8 k^2 \bar{\phi}_0^o |\phi_1^o|^2 \bar{\phi}_1^o r^{-4} \mathbb{E} \\
& + [3 k^2 \bar{\phi}_0^o |\phi_1^o|^2 \bar{\phi}_1^o + k\phi_1^o \bar{\phi}_1^o] r^{-4} + 6\psi_2^o (k\bar{\phi}_0^o \phi_1^o)^2 r^{-3} \mathbb{E}^2 \\
& - 12 k^2 \phi_2^o \bar{\phi}_0^o |\phi_1^o|^2 r^{-3} \mathbb{E} + [2k^2 \phi_2^o \bar{\phi}_0^o |\phi_1^o|^2 + 2k\bar{\phi}_1^o \bar{\phi}_0^o \\
& + 2k\psi_2^o \bar{\phi}_0^o \bar{\phi}_1^o - \frac{4}{3} k \bar{\psi}_0^o \psi_2^o |\phi_1^o|^2 - k\bar{\psi}_0^o |\phi_1^o|^4] r^{-3} \\
& + [6 k^3 \phi_1^o (\bar{\phi}_0^o \phi_1^o)^2 \bar{\phi}_0^o - 22 k^4 (\bar{\phi}_0^o \phi_1^o)^2 |\phi_0^o|^2 |\phi_1^o|^2] r^{-2} \mathbb{E}^2 \\
& + [12 (k\bar{\phi}_0^o \phi_1^o)^2 \mathbb{K} - 4 k\bar{\psi}_0^o \phi_1^o \bar{\phi}_1^o - 198 k^3 (\bar{\phi}_0^o \phi_1^o)^2 |\phi_0^o|^2 |\phi_1^o|^2
\end{aligned}$$

$$\begin{aligned}
& + 12 k^3 \phi_1^0 (\bar{\phi}_0^0 \phi_1^0)^2 \bar{\phi}_0^0 - \frac{38}{3} k^3 \bar{\psi}_0^0 |\phi_0^0|^2 |\phi_1^0|^4 + \frac{185}{3} k^3 \bar{\phi}_0^0 |\phi_0^0|^2 |\phi_1^0|^2 \bar{\phi}_1^0 \\
& + 63 k^3 \bar{\phi}_1^0 (\bar{\phi}_0^0 \phi_1^0)^2 \bar{\phi}_0^0 - 12 k^2 \bar{\phi}_0^0 \phi_1^0 \bar{\phi}_0^0 \bar{\phi}_1^0 + \frac{k^2}{3} \bar{\psi}_0^0 \phi_1^0 |\phi_1^0|^2 \bar{\phi}_0^0 \\
& + 4 k^3 \bar{\phi}_0^0 \bar{\phi}_0^2 (\phi_1^0)^2 |\phi_1^0|^2 + 2 k^3 \bar{\psi}_0^0 \bar{\phi}_0^0 (\phi_1^0)^2 \bar{\phi}_1^0 - 6 k^2 \bar{\phi}_0^0 |\phi_1^0|^2 \bar{\phi}_0^2 \phi_0^0 \\
& - k^2 \bar{\phi}_1^0 |\phi_0^0|^2 \bar{\phi}_0^2 \phi_1^0 + 2 k^2 \bar{\phi}_0^0 \phi_1^0 |\phi_1^0|^2 \bar{\phi}_0^0] r^{-2} + [\bar{\phi}_0^2 K \\
& + 2 k \bar{\psi}_0^0 |\phi_1^0|^2 K - 2 k \bar{\phi}_0^0 \phi_1^0 \bar{\phi}_1^0 K + 5 (k \bar{\phi}_0^0 \phi_1^0)^2 K - 6 k K \bar{\phi}_0^0 \bar{\phi}_1^0 \\
& + 2 k \bar{\phi}_1^0 \bar{\phi}_1^3 \phi_0^0 + 3 k \psi_2^0 \phi_2^0 \bar{\phi}_0^0 - \bar{\psi}_0^0 (\psi_2^0)^2 + \frac{3}{5} k^2 \bar{\psi}_0^2 |\phi_1^0|^4 \\
& - 179 k^4 (\bar{\phi}_0^0 \phi_1^0)^2 |\phi_0^0|^2 |\phi_1^0|^2 + \frac{547}{2} k^3 \bar{\phi}_0^0 |\phi_0^0|^2 |\phi_1^0|^2 \bar{\phi}_1^0 \\
& - \frac{43}{3} k^2 \bar{\phi}_1^0 |\phi_0^0|^2 \bar{\phi}_0^2 \phi_1^0 + 4 k^3 (\bar{\phi}_0^0 \phi_1^0)^2 \phi_1^0 \bar{\phi}_0^0 + 85 k^3 (\bar{\phi}_0^0 \phi_1^0)^2 \bar{\phi}_1^0 \bar{\phi}_0^0 \\
& - \frac{718}{15} k^3 \bar{\psi}_0^0 |\phi_0^0|^2 |\phi_1^0|^4 + k^2 \bar{\phi}_0^0 \phi_1^0 |\phi_1^0|^2 \bar{\phi}_0^0 + 5 k^3 \bar{\phi}_0^0 \bar{\phi}_0^2 (\phi_1^0)^2 |\phi_1^0|^2 \\
& - 18 k^2 \bar{\phi}_0^0 |\phi_1^0|^2 \bar{\phi}_0^2 \phi_0^0 - 8 k^2 \bar{\phi}_0^0 \phi_1^0 \bar{\phi}_0^0 \bar{\phi}_1^0 - k \bar{\phi}_1^0 \bar{\phi}_1^0 \bar{\phi}_0^0 \\
& + 2 k^2 \bar{\psi}_0^0 \bar{\phi}_0^0 (\phi_1^0)^2 \bar{\phi}_1^0 + 7 k^2 \bar{\psi}_0^0 \phi_1^0 (\bar{\phi}_1^0)^2 \bar{\phi}_1^0 - 28 k^2 \bar{\phi}_0^0 \bar{\phi}_1^0 \bar{\phi}_0^0 \bar{\phi}_1^0 \\
& - \frac{19}{3} k^2 \bar{\phi}_0^2 |\phi_1^0|^2 \bar{\phi}_1^0 + \frac{16}{3} k^2 \bar{\psi}_0^0 \phi_1^0 |\phi_1^0|^2 \bar{\phi}_0^0 + 7 k^2 \bar{\psi}_0^0 \bar{\phi}_1^0 |\phi_1^0|^2 \bar{\phi}_0^0 \\
& - k^2 \bar{\psi}_0^0 |\bar{\phi}_1^0|^2 + 2 k \bar{\phi}_0^0 \bar{\phi}_1^2 \phi_1^0 + 3 k^2 \bar{\phi}_1^0 |\phi_1^0|^2 \bar{\phi}_0^2 - \frac{k}{2} \phi_1^0 \bar{\phi}_1^0 \bar{\phi}_0^0 \\
& + \frac{k^2}{2} \phi_0^0 \bar{\phi}_1^0 |\phi_1^0|^2 \bar{\phi}_0^0] r^{-2} + O(r^{-1} E^3) \quad (5.20e)
\end{aligned}$$

B. Tetrad components of the Maxwell tensor:

$$\phi_0 = \phi_0^0 + \phi_0^2 r^2 + O(r^3) \quad (5.21a)$$

$$\begin{aligned}
\phi_1 &= \phi_1^0 r^{-2} - k \phi_1^0 |\phi_0^0|^2 \mathbb{E} + \left[\frac{4}{3} k \phi_1^0 |\phi_0^0|^2 - \frac{1}{2} \bar{\theta} \phi_0^0 \right] - k \bar{\phi}_0^0 \phi_0^2 \phi_1^0 r^2 \mathbb{E} \\
&+ \left[\frac{1}{40} \phi_1^0 |\psi_0^0|^2 + \frac{11}{10} k \bar{\phi}_0^0 \phi_0^2 \phi_1^0 - \frac{3}{20} k \phi_0^0 \bar{\phi}_0^2 \phi_1^0 - \frac{1}{4} \bar{\theta} \phi_0^2 - \frac{17}{120} k^2 \phi_1^0 |\phi_0^0|^4 \right] r^2 \\
&+ O(r^3 \mathbb{E}) \quad (5.21b)
\end{aligned}$$

$$\begin{aligned}
\phi_2 &= -2 k \bar{\phi}_0^0 (\phi_1^0)^2 r^{-2} \mathbb{E} + \bar{\theta} \phi_1^0 r^{-2} + \phi_2^0 r^{-1} + \phi_0^0 (k \bar{\phi}_0^0 \phi_1^0)^2 \mathbb{E}^2 \\
&+ \left[k \phi_1^0 \bar{\phi}_0^0 \bar{\theta} \phi_0^0 - \frac{8}{3} \phi_0^0 (k \bar{\phi}_0^0 \phi_1^0)^2 \right] \mathbb{E} + \left[\frac{1}{2} \bar{\theta}^2 \phi_0^0 + k \bar{\psi}_0^0 \phi_0^0 |\phi_1^0|^2 \right. \\
&- \frac{1}{12} \phi_1^0 \bar{\theta} \psi_0^0 - \frac{5}{2} k \bar{\phi}_0^0 \phi_1^0 \bar{\theta} \phi_0^0 + \frac{k}{2} \bar{\phi}_0^2 (\phi_1^0)^2 + \frac{13}{3} \phi_0^0 (k \bar{\phi}_0^0 \phi_1^0)^2 \\
&\left. - \frac{13}{6} k |\phi_0^0|^2 \bar{\theta} \phi_1^0 \right] + O(r \mathbb{E}) \quad (5.21c)
\end{aligned}$$

C. Spin coefficients:

$$\rho = -r^{-1} + \frac{k}{3} |\phi_0^0|^2 r + \frac{1}{45} [|\psi_0^0|^2 + k^2 |\phi_0^0|^4 + 9 \bar{\phi}_0^0 \phi_0^2 + 9 \phi_0^0 \bar{\phi}_0^2] r^3 + O(r^4) \quad (5.22a)$$

$$\sigma = \frac{1}{3} \psi_0^0 r + \frac{1}{4} \psi_0^1 r^2 + \left[\frac{1}{5} \psi_0^2 + \frac{2}{45} k \psi_0^0 |\phi_0^0|^2 \right] r^3 + O(r^4) \quad (5.22b)$$

$$\begin{aligned}
\alpha &= k \bar{\phi}_0^0 \phi_1^0 r^{-1} \mathbb{E} + \alpha^0 r^{-1} - \frac{k^2}{3} \bar{\phi}_0^0 \phi_1^0 |\phi_0^0|^2 r \mathbb{E} + \frac{1}{2} \left[\frac{k}{3} \alpha^0 |\phi_0^0|^2 \right. \\
&- \frac{1}{3} \bar{\alpha}^0 \bar{\psi}_0^0 + \frac{5}{3} k^2 \bar{\phi}_0^0 \phi_1^0 |\phi_0^0|^2 - \frac{k}{2} \bar{\phi}_0^0 \bar{\theta} \phi_0^0 + k \bar{\phi}_0^2 \phi_1^0 \left. \right] r + O(r^2 \mathbb{E}) \\
&\quad (5.22c)
\end{aligned}$$

$$\begin{aligned}
\beta &= -\bar{\alpha}^0 r^{-1} - \frac{k}{3} \psi_0^0 \bar{\phi}_0^0 \phi_1^0 r \mathbb{E} + \frac{1}{2} \left[\frac{1}{3} \alpha^0 \psi_0^0 - \frac{k}{3} \bar{\alpha}^0 |\phi_0^0|^2 - \frac{1}{4} \bar{\theta} \psi_0^0 \right. \\
&\left. + \frac{5}{6} k \psi_0^0 \bar{\phi}_0^0 \phi_1^0 + \frac{k}{2} \phi_0^2 \bar{\phi}_1^0 \right] r + O(r^2 \mathbb{E}) \quad (5.22d)
\end{aligned}$$

$$\begin{aligned} \tau = & k \phi_0^o \bar{\phi}_1^o r^{-1} \mathbb{E} - \frac{k}{3} [\psi_0^o \bar{\phi}_0^o \phi_1^o + k \phi_0^o \bar{\phi}_1^o |\phi_0^o|^2] r \mathbb{E} + \frac{1}{2} \left[\frac{5}{3} k^2 \phi_0^o \bar{\phi}_1^o |\phi_0^o|^2 \right. \\ & \left. + \frac{5}{6} k \psi_0^o \bar{\phi}_0^o \phi_1^o - \frac{1}{4} \bar{\phi}_0^o \psi_0^o - \frac{k}{2} \phi_0^o \bar{\phi}_0^o + \frac{3}{2} k \phi_0^2 \bar{\phi}_1^o \right] r + o(r^2 \mathbb{E}) \quad (5.22e) \end{aligned}$$

$$\begin{aligned} \lambda = & -(k \bar{\phi}_0^o \phi_1^o)^2 r^{-1} \mathbb{E}^2 + k [\bar{\phi}_0^o \bar{\phi}_1^o \phi_1^o - \frac{1}{3} \bar{\psi}_0^o |\phi_1^o|^2] r^{-1} \mathbb{E} + \lambda^o r^{-1} \\ & + [k \phi_2^o \bar{\phi}_0^o - \frac{1}{3} \bar{\psi}_0^o \psi_2^o - \frac{k}{4} \bar{\psi}_0^1 |\phi_1^o|^2] + \frac{k^3}{3} (\bar{\phi}_0^o \phi_1^o)^2 |\phi_0^o|^2 r \mathbb{E}^2 \\ & + \frac{1}{2} [k^2 \phi_1^o (\bar{\phi}_0^o)^2 \bar{\phi}_0^o - \frac{10}{3} k^3 (\bar{\phi}_0^o \phi_1^o)^2 |\phi_0^o|^2 - \frac{13}{9} k^2 \bar{\psi}_0^o |\phi_0^o|^2 |\phi_1^o|^2 \\ & + \frac{k}{3} \bar{\psi}_0^o \phi_1^o \bar{\phi}_0^o + \frac{k^2}{3} \bar{\phi}_0^o |\phi_0^o|^2 \bar{\phi}_1^o - 2 k^2 \bar{\phi}_0^o \bar{\phi}_1^o (\phi_1^o)^2] r \mathbb{E} \\ & + \frac{1}{2} [k \bar{\phi}_0^2 \bar{\phi}_1^o + \frac{k}{2} \bar{\phi}_0^o \bar{\phi}_1^2 \phi_0^o + \frac{3}{2} k^2 \bar{\phi}_0^o \bar{\phi}_1^2 (\phi_1^o)^2 + \frac{17}{3} k^3 (\bar{\phi}_0^o \phi_1^o)^2 |\phi_0^o|^2 \\ & + \frac{391}{90} k^2 \bar{\psi}_0^o |\phi_0^o|^2 |\phi_1^o|^2 - \frac{1}{3} \bar{\psi}_0^o K - \frac{1}{4} \bar{\psi}_0^1 \psi_2^o - \frac{k}{5} \bar{\psi}_0^2 |\phi_1^o|^2 - \frac{5}{6} k \bar{\psi}_0^o \phi_1^o \bar{\phi}_0^o \\ & - \frac{k}{3} \bar{\psi}_0^o \bar{\phi}_1^o \bar{\phi}_0^o - \frac{k}{12} \bar{\phi}_0^o \phi_1^o \bar{\psi}_0^o - 3 k^2 \phi_1^o (\bar{\phi}_0^o)^2 \bar{\phi}_0^o - 2 k^2 \bar{\phi}_0^o |\phi_0^o|^2 \bar{\phi}_1^o] r \\ & + o(r^2 \mathbb{E}) \quad (5.22f) \end{aligned}$$

$$\begin{aligned} \mu = & -k |\phi_1^o|^2 r^{-3} - \psi_2^o r^{-2} + [k \phi_1^o \bar{\phi}_0^o - 4 k^2 |\phi_0^o|^2 |\phi_1^o|^2] r^{-1} \mathbb{E} + \mu^o r^{-1} \\ & - \frac{k}{3} \psi_2^o |\phi_0^o|^2 + \frac{1}{3} \psi_0^o (k \bar{\phi}_0^o \phi_1^o)^2 r \mathbb{E}^2 + \frac{1}{6} [k^2 \phi_1^o |\phi_0^o|^2 \bar{\phi}_0^o \\ & + k \psi_0^o \bar{\phi}_0^o \bar{\phi}_1^o - 5 \psi_0^o (k \bar{\phi}_0^o \phi_1^o)^2 - 4 k^3 |\phi_0^o|^4 |\phi_1^o|^2 + \frac{3}{2} k \bar{\phi}_0^o \phi_1^o \bar{\phi}_0^o \\ & - \frac{k}{3} |\psi_0^o|^2 |\phi_1^o|^2 - 3 k^2 \phi_0^2 \bar{\phi}_0^o |\phi_1^o|^2] r \mathbb{E} + \frac{1}{6} \left[\frac{1}{4} \bar{\phi}_0^2 \psi_0^o - k |\phi_0^o|^2 K \right. \\ & \left. + \frac{3}{2} k^2 \phi_1^o |\phi_0^o|^2 \bar{\phi}_0^o - \frac{7}{4} k \bar{\phi}_0^o \phi_1^o \bar{\phi}_0^o - k \psi_0^o \bar{\phi}_0^o \bar{\phi}_1^o + \frac{51}{10} k^2 \bar{\phi}_0^o \phi_0^2 |\phi_1^o|^2 \right. \\ & \left. - \frac{39}{10} k^2 \phi_0^o \bar{\phi}_0^2 |\phi_1^o|^2 - k \bar{\phi}_1^o \bar{\phi}_0^2 + \frac{k}{2} \phi_1^o \bar{\phi}_0^2 - k \bar{\phi}_1^o |\phi_0^o|^2 \bar{\phi}_0^o \right] r \\ & + o(r^2 \mathbb{E}^2) \quad (5.22g) \end{aligned}$$

$$\begin{aligned}
\gamma = & -k|\phi_1^0|^2 r^{-3} - \frac{1}{2} \psi_2^0 r^{-2} - k^2 |\phi_0^0|^2 |\phi_1^0|^2 r^{-1} L^2 \\
& + 2k \operatorname{Im}(\bar{\alpha}^0 \bar{\phi}_0^0 \phi_1^0) r^{-1} \mathbb{L} + [k^2 |\phi_0^0|^2 |\phi_1^0|^2 + k \operatorname{Im}(\bar{\phi}_1^0 \bar{\psi}_0^0 + 2 \bar{\alpha}^0 \bar{\phi}_0^0 \phi_1^0)] r^{-1} \\
& + \gamma^0 + \frac{1}{3} [\psi_0^0 (k \bar{\phi}_0^0 \phi_1^0)^2 + k^3 |\phi_0^0|^4 |\phi_1^0|^2] r \mathbb{L}^2 + [\frac{k^2}{4} \phi_1^0 |\phi_0^0|^2 \bar{\psi}_0^0 \\
& + \frac{k^2}{4} \bar{\phi}_1^0 |\phi_0^0|^2 \bar{\psi}_0^0 + \frac{k}{4} \bar{\phi}_0^0 \phi_1^0 \bar{\psi}_0^0 - \frac{5}{3} k^3 |\phi_0^0|^4 |\phi_1^0|^2 - \frac{5}{6} \psi_0^0 (k \bar{\phi}_0^0 \phi_1^0)^2 \\
& - k^2 \bar{\phi}_0^0 \phi_0^2 |\phi_1^0|^2 - \frac{k}{2} \phi_0^0 \bar{\phi}_0^2 |\phi_1^0|^2 + \frac{k}{3} \operatorname{Im}(\bar{\alpha}^0 \bar{\psi}_0^0 \phi_1^0 + k \bar{\alpha}^0 \bar{\phi}_0^0 \phi_1^0 |\phi_1^0|^2)] r L \\
& + [\frac{1}{12} \bar{\psi}_0^2 \psi_0^0 + \frac{k}{4} |\bar{\psi}_0^0|^2 + \frac{3}{20} k |\psi_0^0|^2 |\phi_1^0|^2 + \frac{119}{60} k^3 |\phi_0^0|^4 |\phi_1^0|^2 \\
& + \frac{4}{3} \psi_0^0 (k \bar{\phi}_0^0 \phi_1^0)^2 - \frac{k}{2} \psi_0^0 \bar{\phi}_0^0 \bar{\psi}_0^0 - \frac{11}{12} k^2 \bar{\phi}_1^0 |\phi_0^0|^2 \bar{\psi}_0^0 - \frac{7}{12} k^2 \phi_1^0 |\phi_0^0|^2 \bar{\psi}_0^0 \\
& - \frac{7}{12} k \bar{\phi}_0^0 \phi_1^0 \bar{\psi}_0^0 - \frac{7}{12} k \bar{\phi}_1^0 \bar{\psi}_0^2 - \frac{k}{12} \phi_1^0 \bar{\psi}_0^2 + \frac{7}{20} k^2 \phi_0^0 \bar{\phi}_0^2 |\phi_1^0|^2 \\
& + \frac{67}{20} k^2 \bar{\phi}_0^0 \phi_0^2 |\phi_1^0|^2 + \operatorname{Im}(2 k^2 \alpha^0 \phi_0^0 \bar{\phi}_1^0 |\phi_1^0|^2 + \frac{7}{6} k \alpha^0 \psi_0^0 \bar{\phi}_0^0 \phi_1^0 \\
& - \frac{1}{4} \alpha^0 \bar{\psi}_0^0 - \frac{k}{2} \alpha^0 \phi_0^0 \bar{\psi}_0^0 + \frac{3}{2} k \alpha^0 \phi_0^2 \bar{\phi}_1^0)] r + O(r^2 \mathbb{L}) \quad (5.22h)
\end{aligned}$$

$$\begin{aligned}
v = & 3 k^2 \bar{\phi}_0^0 \phi_1^0 |\phi_1^0|^2 r^{-3} \mathbb{L} + [k^2 \bar{\phi}_0^0 \phi_1^0 |\phi_1^0|^2 - k \bar{\phi}_1^0 \bar{\psi}_0^0] r^{-3} \\
& + 2 k \psi_2^0 \bar{\phi}_0^0 \phi_1^0 r^{-2} \mathbb{L} - \frac{1}{2} [\bar{\psi}_2^0 \psi_0^0 + 2 k \phi_2^0 \bar{\phi}_1^0] r^{-2} \\
& + k^3 \bar{\phi}_0^0 \phi_1^0 |\phi_0^0|^2 |\phi_1^0|^2 r^{-1} \mathbb{L}^3 - k^2 [\bar{\phi}_0^0 (\phi_1^0)^2 \bar{\psi}_0^0 + \bar{\phi}_1^0 |\phi_0^0|^2 \bar{\psi}_0^0 \\
& - 4 k \bar{\phi}_0^0 \phi_1^0 |\phi_0^0|^2 |\phi_1^0|^2 - \frac{1}{3} \bar{\psi}_0^0 \phi_0^0 \bar{\phi}_1^0 |\phi_1^0|^2] r^{-1} \mathbb{L}^2 \\
& - k [2 k \bar{\phi}_1^0 |\phi_0^0|^2 \bar{\psi}_0^0 + 5 k^2 \bar{\phi}_0^0 \phi_1^0 |\phi_0^0|^2 |\phi_1^0|^2 - \bar{\phi}_0^0 \phi_1^0 K - 2 k \bar{\phi}_0^0 (\phi_1^0)^2 \bar{\psi}_0^0 \\
& - \frac{k}{3} \bar{\psi}_0^0 \phi_0^0 \bar{\phi}_1^0 |\phi_1^0|^2] r^{-1} \mathbb{L} - [\bar{\psi}_0 K - 4 k \bar{\phi}_0^0 \phi_1^0 K + 53 k^3 \bar{\phi}_0^0 \phi_1^0 |\phi_0^0|^2 |\phi_1^0|^2
\end{aligned}$$

$$\begin{aligned}
& -\frac{69}{4} k^2 \bar{\phi}_0^\circ |\phi_1^\circ|^2 \bar{\phi}_0^\circ - 5 k^2 \bar{\phi}_0^\circ (\phi_1^\circ)^2 \bar{\phi}_0^\circ - \frac{25}{2} k^2 \bar{\phi}_1^\circ |\phi_0^\circ|^2 \bar{\phi}_1^\circ \\
& + 2 k \bar{\phi}_0^\circ \bar{\phi}_1^\circ + 2 k \bar{\phi}_1^\circ \bar{\phi}_0^\circ + \frac{13}{4} k^2 \bar{\psi}_0^\circ \phi_0^\circ \bar{\phi}_1^\circ |\phi_1^\circ|^2 - \frac{k}{2} \bar{\psi}_0^\circ \phi_1^\circ \bar{\phi}_0^\circ \\
& - \frac{11}{24} k |\phi_1^\circ|^2 \bar{\psi}_0^\circ - \frac{5}{4} k^2 \bar{\phi}_0^\circ \phi_1^\circ |\phi_1^\circ|^2] r^{-1} + O(\epsilon^3) \quad (5.22i)
\end{aligned}$$

D. Metric variables:

$$\begin{aligned}
\omega &= -k \phi_0^\circ \bar{\phi}_1^\circ \bar{\epsilon} + k \phi_0^\circ \bar{\phi}_1^\circ + \frac{1}{24} [\bar{\psi}_0^\circ - 4 k^2 \phi_0^\circ \bar{\phi}_1^\circ |\phi_0^\circ|^2 - \frac{2}{3} k \psi_0^\circ \bar{\phi}_0^\circ \phi_1^\circ \\
&+ 2 k \phi_0^\circ \bar{\phi}_0^\circ - 6 k \phi_0^\circ \bar{\phi}_1^\circ] r^2 + O(r^3 \bar{\epsilon}) \quad (5.23a)
\end{aligned}$$

$$\begin{aligned}
U &= -k |\phi_1^\circ|^2 r^{-2} - \psi_2^\circ r^{-1} + k^2 |\phi_0^\circ|^2 |\phi_1^\circ|^2 \bar{\epsilon}^2 - 2 k^2 |\phi_0^\circ|^2 |\phi_1^\circ|^2 \bar{\epsilon} \\
&+ U^\circ - (\gamma^\circ + \bar{\gamma}^\circ) r + \text{Re} \left[\frac{k^3}{3} |\phi_0^\circ|^4 |\phi_1^\circ|^2 + \frac{1}{18} \psi_0^\circ (k \bar{\phi}_0^\circ \phi_1^\circ)^2 \right. \\
&- \frac{k^2}{6} \bar{\phi}_1^\circ |\phi_0^\circ|^2 \bar{\phi}_0^\circ + \frac{k^2}{2} \phi_0^\circ \bar{\phi}_0^\circ |\phi_1^\circ|^2 - \frac{k}{12} \phi_0^\circ \bar{\phi}_1^\circ \bar{\psi}_0^\circ] r^2 \bar{\epsilon} \\
&+ \text{Re} \left[\frac{2}{3} k \phi_1^\circ \bar{\phi}_0^\circ + \frac{k}{2} \psi_0^\circ \bar{\phi}_0^\circ \bar{\phi}_1^\circ + \frac{k}{2} \bar{\phi}_0^\circ \phi_1^\circ \bar{\psi}_0^\circ + \frac{4}{3} k^2 \bar{\phi}_1^\circ |\phi_0^\circ|^2 \bar{\phi}_0^\circ \right. \\
&- \frac{1}{12} \bar{\psi}_0^\circ \psi_0^\circ - \frac{k}{4} |\bar{\phi}_0^\circ|^2 - \frac{3}{20} k |\psi_0^\circ|^2 |\phi_1^\circ|^2 - \frac{79}{60} k^3 |\phi_0^\circ|^4 |\phi_1^\circ|^2 - \frac{17}{18} \psi_0^\circ (k \bar{\phi}_0^\circ \phi_1^\circ)^2 \\
&- \frac{16}{5} k^2 \phi_0^\circ \bar{\phi}_0^\circ |\phi_1^\circ|^2] r^2 + O(r^3 \bar{\epsilon}) \quad (5.23b)
\end{aligned}$$

$$\begin{aligned}
\xi^A &= \xi^{0A} r^{-1} + \frac{1}{6} [\psi_0^\circ \bar{\xi}^{0A} + k |\phi_0^\circ|^2 \xi^{0A}] r + \frac{1}{12} \psi_0^\circ \bar{\xi}^{0A} r^2 + \frac{1}{20} [\psi_0^{2A} \bar{\xi}^{0A} \\
&+ \frac{7}{18} k^2 |\phi_0^\circ|^4 \xi^{0A} + \frac{7}{18} |\psi_0^\circ|^2 \xi^{0A} + \frac{7}{9} k \psi_0^\circ |\phi_0^\circ|^2 \bar{\xi}^{0A} + k \bar{\phi}_0^\circ \phi_0^{2A} \xi^{0A} \\
&+ k \phi_0^\circ \bar{\phi}_0^{2A} \bar{\xi}^{0A}] r^3 + O(r^4) \quad (5.23c)
\end{aligned}$$

$$\begin{aligned}
X^A &= -2 k \text{Re}(\bar{\phi}_0^\circ \phi_1^\circ \xi^{0A}) r^{-1} \bar{\epsilon} - 2 k \text{Re}(\bar{\phi}_0^\circ \phi_1^\circ \xi^{0A}) r^{-1} + X^{0A} \\
&- \frac{k}{3} \text{Re}[\bar{\psi}_0^\circ \phi_0^\circ \phi_1^\circ \xi^{0A} + k \bar{\phi}_0^\circ \phi_1^\circ |\phi_0^\circ|^2 \xi^{0A}] r \bar{\epsilon} + \frac{1}{4} \text{Re}[6 k \bar{\phi}_0^\circ \phi_1^{2A} \xi^{0A} \\
&+ 8 k^2 \bar{\phi}_0^\circ \phi_1^\circ |\phi_0^\circ|^2 + \frac{14}{3} k \bar{\psi}_0^\circ \phi_0^\circ \bar{\phi}_1^\circ - \bar{\psi}_0^\circ \psi_0^\circ - 2 k \bar{\phi}_0^\circ \bar{\phi}_0^\circ] r + O(r^2 \bar{\epsilon}) \quad (5.23d)
\end{aligned}$$

E. Relationships between the "constants":

$$\omega^0 = \phi_0^1 = 0, \quad \tau^0 = \bar{\alpha}^0 + \beta^0 = 0 \quad (5.24a)$$

$$\xi^{02} = i\xi^{03} = -P(u, \zeta, \bar{\zeta}), \quad P = \bar{P} \quad (5.24b)$$

$$K \equiv \bar{\partial} \bar{\partial} \log P \quad (5.24c)$$

$$\alpha^0 = -\frac{\partial P}{\partial \bar{\zeta}} - \frac{k}{2} \bar{\phi}_0^0 \phi_1^0 \quad (5.24d)$$

$$\lambda^0 = k \bar{\phi}_0^0 \bar{\phi}_1^0 - (k \bar{\phi}_0^0 \phi_1^0)^2 \quad (5.24e)$$

$$x^{02} = \frac{P}{2} (Z^0 + \bar{Z}^0), \quad x^{03} = \frac{P}{2i} (Z^0 - \bar{Z}^0) \quad (5.24f)$$

$$\bar{\partial} Z^0 = 2k \phi_0^0 \bar{\phi}_2^0 - \frac{4}{3} \psi_2^0 \psi_0^0 - k \psi_0^1 |\phi_1^0|^2 \quad (5.24g)$$

$$U^0 = -K + \frac{32}{3} k^2 |\phi_0^0|^2 |\phi_1^0|^2 - 2k \phi_1^0 \bar{\phi}_0^0 - 2k \bar{\phi}_1^0 \phi_0^0 \quad (5.24h)$$

$$\mu^0 = -K + 8 k^2 |\phi_0^0|^2 |\phi_1^0|^2 - 2k \phi_1^0 \bar{\phi}_0^0 - k \bar{\phi}_1^0 \phi_0^0 \quad (5.24i)$$

$$\gamma^0 = -\frac{1}{2} \frac{\dot{P}}{P} + \frac{1}{4} \bar{\partial} Z^0 + \frac{k}{3} \psi_2^0 |\phi_0^0|^2 + \frac{1}{2} \text{Im}(Z^0 \bar{\partial} \log P) \quad (5.24j)$$

$$\psi_2^0 = \bar{\psi}_2^0 \quad (5.24k)$$

$$\bar{\partial} \psi_2^0 = 2k \phi_1^0 \bar{\phi}_2^0 - 2k \psi_2^0 \phi_0^0 \bar{\phi}_1^0 \quad (5.24l)$$

$$\bar{\partial} \phi_1^0 = -3k \phi_0^0 |\phi_1^0|^2 \quad (5.24m)$$

$$\bar{\partial} \phi_0^0 = -k \bar{\phi}_1^0 (\phi_0^0)^2 - \frac{1}{3} \psi_0^0 \phi_1^0 \quad (5.24n)$$

$$\begin{aligned} \bar{\partial} \bar{\partial} \psi_0^0 = & -3 \psi_2^0 \psi_0^1 - \frac{12}{5} k \psi_0^2 |\phi_1^0|^2 + 6 \phi_0^0 \bar{\phi}_1^0 \psi_0^0 \\ & + 2k \phi_0^0 \bar{\phi}_1^0 \bar{\psi}_0^0 + 10k \phi_0^2 \bar{\phi}_1^0 - 6k \bar{\phi}_1^0 \phi_0^2 \\ & - 10k \psi_0^0 \bar{\phi}_1^0 \bar{\psi}_0^0 + \frac{114}{15} k^2 \psi_0^0 |\phi_0^0|^2 |\phi_1^0|^2 \end{aligned} \quad (5.24o)$$

$$\begin{aligned}
\psi_3^0 &= \bar{\theta}K - 3k\bar{\phi}_0^0\phi_1^0K + \frac{3}{2}k\bar{\phi}_1^0\bar{\theta}^2\phi_0^0 + \frac{5}{2}k\bar{\theta}\phi_1^0\bar{\theta}\phi_0^0 \\
&+ 3k^2\psi_0^0\phi_0^0\bar{\phi}_1^0|\phi_1^0|^2 - k^2\bar{\phi}_0^0\phi_1^0|\phi_1^0|^2 - \frac{k}{2}\phi_1^0(\bar{\psi}_0^0\bar{\phi}_1^0) \\
&- 15k^2\bar{\phi}_0^0|\phi_1^0|^2\bar{\theta}\phi_0^0 - \frac{41}{3}k^2\bar{\phi}_1^0|\phi_0^0|^2\bar{\theta}\phi_1^0 - 3k^2\bar{\phi}_0^0(\phi_1^0)^2\bar{\theta}\phi_0^0 \\
&+ \frac{89}{2}k^3\bar{\phi}_0^0\phi_1^0|\phi_0^0|^2|\phi_1^0|^2
\end{aligned} \tag{5.24b}$$

F. Differential equations:

$$\begin{aligned}
\dot{\phi}_1^0 - 2\frac{\dot{P}}{P}\phi_1^0 + \frac{1}{2}[\theta(\phi_1^0\bar{Z}^0) + \bar{\theta}(\phi_1^0Z^0)] &= -\theta\phi_2^0 - \bar{\theta}(\psi_2^0\phi_0^0) \tag{5.25} \\
\dot{\psi}_2^0 - 3\frac{\dot{P}}{P}\psi_2^0 + k|\psi_2^0\phi_0^0|^2 + \frac{3}{4}\psi_2^0[\theta Z^0 + \bar{\theta}Z^0] + \frac{1}{2}[Z^0\bar{\theta}\psi_2^0 + \bar{Z}^0\theta\psi_2^0] \\
&= -\bar{\theta}\bar{\theta}K + 3k[\phi_1^0\bar{\theta}(\bar{\phi}_0^0K) + \bar{\phi}_1^0\bar{\theta}(\phi_0^0K)] + k\phi_2^0\bar{\phi}_2^0 \\
&+ \frac{k}{4}|\phi_1^0|^2[\theta^2\bar{\psi}_0^0 + \bar{\theta}^2\psi_0^0] + \frac{3}{4}k[\phi_1^0\bar{\theta}\bar{\phi}_1^0\bar{\theta}\psi_0^0 + \bar{\phi}_1^0\bar{\theta}\phi_1^0\bar{\theta}\psi_0^0] \\
&+ \frac{k}{2}[\psi_0^0\bar{\phi}_1^0\bar{\theta}^2\phi_1^0 + \bar{\psi}_0^0\phi_1^0\bar{\theta}^2\bar{\phi}_1^0] - \frac{3}{2}k[\theta\bar{\phi}_1^0\bar{\theta}^2\phi_0^0 + \bar{\theta}\phi_1^0\theta^2\bar{\phi}_0^0] \\
&- 16k^2|\psi_0^0|^2|\phi_1^0|^2K + \frac{5}{2}k^2[(\psi_1^0)^2\theta(\bar{\phi}_0^0\bar{\phi}_1^0) + (\bar{\psi}_1^0)^2\bar{\theta}(\phi_0^0\phi_1^0)] \\
&+ 3k^2[(\phi_1^0\bar{\theta}\bar{\phi}_0^0)^2 + (\bar{\phi}_1^0\bar{\theta}\phi_0^0)^2] - \frac{9}{2}k^2|\phi_1^0|^2[\psi_0^0\bar{\phi}_0^0\bar{\theta}\phi_1^0 + \bar{\psi}_0^0\phi_0^0\bar{\theta}\bar{\phi}_1^0] \\
&+ 9k^2|\phi_0^0\bar{\theta}\phi_1^0|^2 + \frac{27}{2}k^2[\phi_0^0\bar{\phi}_1^0\bar{\theta}\bar{\theta}\bar{\theta}\phi_1^0 + \bar{\phi}_0^0\phi_1^0\bar{\theta}\bar{\theta}\bar{\theta}\bar{\phi}_1^0] \\
&+ 18k^2|\phi_1^0\bar{\theta}\phi_0^0|^2 + 3k^2[\bar{\phi}_0^0(\phi_1^0)^2\bar{\theta}^2\bar{\phi}_0^0 + \phi_0^0(\bar{\phi}_1^0)^2\bar{\theta}^2\psi_0^0] \\
&- \frac{9}{4}k^2|\phi_1^0|^2[\psi_0^0\bar{\phi}_1^0\bar{\theta}\bar{\psi}_0^0 + \bar{\phi}_0^0\phi_1^0\bar{\theta}\psi_0^0] - \frac{225}{2}k^2|\phi_0^0|^2|\phi_1^0|^2[\phi_1^0\bar{\theta}\bar{\phi}_0^0 + \bar{\phi}_1^0\bar{\theta}\phi_0^0] \\
&- \frac{15}{2}k^3|\phi_1^0|^2[\psi_0^0(\bar{\phi}_0^0\phi_1^0)^2 + \bar{\psi}_0^0(\phi_0^0\bar{\phi}_1^0)^2] - 15k^3|\phi_1^0|^4[\phi_0^0\bar{\phi}_0^0 + \bar{\phi}_0^0\phi_0^0] \\
&- \frac{21}{2}k^3|\phi_1^0|^2[\psi_0^0(\bar{\phi}_1^0)^2\bar{\theta}\phi_1^0 + \bar{\psi}_0^0(\phi_1^0)^2\bar{\theta}\bar{\phi}_1^0] + 425k^4|\psi_0^0|^4|\phi_1^0|^4
\end{aligned} \tag{5.26}$$

5.2. Analysis of the Einstein-Maxwell Solutions

The results for the Einstein-Maxwell theory obtained in Section 5.1 are obviously many times more complicated and non-linear than those for the previous cases. To extract from these results the equations of motion and the time dependence of the internal structure, we shall therefore have to adopt a linear approximation once more. As in previous chapters, we shall impose the regularity condition $P = P_0(1+I)$ and assume that $\dot{P}_0/P_0$ and I are first-order quantities. In addition, we shall carry out the analysis only to first order in the coupling constant k .

We start by noting from Section 3.2 that in the Reissner-Nordström solution ϕ_1^0 and ψ_2^0 are related to the electric charge and the mass, respectively, by means of the equations $\phi_1^0 = e$ and $\psi_2^0 = -\sqrt{2} k m$. Hence, in the generalization of this solution to that of the full Einstein-Maxwell equations obtained in Section 5.1, we can expect the k -dependence of ϕ_1^0 and ψ_2^0 to be of the form

$$\phi_1^0 = Q + O(k) \quad (5.27)$$

$$\psi_2^0 = kW + O(k^2) \quad (5.28)$$

Next, recalling the spin weights of ϕ_0^0 , ϕ_2^0 , and ψ_0^0 , we write these quantities as

$$\phi_0^0 = \eth A, \quad \phi_2^0 = \eth B, \quad \psi_0^0 = \eth^2 G \quad (5.29)$$

where A , B , and G are all spin-weight zero variables. In our linearization procedure, we shall assume that A , B , and G are also first-order variables like $\dot{P}_0/P_0$ and I .

Now, if we substitute (5.27) into (5.24m) we find that

$$\delta Q = O(k),$$

which implies that $Q = E(u) + O(k)$ or

$$\phi_1^0 = E(u) + O(k) \quad (5.30)$$

Then, substituting $\psi_0^0 = \delta^2 G$ into (5.24o) and using the commutation relation for δ and $\bar{\delta}$, we obtain

$$\bar{\delta}\delta^3 G = \delta^2 \bar{\delta}\delta G + 6K\delta^2 G = O(k).$$

In the linear approximation this becomes

$$\delta_0^2 (\delta_0 \bar{\delta}_0 G + 6G) = O(k), \quad (5.31)$$

which implies that $G = R + O(k)$, where R is an $\ell = 2$ spherical harmonic, i.e.

$$\delta_0 \bar{\delta}_0 R = -6R. \quad (5.32)$$

Thus, ψ_0^0 is approximately given by

$$\psi_0^0 = \delta_0^2 R + O(k). \quad (5.33)$$

Using (5.30) and (5.33) in (5.24n), we then get

$$\mathcal{A}_0^2 \approx -\frac{E}{3} \mathcal{R}_0^2 + O(k), \quad (5.34)$$

the homogeneous part of which has the solution $A = F$ where F is an $\ell = 1$ spherical harmonic. Hence, the complete solution of (5.34) is

$$A \approx F - \frac{1}{3} ER + O(k), \quad (5.35)$$

which means that

$$\phi_0^0 \approx \mathcal{A}_0 F - \frac{1}{3} E \mathcal{R}_0 R + O(k). \quad (5.36)$$

From (5.24g) we find next that

$$\mathcal{Z}^0 = O(k). \quad (5.37)$$

As in the discussion leading to (4.46), the solution to the homogeneous part of (5.37) can be transformed away by means of the coordinate freedom $\zeta' = g(u, \zeta)$. Hence, we have

$$Z^0 = O(k) \quad (5.38)$$

Linearizing (5.25) we then obtain

$$\mathcal{A}_0 \bar{\mathcal{A}}_0 B \approx -\dot{E} + 2E \frac{\dot{P}_0}{P_0} + 2E \dot{t} + O(k) \quad (5.39)$$

If I is assumed to have a specific l -value, i.e. $\mathcal{H}_0 \bar{\mathcal{H}}_0 I = -LI$ with $L = l(l+1)$ we can decompose (5.39) such that

$$B \approx -E \frac{\dot{P}_0}{P_0} - \frac{2E}{L} \dot{I} + O(k) \quad (5.40)$$

and

$$\dot{E} \approx O(k) \quad (5.41)$$

The last equation implies that

$$E \approx e + O(k), \quad \dot{e} = 0, \quad (5.42)$$

where the constant e (assumed to be real) is just the charge of the particle. With this result, (5.30), (5.36), and (5.40) become

$$\hat{\phi}_1 \approx e + O(k) \quad (5.43)$$

$$\phi_0^O \approx \mathcal{H}_0 F - \frac{e}{3} \mathcal{H}_0 R + O(k) \quad (5.44)$$

$$B \approx -e \frac{\dot{P}_0}{P_0} - \frac{2e}{L} \dot{I} + O(k). \quad (5.45)$$

If we then substitute these results together with (5.28) into (5.24) we obtain

$$\mathcal{H}W \approx 2e\mathcal{H}\left(e \frac{\dot{P}_0}{P_0} + 2 \frac{e}{L} \dot{I}\right) + O(k) \quad (5.46)$$

which has the general solution

$$W \approx M(u) + 2e^2 \frac{\dot{P}_0}{P_0} + 4 \frac{e^2}{L} \dot{I} + O(k), \quad (5.47)$$

where M is the solution of the homogeneous part. Thus, ψ_2^0 is given by

$$\psi_2^0 = -k(M + 2e^2 \frac{\dot{P}_0}{P_0} + 4 \frac{e^2}{L} \dot{I}) + O(k^2) \quad (5.48)$$

Finally, we consider the linearization of (5.26). Substituting into this equation the expressions for ϕ_1^0 , ϕ_0^0 , ϕ_2^0 , ψ_0^0 , and ψ_2^0 , we obtain to first order in k

$$\begin{aligned} k \dot{M} + 2ke^2 \left[\frac{\ddot{P}_0}{P_0} - \left(\frac{\dot{P}_0}{P_0} \right)^2 \right] + 4 \frac{ke^2}{L} \ddot{I} - 3k \left(\frac{\dot{P}_0}{P_0} + \frac{\dot{I}}{1+I} \right) (M + 2e^2 \frac{\dot{P}_0}{P_0} \\ + 4 \frac{e^2}{L} \dot{I}) = \mathcal{S}\bar{\mathcal{S}}K - 3ke[\mathcal{S}(K\bar{\mathcal{S}}\bar{A}) + \bar{\mathcal{S}}(K\mathcal{S}A)] \\ - ke^2 \mathcal{S} \left(\frac{\dot{P}_0}{P_0} + \frac{2}{L} \dot{I} \right) \bar{\mathcal{S}} \left(\frac{\dot{P}_0}{P_0} + \frac{2}{L} \dot{I} \right) \\ - \frac{ke^2}{4} (\mathcal{S}^2 \bar{\mathcal{S}}^2 \bar{R} + \bar{\mathcal{S}}^2 \mathcal{S}^2 R) + O(k^2). \end{aligned} \quad (5.49)$$

After using (2.62) and (3.10) and linearizing, this becomes

$$\begin{aligned} k \dot{M} + 2ke^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{I}}{2} \right) + 4 \frac{ke^2}{L} \ddot{I} - 3kM \frac{\dot{P}_0}{P_0} - 3k \dot{M} I \\ = \mathcal{S}_0 \bar{\mathcal{S}}_0 \mathcal{S}_0 \bar{\mathcal{S}}_0 I + 2\mathcal{S}_0 \bar{\mathcal{S}}_0 I - 3ke \mathcal{S}_0 \bar{\mathcal{S}}_0 (F + \bar{F}) \\ + ke^2 \mathcal{S}_0 \bar{\mathcal{S}}_0 (R + \bar{R}) - \frac{ke^2}{4} (\mathcal{S}_0^2 \bar{\mathcal{S}}_0^2 \bar{R} + \bar{\mathcal{S}}_0^2 \mathcal{S}_0^2 R) + O(k^2) \end{aligned} \quad (5.50)$$

Since F and R are spherical harmonics with $l=1$ and $l=2$, respectively, and since $\mathcal{S}_0 \bar{\mathcal{S}}_0 I = -LI$, (5.50) can be simplified further to yield

$$\begin{aligned} k \dot{M} + 2ke^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{I}}{2} \right) + 4 \frac{ke^2}{L} \ddot{I} - 3kM \frac{\dot{P}_0}{P_0} - 3k \dot{M} I \\ = L(L-2)I + 6ke(F + \bar{F}) - 12ke^2(R + \bar{R}) + O(k^2) \end{aligned} \quad (5.51)$$

If we now decompose this into its different l -values, assuming that I is an $l=2$ spherical harmonic, i.e. $L = 6$, then we get

$$l = 0 \quad \dot{M} = 0(k) \quad (5.52)$$

$$l = 1 \quad M \frac{\dot{P}_0}{P_0} = -2e(F+\bar{F}) + \frac{2}{3} e^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right) + 0(k) \quad (5.53)$$

$$l = 2 \quad \frac{2}{3} k e^2 \ddot{I} - 3kM\dot{I} - 24I = -12 k e^2 (R+\bar{R}) + 0(k^2) \quad (5.54)$$

From (5.52) it follows that

$$M = \sqrt{2} m + 0(k), \quad (5.55)$$

where the constant m can be identified with the mass by comparison with the expression for ψ_2^0 in the Schwarzschild solution. Substituting (5.55) into (5.53) and (5.54) we get

$$\sqrt{2} m \frac{\dot{P}_0}{P_0} = -2e(F+\bar{F}) + \frac{2}{3} e^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right) + 0(k) \quad (5.56)$$

$$\frac{2}{3} k e^2 \ddot{I} - 3k\sqrt{2} m \dot{I} - 24I = -12k e^2 (R+\bar{R}) + 0(k^2) \quad (5.57)$$

In the limit $k \rightarrow 0$, (5.56) becomes

$$\sqrt{2} m \frac{\dot{P}_0}{P_0} = -2e(F+\bar{F}) + \frac{2}{3} e^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right). \quad (5.58)$$

The second term on the right has already been shown in (2.68b) to be equivalent to the Abraham radiation reaction force; the first term will now be shown to be equivalent to the Lorentz force.

If the Lorentz-Dirac equation given in (1.8) is expressed in terms of a parameter which is $\sqrt{2}/2$ times the proper time on an arbitrary timelike (regular) world line in Minkowski space, it becomes

$$\sqrt{2} m \ddot{\xi}^\mu = 2e \dot{\xi}_\nu F^{\nu\mu} + \frac{2}{3} e^2 (\ddot{\xi}^\mu + \frac{1}{2} \ddot{\xi}^2 \dot{\xi}^\mu), \quad (5.59)$$

with

$$\dot{\xi}^\mu \dot{\xi}_\mu = 2. \quad (5.60)$$

Multiplying (5.59) by $\dot{\xi}_\mu$ and using (2.46b), we get

$$\sqrt{2} m \frac{\dot{P}_0}{P_0} = 2e F^{\nu\mu} \dot{\xi}_\nu \dot{\xi}_\mu + \frac{2}{3} e^2 \left(\frac{\ddot{P}_0}{P_0} + \frac{\ddot{\xi}^2}{2} \right). \quad (5.61)$$

Now, since $\dot{\xi}_\nu$ can be written in terms of the tetrad vectors $\dot{\xi}_\nu$ and n_ν as

$$\dot{\xi}_\nu = \dot{\xi}_\nu + n_\nu$$

and since $F^{\nu\mu}$ is antisymmetric, then

$$F^{\nu\mu} \dot{\xi}_\nu \dot{\xi}_\mu = F^{\nu\mu} \dot{\xi}_\nu n_\mu = -F^{\mu\nu} \dot{\xi}_\mu n_\nu. \quad (5.62)$$

But from the definition of ϕ_1 given in (2.9), we have

$$F^{\mu\nu} \dot{\xi}_\mu n_\nu = \phi_1 + \bar{\phi}_1.$$

Hence the first term on the right of (5.61) can be expressed as

$$2e F^{\nu\mu} \dot{\xi}_\nu \dot{\xi}_\mu = -2e(\phi_1 + \bar{\phi}_1). \quad (5.63)$$

Now, consider the flat-space limit ($K = 1$, $I = \psi_a = 0$, $a = 0,1,2,3,4$) and the test-charge limit ($\phi_1^0 = 0$) of the Einstein-Maxwell solutions obtained in Section 5.1. In this limiting case the world line $r = 0$ becomes a regular world line in flat space, and the value of ϕ_1 on this line can be seen from (5.21b) to be

$$\phi_1 = -\frac{1}{2} \bar{\gamma}_0 \phi_0^0, \quad (5.64)$$

where ϕ_0^0 , according to (5.24n), satisfies

$$\gamma_0 \phi_0^0 = 0. \quad (5.65)$$

If we again let $\phi_0^0 = \gamma_0 A$, then (5.65) implies that A is an $\ell = 1$ spherical harmonic, i.e. $A = F$. Therefore, (5.63) becomes

$$2eF^{\nu\mu} \xi_{\nu} \ell_{\mu} = e \gamma_0 \bar{\gamma}_0 (F + \bar{F}) = -2e(F + \bar{F}). \quad (5.66)$$

Substituting this back into (5.61) we get precisely (5.58). Thus, we have shown that (5.58) is equivalent to the Lorentz-Dirac equation, with the external field given uniquely by the $\ell = 1$ component of $(\gamma_0 \bar{\phi}_0^0 + \bar{\gamma}_0 \phi_0^0)$.

Finally, we see from (5.57) how the internal degrees of freedom are driven by the incoming gravitational field to lowest order. Without the incoming fields, the two equations (5.56) and (5.57) of course reduce to the linearized RTM equations of Section 3.2 and lead to the solutions with unpleasant runaway features. It would be interesting to look into the higher order extensions of (5.56) and (5.57), but unfortunately we do not yet have a self-consistent approximation procedure for analyzing (5.26) to higher orders.

6.0 SUMMARY AND CONCLUSION

We have presented here a new approach to equations of motion in general relativity, in which motion is analyzed in terms of the structure and behavior of a family of null cones emanating from a special class of singularities, called elementary singularities. By means of this approach we were able to give an intrinsic description of the motion of a singularity in its own space-time, with no reference to a regular background space. The basis for our approach was the condition

$$\rho = -\frac{1}{r} + O(r), \quad \sigma = O(r),$$

which we imposed on a family of null hypersurfaces $u = \text{const}$ in Riemannian space. This condition made it possible to define a fundamental 2-surface (F2S) whose metric is specified by a function $P = P(u, \zeta, \bar{\zeta})$. By assuming that $P = P_0(1+I)$ we were able to give an alternative mode of describing motion, in which $\dot{P}_0/P_0$ is identified with acceleration and I is interpreted as internal degrees of freedom. We then showed that if the Weyl tensor is singular on the line $r = 0$, Einstein's field equations yield differential equations governing the behavior of the F2S, from which one could extract equations of motion for the singularity in terms of $\dot{P}_0/P_0$ as well as equations for the time development of its internal structure I . In particular, we analyzed the motion and structure of singularities in three cases: (1) the Robinson-Trautman solutions and their charged counterparts, (2) a general empty space, and (3) the Einstein-Maxwell theory. To

lowest order several interesting results were obtained, of which the most significant one was the derivation of the Lorentz-Dirac equation, with the Lorentz force appearing in terms of a unique background field and the Abraham radiation reaction force arising unexpectedly without any mass renormalization or ad hoc models. This derivation of the Lorentz-Dirac equation from the Einstein-Maxwell theory plus the discovery of an internal structure I for an elementary singularity constitute the major achievements of our approach to equations of motion.

In conclusion, we would like to point out several directions in which the work presented here could be extended. First, one could study the effects of the non-linear terms on the equations of motion. It would be very interesting, for example, to find out whether the non-linear terms could stabilize the behavior of I in the RTM solutions. Secondly, though it appears that our method does not allow the Weyl tensor to have r^{-4} singularities corresponding to an intrinsic dipole moment and angular momentum, it might still be possible to introduce angular degrees of freedom corresponding to spin or angular momentum by replacing condition (2.70) by

$$\rho = - \frac{1}{r + i J(u, \zeta, \bar{\zeta})} + O(r), \quad \sigma = O(r) \quad .$$

Finally, one could possibly generalize this work to a system of more than one singularity and investigate the interactions between, say, two singularities.

APPENDIX A

We want to prove here (without use of the field equations) that the 2-surfaces, $u = \text{const}$ and $r = \text{const}$, possess a metric given by

$$g_{AB} = -\frac{r^2}{2} g_{AB}^{\circ} + O(r^4), \quad g_{AB}^{\circ} = g_{AB}^{\circ}(u, x^C). \quad (2.71)$$

Since the metric tensor $g_{\mu\nu}$ of space-time satisfies the relation

$$g_{\beta\mu} g^{\mu\alpha} = \delta_{\mu}^{\alpha}, \quad (A.1)$$

it follows for the case $\alpha = A$ and $\beta = B$ that

$$g_{B\mu} g^{\mu A} = g_{B0} g^{0A} + g_{B1} g^{1A} + g_{BC} g^{CA} = \delta_B^A \quad (A.2)$$

But from the form of the metric given in (2.13a) we note that $g^{0A} = 0$ and $g_{B1} = 0$. Hence (A.2) becomes

$$g_{BC} g^{CA} = \delta_B^A, \quad (A.3)$$

which implies that g_{AB} transforms as a 2x2 metric. We also note from (2.13b) that g^{AB} is given by

$$g^{AB} = -(\xi^A \bar{\xi}^B + \bar{\xi}^A \xi^B) \quad (A.4)$$

Now, in Section 4.1 it is shown that the metric equation

$$D\xi^A = \rho \xi^A + \sigma \bar{\xi}^A, \quad (2.16a)$$

together with the conditions, $\rho = -r^{-1} + O(r)$ and $\sigma = O(r)$, leads to the result

$$\xi^A = \xi^{OA} r^{-1} + O(r), \quad (4.13)$$

where $\xi^{OA} = \xi^{OA}(u, x^B)$. Substituting this into (A.4) we obtain

$$g^{AB} = -(\xi^{OA}\bar{\xi}^{OB} + \bar{\xi}^{OA}\xi^{OB}) r^{-2} + O(1) \quad (A.5)$$

or

$$g^{AB} = -2 \overset{\circ}{g}^{AB} r^{-2} + O(1), \quad \overset{\circ}{g}^{AB} = \overset{\circ}{g}^{AB}(u, x^C) \quad (A.6)$$

where

$$\overset{\circ}{g}^{AB} \equiv \frac{(\xi^{OA}\bar{\xi}^{OB} + \bar{\xi}^{OA}\xi^{OB})}{2} \quad (A.7)$$

If we then write (A.6) in matrix form

$$\overset{\circ}{g}^{AB} = \begin{bmatrix} 2 \overset{\circ}{g}^{22} r^{-2} + O(1) & 2 \overset{\circ}{g}^{23} r^{-2} + O(1) \\ 2 \overset{\circ}{g}^{23} r^{-2} + O(1) & 2 \overset{\circ}{g}^{33} r^{-2} + O(1) \end{bmatrix}$$

and take the inverse, which from (A.3) is given by g_{AB} , we get

$$g_{AB} = -\frac{r^2}{2} \overset{\circ}{g}_{AB} + O(r^4), \quad (A.8)$$

where

$$\overset{\circ}{g}_{22} \equiv \frac{\overset{\circ}{g}^{33}}{(\overset{\circ}{g}^{22}\overset{\circ}{g}^{33} - \overset{\circ}{g}^{23}\overset{\circ}{g}^{23})} \quad (A.9a)$$

$$\overset{\circ}{g}_{33} \equiv \frac{\overset{\circ}{g}^{22}}{(\overset{\circ}{g}^{22}\overset{\circ}{g}^{33} - \overset{\circ}{g}^{23}\overset{\circ}{g}^{23})} \quad (\text{A.9b})$$

$$\overset{\circ}{g}_{23} \equiv \frac{-\overset{\circ}{g}^{23}}{(\overset{\circ}{g}^{22}\overset{\circ}{g}^{33} - \overset{\circ}{g}^{23}\overset{\circ}{g}^{23})} \quad (\text{A.9c})$$

This proves Eq. (2.71).

APPENDIX B

In Chapters 4.0 and 5.0 it was shown that the condition (2.70), namely, $\rho = -r^{-1} + O(r)$ and $\sigma = O(r)$, led to the result that ψ_0 and ϕ_0 must be regular functions near $r = 0$. Here we shall consider the inverse problem of finding ρ and σ under the assumption that $\psi_0 = O(1)$ and $\phi_0 = O(1)$.

Consider the radial equations (2.18a,b) for ρ and σ ,

$$D\rho = \rho^2 + \sigma \bar{\sigma} + k \phi_0 \bar{\phi}_0 \quad (\text{B.1a})$$

$$D\sigma = 2\rho \sigma + \psi_0 \quad (\text{B.1b})$$

If we let

$$\underline{H} = \begin{bmatrix} \rho & \sigma \\ \bar{\sigma} & \rho \end{bmatrix} \quad \underline{J} = \begin{bmatrix} k\phi_0 \bar{\phi}_0 & \psi_0 \\ \psi_0 & k\phi_0 \bar{\phi}_0 \end{bmatrix}, \quad (\text{B.2})$$

we can express (2.18a) and (2.18b) in the form

$$D\underline{H} = \underline{H}^2 + \underline{J}. \quad (\text{B.3})$$

Putting $s = -1/r$ and $d = d/ds$, this matrix equation becomes

$$d\underline{H} = -s^{-2} (\underline{H}^2 + \underline{J}). \quad (\text{B.4})$$

To solve for $\underline{H}$ asymptotically for large s , we set

$$\underline{H} = s^2 (d\underline{Y})\underline{Y}^{-1} - \underline{I} s, \quad (\text{B.5})$$

where $\underline{I}$ is the unit matrix and

$$\underline{Y} = \begin{bmatrix} y_1 & y_2 \\ \bar{y}_1 & \bar{y}_2 \end{bmatrix} \quad (\text{B.6})$$

is a non-singular solution of

$$s^2 \frac{d\underline{Y}}{ds} = (\underline{H} + \underline{I} s) \underline{Y} \quad (\text{B.7})$$

and hence satisfies

$$d^2 \underline{Y} = - \underline{G} \underline{Y}, \quad \underline{G} = s^{-4} \underline{J}. \quad (\text{B.8})$$

Now, by a theorem from the theory of differential equations,⁴⁵ if the matrix elements G_{AB} of $\underline{G}$ satisfy

$$\int s |G_{AB}| ds = \hat{O}(1) \quad (\text{B.9})$$

then the asymptotic behavior of the solutions to (B.8) is given by*

$$d\underline{Y} = \underline{A} + \hat{O}(1) \quad (\text{B.10})$$

$$\underline{Y} = s \underline{A} + \hat{O}(s), \quad (\text{B.11})$$

where $\underline{A}$ is a constant matrix. Since $\psi_0, \phi_0 = \hat{O}(1)$ by assumption, then $\underline{G} = \hat{O}(s^{-4})$ and so the theorem applies. Substituting (B.11) and $\underline{G} = \hat{O}(s^{-4})$ into the right side of (B.8) we get

$$* \overline{f(s, u, x^A) = \hat{O}[g(s)]} \text{ means } \lim_{s \rightarrow \infty} \frac{f(s, u, x^A)}{g(s)} = 0 \text{ for each } u, x^A.$$

$$d^2 \underline{Y} = -s \underline{G} \underline{A} + \hat{O}(s^{-3}) = \hat{\underline{O}}(s^{-3}) .$$

Then integrating this twice and comparing with (B.10), we obtain

$$d\underline{Y} = \underline{A} + \hat{\underline{O}}(s^{-2}) \quad (\text{B.12})$$

$$\underline{Y} = s \underline{A} + \underline{B} + \hat{\underline{O}}(s^{-1}) . \quad (\text{B.13})$$

We can now go back to (B.5) and solve for $\underline{H}$. Thus, we have

$$\begin{bmatrix} \rho & \sigma \\ \bar{\sigma} & \rho \end{bmatrix} = \frac{s^2}{|\underline{Y}|} \begin{bmatrix} dy_1 & dy_2 \\ d\bar{y}_1 & d\bar{y}_2 \end{bmatrix} \begin{bmatrix} \bar{y}_2 & -y_2 \\ -\bar{y}_1 & y_1 \end{bmatrix} - \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix}$$

which yields

$$\rho = \frac{s^2(\bar{y}_2 dy_1 - \bar{y}_1 dy_2) - (y_1 \bar{y}_2 - y_2 \bar{y}_1)s}{y_1 \bar{y}_2 - y_2 \bar{y}_1} \quad (\text{B.14})$$

$$\sigma = \frac{s^2(y_1 d\bar{y}_2 - y_2 d\bar{y}_1)}{y_1 \bar{y}_2 - y_2 \bar{y}_1} \quad (\text{B.15})$$

Using (B.12) and (B.13) and letting $|\underline{A}| = \det(\underline{A})$, these become

$$\rho = \frac{(B_2 \bar{A}_1 - B_1 \bar{A}_2)s^2 + \hat{O}(s)}{|\underline{A}| s^2 + (A_1 \bar{B}_2 - A_2 \bar{B}_1 + \bar{A}_2 B_1 - \bar{A}_1 B_2)s + \hat{O}(1)} \quad (\text{B.16})$$

$$\sigma = \frac{(B_1 A_2 - B_2 A_1)s^2 + \hat{O}(s)}{|\underline{A}| s^2 + (A_1 \bar{B}_2 - A_2 \bar{B}_1 + \bar{A}_2 B_1 - \bar{A}_1 B_2)s + \hat{O}(1)} \quad (\text{B.17})$$

Let us first consider the general case, $A \neq 0$, $|\underline{A}| \neq 0$, $\underline{B} \neq 0$. Then (B.16) and (B.17) yield

$$\rho = \frac{(B_2 \bar{A}_1 - B_1 \bar{A}_2)}{|\underline{A}|} + \hat{O}(s^{-1}) \quad (\text{B.18})$$

$$\sigma = \frac{(B_1 A_2 - B_2 A_1)}{|\underline{A}|} + \hat{O}(s^{-1}) \quad (\text{B.19})$$

or

$$\rho = \frac{C}{|\underline{A}|} + O(r), \quad \sigma = \frac{E}{|\underline{A}|} + O(r) \quad (\text{B.20})$$

where

$$C \equiv (B_2 \bar{A}_1 - B_1 \bar{A}_2), \quad E \equiv (B_1 A_2 - B_2 A_1) \quad (\text{B.21})$$

Now, when $\psi_0 = \phi_0 = 0$ the solutions of (2.18a) and 2.18b) are

$$\rho = \frac{\rho^0 - r}{(\rho^0 - r)^2 - \sigma^0 \bar{\sigma}^0}, \quad \sigma = \frac{\sigma^0}{(\rho^0 - r)^2 - \sigma^0 \bar{\sigma}^0}, \quad (\text{B.22})$$

for the case $\rho = \bar{\rho} \neq 0$, $\sigma \neq 0$, $\rho^2 \neq \sigma \bar{\sigma}$ (see Ref. 37). The null geodesics described by (B.22) are what are known as "spherical rays". For small values of r , (B.22) can be expressed as

$$\rho = \frac{\rho^0}{\rho^{02} - \sigma^0 \bar{\sigma}^0} + O(r), \quad \sigma = \frac{\sigma^0}{\rho^{02} - \sigma^0 \bar{\sigma}^0} + O(r) \quad (\text{B.23})$$

which are of the same form as (B.20). Therefore, for ψ_0 and ϕ_0 regular, the general solutions of (B.1a) and (B.1b) correspond to spherical rays near $r = 0$.

Next, we consider three special cases, one of which will lead to the choice we have adopted in our work, namely, $\rho = -r^{-1} + O(r)$ and $\sigma = O(r)$.

I. $\underline{B} = 0$, $\underline{A} \neq 0$, $|\underline{A}| \neq 0$: For this special case, (B.16) and (B.17) yield

$$\rho = \frac{\hat{O}(s)}{|\underline{A}| s^2 + \hat{O}(1)}, \quad \sigma = \frac{\hat{O}(s)}{|\underline{A}| s^2 + \hat{O}(1)} \quad (\text{B.24})$$

i.e.

$$\rho = \hat{O}(s^{-1}) = O(r), \quad \sigma = \hat{O}(s^{-1}) = O(r). \quad (\text{B.25})$$

Since $\rho, \sigma \rightarrow 0$ as $r \rightarrow 0$, this case therefore corresponds locally to "plane rays".³⁷

II. $\underline{B} \neq 0$, $\underline{A} \neq 0$, $|\underline{A}| = 0$: Using (B.21) in (B.16) and (B.17), we get

$$\rho = \frac{C s^2 + \hat{O}(s)}{(\bar{C} - C)s + \hat{O}(1)}, \quad \sigma = \frac{E s^2 + \hat{O}(s)}{(\bar{C} - C)s + \hat{O}(1)} \quad (\text{B.26})$$

which yield

$$\rho = \frac{Cs}{\bar{C} - C} + \hat{O}(1), \quad \sigma = \frac{Es}{\bar{C} - C} + \hat{O}(1) \quad (\text{B.27})$$

Since $\rho = \bar{\rho}$, we must have $C = -\bar{C}$, i.e. $C = i a$, where a is real.

Moreover, since $|\underline{A}| = 0$ or $A_1 \bar{A}_2 - A_2 \bar{A}_1 = 0$ we can write E as

$$E = B_1 \frac{A_1}{\bar{A}_1} \bar{A}_2 - B_2 \frac{A_2}{\bar{A}_2} A_1 = -\frac{A_1}{\bar{A}_1} C = -i a e^{2i\alpha} \quad (\text{B.28})$$

where $A_1 = |A_1| e^{i\alpha}$. Substituting these into (B.27) we obtain

$$\rho = -\frac{1}{2} r^{-1} + O(1), \quad \sigma = \frac{e^{2i\alpha}}{2} r^{-1} + O(1) \quad (\text{B.29})$$

By means of a spatial rotation $\ell^{\mu'} = \ell^\mu$, $n^{\mu'} = n^\mu$, $m^{\mu'} = m^\mu e^{-i\alpha}$, we can set $\sigma = \frac{1}{2} r^{-1} + O(1)$. Then, if we let

$$\rho = -\frac{1}{2} r^{-1} + g, \quad \sigma = \frac{1}{2} r^{-1} + h,$$

where $g, h = O(1)$, and substitute these into (B.1a) and (B.1b), we get

$$Dg + r^{-1} g - \frac{1}{2}(h + \bar{h})r^{-1} = O(1) \quad (\text{B.30a})$$

$$Dh + r^{-1} h - g r^{-1} = O(1) \quad (\text{B.30b})$$

Remembering that g is real because ρ is real, we can combine (B.30a) and (B.30b) to obtain

$$D(h + \bar{h} + 2g) = O(1),$$

which implies that $h + \bar{h} = -2g + O(r)$. Putting this back into (B.30a) we then get

$$Dg + 2r^{-1}g = O(1),$$

with the solution $g = O(r)$. If we now substitute this into (B.30b) and integrate, we also find that $h = O(r)$. Hence, ρ and σ become

$$\rho = -\frac{1}{2r} + O(r), \quad \sigma = \frac{1}{2r} + O(r) \quad (B.31)$$

Locally, these expressions for ρ and σ satisfy $\rho^2 = \sigma\bar{\sigma}$, and so the case $|\underline{A}| = 0$ corresponds to "cylindrical rays" near $r = 0$.³⁷

III. $\underline{B} \neq 0, \underline{A} = 0$: For this case we find from (B.16) and (B.17) that

$$\rho = \hat{O}(s) = O(r^{-1}), \quad \sigma = \hat{O}(s) = O(r^{-1}) \quad (B.32)$$

Substituting this into $\underline{H}$, we get $\underline{H} = \underline{O}(r^{-1})$, and assuming that $\underline{H}$ is non-singular, i.e. $\rho^2 \neq \sigma\bar{\sigma}$, we can express (B.3) as

$$D\underline{H}^{-1} = -\underline{I} - \underline{H}^{-1} \underline{J} \underline{H}^{-1} \quad (B.33)$$

But since $\underline{H}^{-1} = \underline{O}(r)$ and $\underline{J} = \underline{O}(1)$, (B.33) becomes

$$D\underline{H}^{-1} = -\underline{I} + \underline{O}(r^2), \quad (B.34)$$

which can be integrated to yield

$$\underline{H}^{-1} = -\underline{I}r + \underline{O}(r^3), \quad (B.35)$$

the constant of integration being zero because $\underline{H}^{-1} = O(r)$. Taking the inverse of (B.35), we then obtain

$$\underline{H} = -\underline{I}r^{-1} + \underline{O}(r), \quad (B.36)$$

i.e.

$$\rho = -\frac{1}{r} + o(r), \quad \sigma = o(r) \quad (\text{B.37})$$

which is precisely the condition we have adopted in our approach to equations of motion. Since $\rho \neq 0$, $\sigma = 0$ near $r = 0$ according to (B.37), the case $\underline{A} = 0$ therefore corresponds locally to shear-free spherical rays.³⁷

APPENDIX C

In Chapter 4.0 and 5.0 we adopted the assumption $\psi_1^0 = 0$. We now show that at least for empty space the conditions $\rho = -r^{-1} + O(r)$ and $\sigma = O(r)$ lead to the result $\psi_1^0 = 0$.

Consider the integration of the radial equations in Section 4.1 without the assumption $\psi_1^0 = 0$. Starting with (4.20a), i.e.

$$D\psi_1 + 4r^{-1}\psi_1 = O(r^{-2}) \rightarrow \psi_1 = \psi_1^0 r^{-4} + O(r^{-1}) \quad (C.1)$$

we go on to (2.18, d,c,e,)' and (2.16b)' and obtain

$$(2.18d)' \quad D\beta + r^{-1}\beta = \psi_1^0 r^{-4} + O(r^{-1}) \rightarrow \beta = -\frac{1}{2}\psi_1^0 r^{-3} + \beta^0 r^{-1} + O(1) \quad (C.2)$$

$$(2.18c)' \quad D\alpha + r^{-1}\alpha = -\frac{1}{6}\bar{\psi}_0^0 \psi_1^0 r^{-2} - \frac{1}{8}\bar{\psi}_0^1 \psi_1^0 r^{-1} + O(1) + \alpha = -\frac{1}{6}\bar{\psi}_0^0 \psi_1^0 r^{-1} \log r + \alpha^0 r^{-1} - \frac{1}{8}\bar{\psi}_0^1 \psi_1^0 + O(r) \quad (C.3)$$

$$(2.18e) \quad \tau = \bar{\alpha} + \beta = -\frac{1}{2}\psi_1^0 r^{-3} - \frac{1}{6}\psi_0^0 \bar{\psi}_1^0 r^{-1} \log r + O(1), \quad (C.4)$$

where we have set $\tau^0 = \bar{\alpha}^0 + \beta^0 = 0$ by means of a null rotation.

$$(2.16b)' \quad D\omega + r^{-1}\omega = \frac{1}{2}\psi_1^0 r^{-3} + O(r^{-1} \log r) \rightarrow \omega = -\frac{1}{2}\psi_1^0 r^{-2} + \omega^0 r^{-1} + O(\log r) \quad (C.5)$$

From (2.17b), however, we find that

$$\omega^0 = 0, \quad \alpha^0 = -\frac{\partial \rho}{\partial \xi} + \frac{1}{4} \bar{\psi}_0^0 \psi_1^0. \quad (C.6)$$

Using this to integrate the radial equations further, we get the results:

$$\psi_1 = \psi_1^0 r^{-4} - \frac{1}{6} \psi_1^1 \bar{\psi}_1^0 r^{-1} + \frac{1}{6} \psi_1^0 |\psi_0^0|^2 \log r + O(1) \quad (C.7a)$$

$$\alpha = -\frac{1}{6} \bar{\psi}_0^0 \psi_1^0 r^{-1} \log r + \alpha^0 r^{-1} - \frac{1}{8} \bar{\psi}_0^1 \psi_1^0 + O(r) \quad (C.7b)$$

$$\beta = -\frac{1}{2} \psi_1^0 r^{-3} - \bar{\alpha}^0 r^{-1} - \frac{1}{6} \psi_1^1 \bar{\psi}_1^0 + O(r \log r) \quad (C.7c)$$

$$\tau = -\frac{1}{2} \psi_1^0 r^{-3} - \frac{1}{6} \psi_0^0 \bar{\psi}_1^0 r^{-1} \log r - \frac{7}{24} \psi_0^1 \bar{\psi}_1^0 + O(r \log r) \quad (C.7d)$$

$$\omega = -\frac{1}{2} \psi_1^0 r^{-2} + \frac{1}{6} \psi_0^0 \bar{\psi}_0^1 \log r - \frac{1}{3} \psi_0^0 \bar{\psi}_1^0 + \frac{1}{12} \psi_0^1 \bar{\psi}_1^0 r + O(r^2) \quad (C.7e)$$

Then integrating (2.16c) we obtain

$$\begin{aligned} X^A &= \frac{1}{3} \operatorname{Re} (\bar{\psi}_1^0 \xi^{0A}) r^{-3} + \frac{1}{3} \operatorname{Re} (\bar{\psi}_0^0 \psi_1^0 \xi^{0A}) r^{-1} \log r \\ &\quad - \frac{1}{6} \operatorname{Re} (\bar{\psi}_0^0 \psi_1^0 \xi^{0A}) r^{-1} - \frac{2}{3} \operatorname{Re} (\bar{\psi}_0^1 \psi_1^0 \xi^{0A}) \log r + X^{0A} \\ &\quad + O(r \log r) \end{aligned}$$

The next set of radial equations (2.18g)', (2.18h)', and (2.20b)' can be integrated similarly by using the lemma given in Section 4.1 to show that when $\psi_1^0 \neq 0$, $\lambda = O(r^{-4})$, $\mu = O(r^{-6})$ and $\psi_2 = O(r^{-7})$.

Thus, we have the following results:

$$\lambda = -\frac{1}{36} \bar{\psi}_0^0 |\psi_1^0|^2 r^{-3} - \frac{1}{24} \bar{\psi}_0^1 |\psi_1^0|^2 r^{-2} + O[r^{-1} (\log r)^2] \quad (\text{C.7g})$$

$$\begin{aligned} \mu = & \frac{1}{6} |\psi_1^0|^2 r^{-5} - \frac{1}{6} \bar{\psi}_0^0 (\psi_1^0)^2 r^{-3} \log r - \frac{1}{2} [\bar{\psi}_0^1 \psi_1^0 - \frac{1}{3} \bar{\psi}_0^0 (\psi_1^0)^2] r^{-3} \\ & + \frac{1}{6} \text{Re} [\psi_0^1 (\bar{\psi}_1^0)^2] r^{-2} \log r + \left\{ \frac{1}{6} \text{Re} [\psi_0^1 (\bar{\psi}_1^0)^2] - \psi_2^0 \right\} r^{-2} \\ & - \frac{1}{6} [\bar{\psi}_0^1 (\psi_1^0)^2 + \frac{1}{2} |\psi_0^0|^2 |\psi_1^0|^2] r^{-1} \log r + \mu^0 r^{-1} + O(\log r) \end{aligned} \quad (\text{C.7h})$$

$$\begin{aligned} \psi_2 = & -\frac{2}{3} |\psi_1^0|^2 r^{-6} + \frac{1}{3} \bar{\psi}_0^0 (\psi_1^0)^2 r^{-4} \log r + [\bar{\psi}_0^1 \psi_1^0 - \frac{1}{2} \bar{\psi}_0^0 (\psi_1^0)^2] r^{-4} \\ & - \frac{1}{6} \text{Re} [\psi_0^1 (\bar{\psi}_1^0)^2] r^{-3} \log r + \psi_2^0 r^{-3} - \frac{1}{6} [\bar{\psi}_0^1 \psi_1^0] \\ & + \frac{7}{15} |\psi_0^0|^2 |\psi_1^0|^2 r^{-2} + O(r^{-1} \log r) \end{aligned} \quad (\text{C.7i})$$

Finally, the integration of (2.16d) and (2.18f) yields

$$U = -\frac{1}{12} |\psi_1^0|^2 r^{-4} + O(r^{-2} \log r) \quad (\text{C.7j})$$

$$\gamma = \frac{1}{12} |\psi_1^0|^2 r^{-5} + O(r^{-3} \log r), \quad (\text{C.7k})$$

where only the leading terms have been written out explicitly because the other terms will not be required in what follows.

Let us substitute the results (C.7 a-k) together with the expression for ρ , σ and ξ^A given in (4.7), (4.8), and (4.15) into

the non-radial equation (2.19g), namely,

$$U D\sigma + \dot{\sigma} + X^A \sigma_{,A} - \omega D\tau - \xi^A \tau_{,A} + 2\tau\beta + (\mu + \bar{\gamma} - 3\gamma)\sigma + \bar{\lambda}\rho = 0 \quad .$$

Taking only the leading terms we obtain

$$\frac{5}{4} (\psi_1^0)^2 r^{-6} + O(r^{-4} \log r) = 0, \quad (C.8)$$

which implies that $\psi_1^0 = 0$. Therefore, under the assumption that $\rho = -r^{-1} + O(r)$ and $\sigma = O(r)$, the coefficient of r^{-4} in the Weyl tensor vanishes, which means that the elementary singularity cannot have an intrinsic mass dipole moment and angular momentum. It is our guess that this result for empty space also holds in the Einstein-Maxwell theory.

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